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Minimax Rates and Improved Spectral Convergence for Multi-Manifold Clustering

Haoyu Chen* HAOYU.CHEN@UTAH.EDU
Department of Mathematics
University of Utah
Salt Lake City, UT 84112, USA

Chenghui Li* CLI539@WISC.EDU
Department of Statistics
University of Wisconsin Madison
Madison, WI 53706, USA

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Abstract

We study eigenpair estimation for the tensorized, density-weighted Laplace–Beltrami operator from n independent data points sampled from a multi-manifold mixture distribution where manifolds may intersect. Let d denote the largest intrinsic dimension of the manifold components and $s \geq 2$ the smoothness of the sampling densities. For simple, nontrivial eigenpairs, we establish the minimax rate $n^{-s/(d+2s)}$ for approximating the eigenvalue and eigenfunction of the tensorized Laplace–Beltrami operator, and show that a continuum plug-in estimator based on kernel density estimation attains the same rate when the supporting manifolds are known. We further analyze a manifold-agnostic graph Laplacian estimator and prove that its eigenpairs converge to those of the tensorized Laplace–Beltrami operator at rate $n^{-1/(2.5d-1)}$, up to logarithmic factors, improving upon previously known spectral convergence rates for multi-manifold learning. Numerical experiments on synthetic multi-manifold data corroborate the rates guaranteed by our theorems. These results characterize the information-theoretic difficulty of multi-manifold spectral estimation and quantify the additional cost of learning without oracle component information.

Keywords: manifold learning, multi-manifold clustering, harmonic analysis

1 Introduction

Multi-manifold clustering (MMC) is the unsupervised learning problem of partitioning observations generated from a union of low-dimensional manifolds according to their latent manifold membership. A central challenge is that the manifold components need not be convex or well separated in the ambient space: individual components may exhibit nonlinear intrinsic geometries, and distinct manifolds may approach or intersect one another. This problem captures heterogeneous geometric structure that arises naturally in motion segmentation and optical flow (Costeira and Kanade, 1998; Vidal, 2011; Elhamifar and Vidal, 2013), face and image analysis (Ho et al., 2003; Wright et al., 2008; Liu et al., 2012), dynamic pattern recognition (Tomasi and Kanade, 1992; Vidal et al., 2005), and document

*. Equal contribution.

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classification (Belkin and Niyogi, 2001; Ji et al., 2017). Identifying the manifold components is often a prerequisite for downstream tasks such as dimension reduction, geometric representation, classification, and componentwise inference.

Most algorithms for MMC, such as that of Goh and Vidal (2007), assume that manifold components do not intersect. In contrast, methods that address the setting considered here (Wang et al., 2011; Arias-Castro et al., 2013; Wang et al., 2015; Babaeian, 2018; Trillos et al., 2023; Chen et al., 2025) exploit local tangent information, constrained paths, or higher-order simplicial geometry to suppress spurious cross-manifold connections while preserving connectivity within each component. Nevertheless, designing computationally practical algorithms that provably remove spurious cross-manifold connections without distorting the within-manifold geometry therefore remains challenging.

Quantitative finite-sample guarantees are scarce among the existing methods tailored to intersecting manifolds. Despite their substantially different constructions, Arias-Castro et al. (2013) and Chen et al. (2025) identified the same critical local scale of order $(\log n/n)^{1/d}$ for successfully discriminating intersecting manifold components with high probability under small noise perturbations; here d denotes the largest intrinsic dimension among the manifold components. By contrast, the probability bound in Theorem 3.1 of Wang et al. (2015) leads to the slower-shrinking neighborhood scale $(\log n/n)^{1/(d+2)}$. The curvature-constrained path method of Babaeian (2018) provides a qualitative asymptotic guarantee for selecting landmarks away from manifold intersections, but no explicit sample-size rate for the final clustering error. Bungert and Slepčev (2026) establishes Γ-convergence result for discrete normalized Dirichlet energy without providing an explicit convergence rate for spectral convergence, in which work the normalized graph Laplacian is considered. The most directly comparable result is Trillos et al. (2023), which establishes spectral convergence of an angle-constrained graph Laplacian toward the tensorized Laplace–Beltrami operator, with eigenvalue and eigenfunction errors decaying at rates $n^{-1/(3d-1)}$ and $n^{-1/(6d-2)}$, respectively, up to logarithmic factors. Nevertheless, the information-theoretic limits of multi-manifold spectral estimation remain uncharacterized, and the existing spectral convergence rates remain substantial room for improvement.

To formalize the problem, we assume that the data are sampled from a mixture distribution supported on $\mathcal{M} := \cup_{j=1}^{N} \mathcal{M}_j$, where $\mathcal{M}_j$ is a d_j-dimensional smooth, compact, and connected Riemannian manifold without boundary embedded in $\mathbb{R}^D$; the distinct components $\mathcal{M}_j$ may intersect. We set $d := \max_{j\in[N]} d_j \ll D$. The goal of MMC is to recover the latent assignment of each sample to its generating manifold component.

Following Trillos et al. (2023); Bungert and Slepčev (2026), we take the tensorized Laplace-Beltrami operator $\Delta_{\mathcal{M}}$, which acts independently on individual manifold components, as a natural spectral target for MMC. Since each $\mathcal{M}_j$ is connected, the nullspace of $\Delta_{\mathcal{M}}$ is spanned by the component indicator functions and thus encodes the latent membership information of $\mathcal{M}_j$. We formally define this operator and describe its spectral properties in Section 1.1. In this work, we characterize the convergence rates for the eigenvalues and eigenfunctions of $\Delta_{\mathcal{M}}$.

Our first contribution establishes, to our knowledge, the first minimax lower bound for spectral estimation in MMC; see Theorem 4. For a simple, nontrivial eigenpair of $\Delta_{\mathcal{M}}$, this lower bound scales as $n^{-s/(d+2s)}$ where s denotes the smoothness of the sampling densities. In particular, when $s = 2$, this rate coincides with the minimax rate established

by Trillos et al. (2026) for a single d-dimensional flat torus. In a related setting, Chaubet and Divol (2025) consider a different partial differential operator on a given single manifold and establish the minimax rates for eigenvalue and eigenfunction estimation in the L^2 norm.

Our second contribution establishes that this lower bound is sharp when the manifold is known; see Theorem 6. In the oracle setting, a continuum plug-in estimator based on kernel density estimation (KDE) attains the matching rate. Knowledge of the manifold components enables componentwise estimation even when the manifolds intersect. However, since the manifold components are unknown in the original MMC problem, the result serves as a statistical benchmark rather than a guarantee for a practically implementable estimator.

Our third contribution sharpens the spectral convergence guarantees of Trillos et al. (2023) for the annular proximity graph with angle constraints; see Theorem 7 and see Section C.1 for a description of the graph construction. This practical graph-based estimator requires knowledge of the manifolds, and its construction preserves *within-manifold connectivity* and suppresses *spurious cross-manifold connections*—the two sufficient conditions for spectral convergence. Our key refinement is a direct residual comparison: using self-adjointness, we test the graph eigenvalue equation against the target population eigenfunction and decompose the resulting operator error into within- and between-manifold contributions. The analysis accommodates intersecting components of possibly different intrinsic dimensions, with the rate governed by the largest dimension d. For $d \geq 2$, the eigenpair errors both decay at the rate $n^{-1/(2.5d-1)}$, up to logarithmic factors. Although slower than the oracle rate, this guarantee holds without oracle component information.

Our fourth contribution is numerical: we corroborate the theoretical predictions in Section 4. We examine the minimax lower bound through the oracle KDE estimator and the practical upper bound through the graph-based estimator. Samples are drawn from intersecting flat tori with nonuniform sampling densities in dimensions $d \in \{1, 2, 3\}$, and errors are reported across multiple eigenpairs. The observed oracle convergence rates closely match the minimax rates predicted by the lower bound, while the graph-based experiments corroborate the practical upper bound and suggest room for further improvement. Together, our results establish a sharp oracle benchmark for MMC estimation, characterize the intrinsic statistical difficulty of multi-manifold spectral estimation, and quantify the gap between this benchmark and what the state-of-the-art MMC algorithm achieves.

1.1 Notation and Assumptions

For $N \in \mathbb{N}$, write $[N] := \{1, \ldots, N\}$. Throughout, ℓ indexes eigenpairs and $j \in [N]$ indexes the manifold components. Set $d_j := \dim(\mathcal{M}_j)$ and $d := \max_{j\in[N]} d_j$. While the components may intersect in the ambient space, functions and differential operators on $\mathcal{M}$ are understood componentwise, or equivalently on the abstract disjoint union $\sqcup_{j=1}^{N} \mathcal{M}_j$. Thus, a function f_ℓ on $\mathcal{M}$ is identified with a tuple $f_\ell = (f_{\ell,1}, \ldots, f_{\ell,N})$, where $f_{\ell,j} : \mathcal{M}_j \to \mathbb{R}$ denotes its j-th component. The following geometric assumption is imposed throughout.

Assumption 1 (Manifold geometry) *There exist uniform positive constants such that every component $\mathcal{M}_j$ has injectivity radius bounded below, sectional curvature bounded in absolute value, and Riemannian volume bounded both above and below. We denote by* $\mathbf{M}$ *the collection of all multi-manifold configurations satisfying these bounds.*

If the manifold components ever intersect, we assume the intersections are nontrivial.

Assumption 2 (Component intersections) *For every $j \neq k$, the intersection $\mathcal{M}_{jk} := \mathcal{M}_j \cap \mathcal{M}_k$ is either empty or a smooth submanifold of dimension $d_{jk} < \min\{d_j, d_k\}$. For $x \in \mathcal{M}_{jk}$, let $T_x\mathcal{M}_{jk}^{\perp,j}$ and $T_x\mathcal{M}_{jk}^{\perp,k}$ denote the orthogonal complements of $T_x\mathcal{M}_{jk}$ in $T_x\mathcal{M}_j$ and $T_x\mathcal{M}_k$. There exists a fixed $\beta < \pi/2$ such that for every $\mathcal{M}_{jk}$ and every $x \in \mathcal{M}_{jk}$*

$$\sup_{v \in T_x\mathcal{M}_{jk}^{\perp,j}, \tilde{v} \in T_x\mathcal{M}_{jk}^{\perp,k}, \|v\|_g = \|\tilde{v}\|_g = 1} \left| \angle(v, \tilde{v}) - \frac{\pi}{2} \right| \leq \beta.$$

For each component, let $d\mathrm{vol}_g$ denote the associated Riemannian volume measure, and define the probability measure $d\mu_j := \rho_j \, d\mathrm{vol}_g$. The global sampling measure is the mixture

$$\mu := \sum_{j=1}^{N} w_j \mu_j, \qquad w_j > 0, \qquad \sum_{j=1}^{N} w_j = 1.$$

We write $w_{\min} := \min_j w_j$. The following regularity condition on ρ_j is imposed throughout.

Assumption 3 (Density class) *For an integer $s \geq 2$, let $\mathcal{P}_{\mathcal{M}_j}^{(s)}$ be the collection of strictly positive probability densities $\rho_j \in C^s(\mathcal{M}_j)$ satisfying*

$$0 < \rho_{\min} \leq \rho_j(x) \leq \rho_{\max}, \qquad \int_{\mathcal{M}_j} \rho_j d\mathrm{vol}_g = 1, \qquad \|\nabla_g^k \rho_j\|_{L^\infty(\mathcal{M}_j)} \leq C_\rho^{(k)},$$

for $k = 0, \ldots, s$. The corresponding multi-manifold density class is $\mathcal{P}_{\mathcal{M}}^{(s)} := \prod_{j=1}^{N} \mathcal{P}_{\mathcal{M}_j}^{(s)}$.

Let $\mathcal{X}_n := \{X_1, \ldots, X_n\}$ be an independent sample from μ, and we write

$$\mathcal{X}_{n,j} := \mathcal{X}_n \cap \mathcal{M}_j, \qquad \widehat{n}_j := |\mathcal{X}_{n,j}|, \qquad n_j := n w_j.$$

Unless otherwise specified, probability $\mathbb{P}$ and expectation $\mathbb{E}$ are taken with respect to the joint distribution of $\mathcal{X}_n$.

For $x, y \in \mathcal{M}_j$, let $d_g(x, y)$ denote their geodesic distance and let $\|x - y\|$ denote their ambient Euclidean distance. We write $T_x\mathcal{M}_j$ for the tangent space at x, and $\langle \cdot, \cdot \rangle_g$ and $\| \cdot \|_g$ for the Riemannian inner product and norm. The Euclidean and geodesic balls of radius r centered at x are denoted by $B(x, r)$ and $B_g(x, r)$, respectively. We use $\exp_x$ for the Riemannian exponential map and write $\log_x(y) := \exp_x^{-1}(y)$. The symbols ∇_g, ∇_g^k, and div_g denote the Riemannian gradient, the k-th covariant derivative, and the Riemannian divergence, respectively.

For componentwise functions f_ℓ, h_ℓ with components $f_{\ell,j}, h_{\ell,j} : \mathcal{M}_j \to \mathbb{R}$, define the standard Riemannian L^2 and H^1 norm by

$$\|f_{\ell,j}\|_{L^2(\mathcal{M}_j)}^2 := \langle f_{\ell,j}, f_{\ell,j} \rangle_{L^2(\mathcal{M}_j)}, \qquad \|f_{\ell,j}\|_{H^1(\mathcal{M}_j)}^2 := \|f_{\ell,j}\|_{L^2(\mathcal{M}_j)}^2 + \|\nabla_g f_{\ell,j}\|_{L^2(\mathcal{M}_j)}^2$$

with $\langle f_{\ell,j}, h_{\ell,j} \rangle_{L^2(\mathcal{M}_j)} := \int_{\mathcal{M}_j} f_{\ell,j} h_{\ell,j} \, d\,\mathrm{vol}_g$. The unweighted global counterparts are defined by

$$\|f_{\ell,j}\|_{L^2(\mathcal{M})}^2 := \sum_{j=1}^{N} \|f_{\ell,j}\|_{L^2(\mathcal{M}_j)}^2, \qquad \|f_\ell\|_{H^1(\mathcal{M})}^2 := \sum_{j=1}^{N} \|f_{\ell,j}\|_{H^1(\mathcal{M}_j)}^2.$$

The density-weighted componentwise and global L^2-inner products are defined by

$$\langle f_{\ell,j}, h_{\ell,j}\rangle_{L^2(\mu_j)} = \int_{\mathcal{M}_j} f_{\ell,j} h_{\ell,j} d\mu_j, \qquad \langle f_\ell, h_\ell\rangle_{L^2(\mu)} := \sum_{j=1}^N w_j \langle f_{\ell,j}, h_{\ell,j}\rangle_{L^2(\mu_j)}.$$

The corresponding norms are defined analogously. Unless a measure is displayed explicitly, norms default to the Riemannian volume measure $d\mathrm{vol}_g$.

For functions $u, v : \mathcal{X}_{n,j} \to \mathbb{R}$, define the empirical inner product and norm by

$$\langle u, v\rangle_{L^2(\mathcal{X}_{n,j})} := \frac{1}{\widehat{n}_j} \sum_{x\in\mathcal{X}_{n,j}} u(x)v(x), \qquad \|u\|^2_{L^2(\mathcal{X}_{n,j})} := \langle u, u\rangle_{L^2(\mathcal{X}_{n,j})}.$$

The global empirical quantities on $\mathcal{X}_n$ are defined analogously, normalized by n instead.

For each component, define the density-weighted Laplace–Beltrami operator

$$\Delta_{\mathcal{M}_j} f_{\ell,j} \;:=\; -\frac{1}{\rho_j} \operatorname{div}_g\big(\rho_j^2 \nabla_g f_{\ell,j}\big), \qquad f_{\ell,j} : \mathcal{M}_j \to \mathbb{R}. \tag{1}$$

The population operator is the weighted direct sum $\Delta_{\mathcal{M}} := \bigoplus w_j \Delta_{\mathcal{M}_j}$, or equivalently,

$$\Delta_{\mathcal{M}} f_\ell := (w_1 \Delta_{\mathcal{M}_1} f_{\ell,1}, w_2 \Delta_{\mathcal{M}_2} f_{\ell,2}, \ldots, w_N \Delta_{\mathcal{M}_N} f_{\ell,N}).$$

Throughout, (λ_ℓ, f_ℓ) denotes the ℓ-th population eigenpair of $\Delta_{\mathcal{M}}$, so an equivalence between the global eigen-equation and the system of componentwise equations is characterized by

$$\Delta_{\mathcal{M}} f_\ell = \lambda_\ell f_\ell, \qquad \Longleftrightarrow \qquad w_j \Delta_{\mathcal{M}_j} f_{\ell,j} = \lambda_\ell f_{\ell,j}, \quad j \in [N].$$

For each $j \in [N]$, set $\lambda_{\ell,j} := \lambda_\ell / w_j$ so that whenever $f_{\ell,j} \not\equiv 0$, the pair $(\lambda_{\ell,j}, f_{\ell,j})$ is a component eigenpair of $\Delta_{\mathcal{M}_j}$. For the eigenvalues, the spectrum of $\Delta_{\mathcal{M}}$ is the multiset union of the weighted component spectra, counting multiplicity and arranged in nondecreasing order. Since each $\mathcal{M}_j$ is connected, each $\Delta_{\mathcal{M}_j}$ has a one-dimensional nullspace of constant functions associated with eigenvalue zero. Consequently

$$0 = \lambda_1 = \cdots = \lambda_N < \lambda_{N+1} \le \lambda_{N+2} \le \cdots .$$

For the eigenfunctions, we normalize f_ℓ according to

$$\|f_\ell\|^2_{L^2(\mu)} = \sum_{j=1}^N w_j \|f_{\ell,j}\|^2_{L^2(\mu_j)} = 1.$$

Elliptic regularity for the componentwise uniformly elliptic equations on compact manifolds guarantees that $f_{\ell,j} \in C^2(\mathcal{M}_j)$ for all $j \in [N]$. The energy identity

$$\lambda_{\ell,j} \|f_{\ell,j}\|^2_{L^2(\mu_j)} = \big\langle f_{\ell,j}, \Delta_{\mathcal{M}_j} f_{\ell,j}\big\rangle_{L^2(\mu_j)} = \int_{\mathcal{M}_j} \rho_j^2 \|\nabla_g f_{\ell,j}\|_g^2 \, d\,\mathrm{vol}_g \tag{2}$$

together with the density bounds, yields

$$\|\nabla_g f_{\ell,j}\|_{L^2(\mathcal{M}_j)} \lesssim \lambda_{\ell,j}^{1/2} \|f_{\ell,j}\|_{L^2(\mathcal{M}_j)} \lesssim w_j^{-1/2} \lambda_{\ell,j}^{1/2}, \tag{3}$$

while the corresponding second-order elliptic estimate gives

$$\|\Delta_g f_{\ell,j}\|_{L^2(\mathcal{M}_j)} + \|\nabla_g^2 f_{\ell,j}\|_{L^2(\mathcal{M}_j)} \lesssim (1+\lambda_{\ell,j})\|f_{\ell,j}\|_{L^2(\mathcal{M}_j)} \lesssim w_j^{-1/2}(1+\lambda_{\ell,j}). \tag{4}$$

Here, the implicit constants depend only on the uniform geometric and density-class parameters. Since $w_{\min} > 0$, these weight factors are uniformly bounded and may be absorbed into the implicit constants. By the norm equivalence stated above, the same estimates hold with the corresponding weighted $L^2(\mu_j)$ norms. Finally, since the components are compact and $N < \infty$, the preceding C^2-regularity implies

$$C_f^{(\nu)} := \max_{j\in[N]} \|\nabla_g^\nu f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)} < \infty, \qquad \text{for } \nu = 0,1,2. \tag{5}$$

For quantitative L^∞ estimates for Laplace–Beltrami eigenfunctions, see Sogge (2001).

Define the index set associated with λ_ℓ and the corresponding population eigenspace by

$$[\ell]_\lambda := \{k : \lambda_k = \lambda_\ell\}, \qquad E_\ell := \ker(\Delta_\mathcal{M} - \lambda_\ell I) = \operatorname{span}\{f_k : k \in \mathcal{I}_\ell\}.$$

The eigengap separating E_ℓ from the remainder of the spectrum is defined by

$$\gamma_\ell := \operatorname{dist}(\lambda_\ell, \operatorname{spec}(\Delta_\mathcal{M}) \setminus \{\lambda_\ell\}) = \min_{k\notin[\ell]_\lambda} |\lambda_k - \lambda_\ell|. \tag{6}$$

Simple eigenfunctions are compared modulo sign, whereas for a repeated eigenvalue, comparison with an estimated eigenfunction is understood through the optimal alignment

$$\operatorname{dist}(\widehat{f}_\ell, E_\ell) := \inf_{f\in E_\ell,\, \|f\|_{L^2(\mu)}=1} \|\widehat{f}_\ell - f\|.$$

Here and below, we use $(\widehat{\lambda}_\ell, \widehat{f}_\ell)$ to denote the empirical estimate of (λ_ℓ, f_ℓ).

For a symmetric weighted graph on $\mathcal{X}_n$, let $W = (W_{ik})_{i,k=1}^n$ denote its weight matrix and let D be the diagonal matrix with $D_{ii} := \sum_{k=1}^n W_{ik}$. We write $\mathcal{L} := D - W$ for its graph Laplacian. The graph Laplacian operator $\mathcal{L}_{n,\varepsilon} : L^2(\mathcal{X}_n) \to L^2(\mathcal{X}_n)$ can be defined as

$$\mathcal{L}_{n,\varepsilon} f(x) := \frac{2}{C_d n \varepsilon^{d+2}} \sum_{y\in\mathcal{X}_n} W_{x\sim y}(f(x) - f(y)).$$

In the practical upper bound, W is constructed from the annular proximity graph with angle constraints; its precise definition is given in Appendix C.1.

For the graph construction, let $\eta : [0,\infty) \to [0,\infty)$ be a bounded, non-increasing, Lipschitz kernel profile supported on $[0,1]$, and define $\eta_h(t) := \eta(t/h)$ for bandwidth $h > 0$. For an intrinsic dimension d_j, define

$$\sigma_{\eta,d_j} := \int_{\mathbb{R}^{d_j}} z_1^2 \eta(\|z\|) dz = \frac{1}{d_j} \int_{\mathbb{R}^{d_j}} \|z\|^2 \eta(\|z\|)\, dz. \tag{7}$$

For an eigenpair supported on $\mathcal{M}_j$, we abbreviate σ_{η,d_j} as σ_η when the dimension is clear. In this work, we consider $\eta(t) = 1_{t\le 1}$.

For two nonnegative quantities a and b, we write $a \lesssim b$ if $a \le Cb$ for a constant $C > 0$ independent of the sample size and all asymptotically vanishing parameters. We write $a \gtrsim b$ if $b \lesssim a$, and $a \asymp b$ if both inequalities hold. The symbols $C, c > 0$ denote generic constants whose values may change from line to line and may depend only on the fixed geometric and density-class parameters, the kernel profile, and the eigenvalue index under consideration.

1.2 Main Results

We first study the minimax estimation of a nontrivial eigenpair of $\Delta_{\mathcal{M}}$. The analysis consists of an information-theoretic lower bound and a matching oracle upper bound. To establish the lower bound over $\mathcal{P}_{\mathcal{M}}^{(s)}$ for an integer $s \geq 2$, we restrict attention to a *simple, nontrivial* global eigenpair (λ_ℓ, f_ℓ) with $\ell \geq N+1$. Simplicity of λ_ℓ ensures a strictly positive eigengap $\gamma_\ell > 0$. The direct-sum structure implies that there exists a unique $j \in [N]$ such that

$$f_{\ell,j} \not\equiv 0, \qquad \text{and} \qquad f_{\ell,k} = 0 \quad \text{for every } k \neq j.$$

In this case, we say f_ℓ is componentwise supported on $\mathcal{M}_j$. Our first theorem gives a lower bound over the full model class $\mathbf{M}$ and applies to every measurable estimator based on $\mathcal{X}_n$.

Theorem 4 (Spectral convergence lower bound) *Suppose Assumptions 1 and 3 hold with $\rho \in \mathcal{P}_{\mathcal{M}}^{(s)}$ for an integer $s \geq 2$. For $\ell \geq N+1$, let (λ_ℓ, f_ℓ) be a simple, nontrivial eigenpair of $\Delta_{\mathcal{M}}$, componentwise supported on $\mathcal{M}_j$. Then there exists $c_\ell > 0$, depending on the geometric and density-class parameters, the index ℓ, and the eigengap γ_ℓ, such that*

$$\inf_{(\widehat{\lambda}_\ell, \widehat{f}_\ell)} \sup_{\mathcal{M} \in \mathbf{M},\, \rho \in \mathcal{P}_{\mathcal{M}}^{(s)}} \mathbb{E}\left[|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M})} \right] \geq c_\ell \max_j \sqrt{\lambda_\ell\, w_j}\, n_j^{-s/(d_j+2s)}, \tag{8}$$

where the infimum on the left is taken over all measurable estimators based on $\mathcal{X}_n$ and the maximum on the right is taken over all admissible supporting components.

The H^1-based eigenfunction loss arises naturally from our lower-bound construction and is consistent with the single-manifold minimax formulation. Replacing it with the L^2-loss gives a slightly weaker eigenfunction-separation rate.

Corollary 5 (L^2 spectral lower bound) *Under the assumptions of Theorem 4, one has*

$$\inf_{(\widehat{\lambda}_\ell, \widehat{f}_\ell)} \sup_{\mathcal{M} \in \mathbf{M},\, \rho \in \mathcal{P}_{\mathcal{M}}^{(s)}} \mathbb{E}\left[|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{L^2(\mathcal{M})} \right] \geq c_\ell \max_j \sqrt{\lambda_\ell\, w_j}\, n_j^{-(s+1)/(d_j+2s)}.$$

The lower-bound rate is attained by an oracle plug-in eigenpair estimator based on kernel density estimation (KDE), which is briefly reviewed in Section A.2.

Theorem 6 (Oracle plug-in upper bound) *Suppose Assumptions 1 and 3 hold with $\rho \in \mathcal{P}_{\mathcal{M}}^{(s)}$ for an integer $s \geq 2$. Let (λ_ℓ, f_ℓ) be a simple, nontrivial global eigenpair of $\Delta_{\mathcal{M}}$ componentwise supported on $\mathcal{M}_j$. Assume that $\mathcal{M}_j$ is known and that the samples in $\mathcal{X}_{n,j}$ can be identified. Let $\widehat{\rho}_j$ be the KDE from the $\widehat{n}_j$ observed samples on $\mathcal{M}_j$, with bandwidth*

$$h_j \asymp (n_j \vee 1)^{-1/(2s+d_j)}.$$

Let $(\widehat{\lambda}_\ell, \widehat{f}_\ell)$ be the plug-in eigenpair obtained by replacing ρ_j with $\widehat{\rho}_j$ in the continuum eigenvalue problem. Then there exists $C_\ell > 0$, depending only on the geometric and density-class parameters, the index ℓ, and the eigengap γ_ℓ, such that

$$\mathbb{E}\left[|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M})} \right] \leq C_\ell \max_j n_j^{-s/(d_j+2s)}.$$

We next turn to the practical setting in which neither the component manifolds nor the latent component labels are known. The main difficulty arises near manifold intersections, where ambient proximity alone may create spurious edges between points sampled from different components. To this end, we apply the annular proximity graph with angle constraints (Section C.1) for graph construction. We establish a sharper spectral convergence rate for the resulting graph Laplacian.

Unlike the population operator $\Delta_{\mathcal{M}}$, the finite-sample graph Laplacian need not have N exact zero eigenvalues when between-manifold connections are present. Accordingly, rather than restricting to the nontrivial range $\ell \geq N+1$ considered in the lower-bound analysis, we state the practical upper-bound result for any fixed spectral index $\ell \geq 1$. We impose neither a simplicity assumption on λ_ℓ nor the requirement that f_ℓ be componentwise supported on a single manifold. We only assume, in addition to Assumption 3, that $\rho \in C^{2,\alpha}$ for some Hölder exponent $\alpha > 0$ so that

$$B_\ell := \max_{j\in[N]} \left(\sum_{\nu=0}^{3} \|\nabla_g^\nu f_\ell\|^2_{L^\infty(\mathcal{M}_j)} \right)^{1/2} < \infty. \tag{9}$$

Theorem 7 (Graph-Laplacian spectral convergence) *Suppose Assumptions 1 and 3 hold with $\rho \in C^{2,\alpha}$, and recall that $d := \max_{j\in[N]} d_j \geq 2$. Fix $\ell \geq 1$ and let $(\widehat{\lambda}_\ell, \widehat{f}_\ell)$ be the corresponding normalized eigenpair of the graph Laplacian $\mathcal{L}$ based on the sample $\mathcal{X}_n$. Then there exists a normalized population eigenpair (λ_ℓ, f_ℓ) such that, provided $B_\ell < \infty$,*

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \leq C_\ell B_\ell \Big(\frac{\log n}{n}\Big)^{\frac{1}{2.5d-1}}, \qquad \|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} \leq C_\ell B_\ell \Big(\frac{\log n}{n}\Big)^{-\frac{1}{2.5d-1}} \log(n),$$

with probability at least $1 - Cn^{-2}$. Here the constant $C_\ell > 0$ depends only on the geometry class $\mathbf{M}$, the intersection parameters, and the index ℓ.

Proof See Section 3. ∎

The remainder of the paper is organized as follows. Section 2 establishes the minimax lower bound via Fano's method and concludes with the oracle plug-in KDE upper bound. Section 3 analyzes the graph Laplacian by controlling its within- and between-manifold contributions. Section 4 numerically tests both the minimax lower bound and the practical upper bound on synthetic data sampled from two intersecting flat tori with prescribed smooth densities. Section 5 concludes with directions for future work.

2 Minimax Lower Bound and an Oracle Upper Bound

This section establishes the minimax lower bound in Theorem 4 and the matching oracle upper bound in Theorem 6. We apply Fano's method to construct the lower bound where a brief review of Fano's method is given in Appendix A.1; see Wainwright (2019) for a comprehensive treatment of minimax lower bounds. The oracle upper bound is achieved by the KDE estimator, which is presented in Section A.2.

For each $\rho \in \mathcal{P}_{\mathcal{M}}^{(s)}$ with $s \geq 2$, our parameter of interest is the selected eigenpair

$$\theta(\rho) := (\lambda_\ell(\rho), f_\ell(\rho)) \in \mathbb{R} \times H^1(\mathcal{M}), \qquad \ell \geq N+1.$$

Given $\mathcal{X}_n \sim \mu_\rho^{\otimes n}$, we consider an arbitrary measurable estimator

$$\widehat{\theta}(\mathcal{X}_n) := (\widehat{\lambda}_\ell(\mathcal{X}_n), \widehat{f}_\ell(\mathcal{X}_n)) \in \mathbb{R} \times H^1(\mathcal{M}).$$

We measure the estimation error using

$$d_{\mathrm{eig}}\big(\theta(\rho), \widehat{\theta}(\mathcal{X}_n)\big) := |\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M})}, \qquad \ell \geq N+1.$$

Dependence on ρ and $\mathcal{X}_n$ is suppressed when clear. Since the target eigenfunction is componentwise supported on $\mathcal{M}_j$, setting the remaining components of any estimated eigenfunction to zero cannot increase the global H^1-loss. It therefore suffices to consider estimators $\widehat{f}_\ell$ satisfying $\widehat{f}_{\ell,k} \equiv 0$ for every $k \neq j$, in which case

$$d_{\mathrm{eig}}(\theta(\rho), \widehat{\theta}(\mathcal{X}_n)) = |\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M}_j)}.$$

For any subclass $\mathcal{Q} \subseteq \mathcal{P}_{\mathcal{M}}^{(s)}$, define the minimax risk

$$\mathfrak{M}_n\big(\theta(\mathcal{Q}); d_{\mathrm{eig}}\big) := \inf_{\widehat{\theta}} \sup_{\rho \in \mathcal{Q}} \mathbb{E}_{\mathcal{X}_n \sim \mu_\rho^{\otimes n}} \big[d_{\mathrm{eig}}\big(\theta(\rho), \widehat{\theta}(\mathcal{X}_n)\big)\big],$$

where the infimum is taken over all measurable estimators. This quantity is the smallest worst-case eigenpair estimation error achievable from n independent observations. Fano's method lower bounds this risk by reducing estimation to a finite multi-hypothesis testing problem. Specifically, we construct, via perturbations of the density on $\mathcal{M}_j$, a finite family

$$\mathcal{P}_M := \{\rho^{(1)}, \ldots, \rho^{(M)}\} \subset \mathcal{P}_{\mathcal{M}}^{(s)}$$

while keeping the densities on all remaining components fixed. The construction is designed to satisfy the two conditions (80) and (81) required by Fano's method. Fano's inequality then gives a minimax lower bound. As $\mathcal{P}_M \subset \mathcal{P}_{\mathcal{M}}^{(s)}$, the resulting lower bound also applies to the full density class. The perturbation amplitudes are additionally chosen small enough that the target eigenpair remains simple and satisfies the uniform eigengap condition (6) throughout the constructed family. The remainder of the lower-bound proof constructs these perturbations and verifies the required spectral-separation and KL-divergence bounds. The construction of the distribution family is presented in Section 2.1, and verification of Fano's conditions is carried out in Section 2.2. The proof of Theorem 4, Corollary 5, and a matching oracle upper bound specified in Theorem 6 are completed in Section 2.3.

2.1 Distribution Construction

Throughout, $m \in \mathbb{N}$ is a sufficiently large scale parameter and $r := m^{-1}$. We fix a reference density tuple $\rho \in \mathcal{P}_{\mathcal{M}}^{(s)}$ satisfying the defining bounds with uniform slack throughout.

2.1.1 Local Geometric Decomposition

For m sufficiently large, $r = m^{-1}$ lies below the injectivity radius of $\mathcal{M}_j$ and geodesic balls of radius r are contained in normal coordinate neighborhoods. Choose a maximal r-separated set of points $\{p_i\}_{i=1}^K \subset \mathcal{M}_j$ so that

$$d_g(p_i, p_{i'}) \geq r \quad \text{for all } i \neq i', \qquad \text{and} \qquad \mathcal{M}_j = \bigcup_{i=1}^K B_g(p_i, r).$$

Uniform volume comparison and packing estimates give $K \asymp r^{-d_j} = m^{d_j}$, where the implicit constants depend only on the geometric bounds of $\mathcal{M}_j$. Assigning equidistant points to one of the minimizing centers, define the associated Voronoi cells by

$$\Omega_i := \big\{x \in \mathcal{M}_j : \ d_g(x, p_i) \leq d_g(x, p_{i'}) \text{ for all } i' \neq i\big\}.$$

The cells $\{\Omega_i\}_{i=1}^K$ form a measurable partition of $\mathcal{M}_j$ and satisfy, uniformly in i,

$$B_g(p_i, r/2) \subset \Omega_i \subset B_g(p_i, r), \qquad \mathrm{vol}_g(\Omega_i) \asymp r^{d_j}, \qquad \mathrm{diam}_g(\Omega_i) \lesssim r.$$

Perturbation templates constructed below have pairwise disjoint supports within these cells.

For sufficiently large m, the exponential map defines a normal-coordinate chart

$$\exp_{p_i} : B(0, r) \subset T_{p_i}\mathcal{M}_j \simeq \mathbb{R}^{d_j} \longrightarrow B_g(p_i, r). \tag{10}$$

All local constructions are performed in these charts, allowing Euclidean constructions in the coordinate domain to be transferred to $\mathcal{M}_j$ with uniformly controlled geometric distortion.

2.1.2 Construction of Perturbation Templates

We first construct a mean-zero Euclidean profile whose directional derivative has opposite signs on two separated regions. Fix $e \in \mathbb{S}^{d_j-1}$ and write $u = te + v$, where $t \in \mathbb{R}$ and $v \in e^\perp$. Choose an even function $q \in C_c^\infty((-a, a))$ such that

$$\int_{\mathbb{R}} q(t)dt = 0, \qquad \text{and} \qquad q'(t_0) > 0$$

for some $t_0 \in (0, a)$. Such a function exists: take any nonzero even function $\varphi \in C_c^\infty((-a, a))$, set $q = \varphi''$, and, if necessary, replace q by $-q$. Let $\zeta \in C_c^\infty(e^\perp)$ be nonnegative and strictly positive near the origin, with the supports of q and ζ chosen sufficiently small that

$$\phi(te + v) := q(t)\zeta(v) \in C_c^\infty\left(B(0, 1/4)\right).$$

By Fubini's theorem, ϕ has mean zero. Since q is even, q' is odd and so $q'(-t_0) < 0$. The identity $\partial_e\phi(te + v) = q'(t)\zeta(v)$, along with continuity, yields constants $b_0, \tau_0 > 0$ such that, with $u_0 := t_0 e$ and $B(\pm u_0, \tau_0) \subset B(0, 1/4)$,

$$\partial_e\phi(u) \geq b_0 \quad \text{on } B(u_0, \tau_0), \qquad \partial_e\phi(u) \leq -b_0 \quad \text{on } B(-u_0, \tau_0). \tag{11}$$

Thus, the gradient of ϕ has uniformly opposite components along e on two separated regions. After lifting to $\mathcal{M}_j$, these regions provide witness sets for the spectral separation argument.

For each $i \in [K]$, let $R_i \in O(d_j)$ and define $\phi_i(u) := \phi(R_i u)$. On the spectrally active patches, the rotations R_i will be chosen in Lemma 8 to align the distinguished direction of ϕ_i with the eigenfunction's gradient $\nabla_g f_\ell$. Define the raw template $\tilde{a}_i : \mathcal{M}_j \to \mathbb{R}$ by

$$\tilde{a}_i(x) := \frac{1}{\rho_j(x)} \phi_i\big(m \exp_{p_i}^{-1}(x)\big) \cdot \mathbf{1}_{B_g(p_i, r/4)}(x).$$

Let a_i denote the weighted mean-zero correction of $\tilde{a}_i$ constructed in Appendix B.1. The correction is supported away from the witness regions, and the resulting templates satisfy

$$\int_{\mathcal{M}_j} a_i \, d\mu_j = 0, \qquad \operatorname{supp}(a_i) \subset B_g(p_i, r/2) \subset \Omega_i, \qquad \|\nabla_g^\nu a_i\|_{L^\infty(\mathcal{M}_j)} \lesssim m^\nu, \quad \nu = 0, \dots, s. \tag{12}$$

In particular, the templates a_i have pairwise disjoint supports. The mean-zero property preserves the normalization of the perturbed densities, while the derivative bounds ensure that multiplication by m^{-s} preserves the prescribed s-th order regularity. Moreover,

$$\|a_i\|_{L^2(\mathcal{M}_j)}^2 \le \|a_i\|_{L^\infty(\mathcal{M}_j)}^2 \operatorname{vol}_g(\operatorname{supp}(a_i)) \lesssim \operatorname{vol}_g(\Omega_i) \lesssim m^{-d_j}. \tag{13}$$

Note that the Euclidean profile ϕ and the estimates in (12) are independent of f_ℓ. Only the rotations R_i on the active patches are selected using the gradient direction of the fixed reference eigenfunction. Lemma 8 shows that this selection yields a positive fraction of patches on which the perturbations interact with f_ℓ in a sign-definite manner.

2.1.3 The Perturbed Density Family

For each configuration $\mathbf{c} = (c_1, \dots, c_K) \in \mathcal{C} := \{-1, 1\}^K$, define the component densities

$$\rho_j^{(\mathbf{c})}(x) := \rho_j(x) \left(1 + m^{-s} \sum_{i=1}^K c_i a_i(x)\right), \qquad \rho_k^{(\mathbf{c})} := \rho_k \quad (k \neq j). \tag{14}$$

Write $\rho^{(\mathbf{c})} := \left(\rho_1^{(\mathbf{c})}, \dots, \rho_N^{(\mathbf{c})}\right)$ for the resulting density tuple, and let

$$\mu^{(\mathbf{c})} := \sum_{k=1}^N w_k \mu_k^{(\mathbf{c})}, \qquad d\mu_k^{(\mathbf{c})} := \rho_k^{(\mathbf{c})} d\operatorname{vol}_g,$$

be its associated mixture distribution. Recall that we fix the mixture weights throughout.

One can verify that $\rho^{(\mathbf{c})}$ remains valid in the sense that $\{\rho^{(\mathbf{c})} : \mathbf{c} \in \mathcal{C}\} \subset \mathcal{P}_{\mathcal{M}}^{(s)}$. Indeed, the mean-zero condition in (12) gives $\int_{\mathcal{M}_j} \rho_j^{(\mathbf{c})} d\operatorname{vol}_g = 1$. Since the templates have pairwise disjoint supports, at most one a_i contributes at any point. Thus for sufficiently large m, $\|\nabla_g^\nu(\rho_j^{(\mathbf{c})} - \rho_j)\|_{L^\infty(\mathcal{M}_j)} \lesssim m^{\nu - s}$ for each $\nu = 0, \dots, s$. The perturbed densities are therefore strictly positive and normalized, and satisfy $\rho^{(\mathbf{c})} \in \mathcal{P}_{\mathcal{M}}^{(s)}$. The scaling m^{-s} simultaneously controls the s-th order regularity and keeps the corresponding sampling measures close in KL divergence, while the oscillations at scale m^{-1} generate the spectral separation below. After identifying the spectrally active patches, Section 2.2 selects a Hamming-separated codebook $\mathcal{T} \subset \mathcal{C}$. Fano's method is then applied to the finite family $\mathcal{P}_M = \{\rho^{(\mathbf{c})} : \mathbf{c} \in \mathcal{T}\}$.

2.2 Verification of Fano Conditions

We verify the two conditions (80) and (81) for the perturbation family (14). We first establish pairwise separation of the induced eigenpairs under d_{eig}. The argument proceeds by identifying a positive fraction of patches on which the reference eigenfunction has nonvanishing gradient, aligning the local perturbations with that gradient, and then converting the resulting local forcing into a global eigenpair separation.

2.2.1 Separation in the eigenpair metric

Lemma 8 (Spectral visibility and alignment on active patches) *Let (λ_ℓ, f_ℓ) be a simple, nontrivial global eigenpair componentwise supported on $\mathcal{M}_j$. There exist constants $m_0 \in \mathbb{N}$ and $c_0, c_1, \epsilon_0 > 0$, independent of m, and an index set $\mathcal{I} \subset [K]$ with*

$$|\mathcal{I}| \geq \epsilon_0 m^{d_j}, \tag{15}$$

such that, for every $m \geq m_0$ and $i \in \mathcal{I}$, there are measurable witness sets $\mathcal{N}_i^{\pm} \subset \Omega_i$ satisfying

$$\mathrm{vol}_g(\mathcal{N}_i^{\pm}) \geq c_0 m^{-d_j}, \qquad \pm\big\langle \nabla_g(\rho_j^2 a_i), \nabla_g f_{\ell,j} \big\rangle_g \geq c_1 m \sqrt{\lambda_{\ell,j}} \quad \text{on } \mathcal{N}_i^{\pm}. \tag{16}$$

Proof We first identify a positive-volume region on which the reference eigenfunction has a quantitatively nonvanishing gradient. Given that the eigenpair is simple and nontrivial, the energy identity (2) implies $\lambda_{\ell,j} > 0$. Fix $\kappa > 0$ and define the high-gradient set with associated active index set

$$S := \Big\{ x \in \mathcal{M}_j : \rho_j(x) \|\nabla_g f_{\ell,j}(x)\|_g \geq \kappa \sqrt{\lambda_{\ell,j}} \Big\}, \qquad \mathcal{I} := \big\{ i \in [K] : \Omega_i \cap S \neq \varnothing \big\}.$$

Choose κ small enough that $\kappa^2 \,\mathrm{vol}_g(\mathcal{M}_j) \leq 1/2$. The uniform C^1 bound (5) on $f_{\ell,j}$ implies

$$\mathrm{vol}_g(S) \geq \frac{\lambda_{\ell,j}}{2\rho_{\max}^2 \|\nabla_g f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)}^2} > 0. \tag{17}$$

Since the cells form a partition and $S \subset \bigcup_{i \in \mathcal{I}} \Omega_i$, the uniform cell-volume bound gives

$$0 < \mathrm{vol}_g(S) \leq \sum_{i \in \mathcal{I}} \mathrm{vol}_g(\Omega_i) \leq C_\Omega |\mathcal{I}| m^{-d_j}.$$

Rearranging terms gives (15). Thus, a positive fraction of the $K \asymp m^{d_j}$ patches intersect the region on which the eigenfunction gradient is nontrivial. Indeed, this eigenfunction gradient lower bound propagates across the entire patch. To see this, fix $i \in \mathcal{I}$ and choose a reference point $x_i \in \Omega_i \cap S$ where the eigenfunction gradient satisfies the lower bound. Set

$$G(x) := \rho_j(x) \|\nabla_g f_{\ell,j}(x)\|_g, \qquad x \in \Omega_i.$$

The C^1 bound on ρ_j and the C^2 bound on $f_{\ell,j}$ (5) imply that G is uniformly Lipschitz. Along a minimizing unit-speed geodesic, the fundamental theorem of calculus gives

$$\big| \|\nabla_g f_{\ell,j}(x)\|_g - \|\nabla_g f_{\ell,j}(y)\|_g \big| \leq \|\nabla_g^2 f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)} d_g(x,y),$$

and therefore

$$|G(x) - G(y)| \le L d_g(x,y), \qquad L := \rho_{\max}\|\nabla_g^2 f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)} + \|\nabla_g \rho_j\|_{L^\infty(\mathcal{M}_j)}\|\nabla_g f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)},$$

where L is independent of m and of the patch index. Since $\Omega_i \subset B_g(p_i, r)$ and $x_i \in \Omega_i$,

$$d_g(x, x_i) \le d_g(x, p_i) + d_g(p_i, x_i) \le 2r = \frac{2}{m}, \qquad x \in \Omega_i.$$

Consequently,

$$G(x) \ge G(x_i) - \frac{2L}{m} \ge \kappa\sqrt{\lambda_{\ell,j}} - \frac{2L}{m}.$$

After enlarging m_0 so that $\frac{2L}{m_0} \le \frac{\kappa}{2}\sqrt{\lambda_{\ell,j}}$ for all $m \ge m_0$, we obtain

$$G(x) = \rho_j(x)\|\nabla_g f_{\ell,j}(x)\|_g \ge \frac{\kappa}{2}\sqrt{\lambda_{\ell,j}}, \qquad x \in \Omega_i. \tag{18}$$

In particular, the eigenfunction at the patch center p_i has nontrivial gradient. We may therefore define the unit gradient direction at p_i by

$$e_i(p_i) := \frac{\nabla_g f_{\ell,j}(p_i)}{\|\nabla_g f_{\ell,j}(p_i)\|_g} \in T_{p_i}\mathcal{M}_j.$$

We rotate the Euclidean profile ϕ so that its distinguished direction e agrees with $e_i(p_i)$, ensuring that the eigenfunction gradient interacts productively with the perturbation a_i. Identify $T_{p_i}\mathcal{M}_j$ isometrically with $\mathbb{R}^{d_j}$ and choose $R_i \in O(d_j)$ such that $R_i e_i(p_i) = e$. Set

$$\phi_i(u) := \phi(R_i u), \qquad u_0^{(i)} := R_i^\top u_0.$$

For notational simplicity, we relabel u_0 for $u_0^{(i)}$. By the chain rule,

$$\partial_{e_i(p_i)}\phi_i(u) = \big\langle \nabla\phi(R_i u), R_i e_i(p_i)\big\rangle = \partial_e \phi(R_i u).$$

Therefore, the directional-witness property of ϕ from Equation (10) becomes

$$\partial_{e_i(p_i)}\phi_i(u) \ge b_0 \quad \text{on } B(u_0, \tau_0), \qquad \partial_{e_i(p_i)}\phi_i(u) \le -b_0 \quad \text{on } B(-u_0, \tau_0).$$

To transfer these two regions to the manifold at scale m^{-1}, define

$$U_i^\pm := B\Big(\pm\frac{u_0}{m}, \frac{\tau_0}{m}\Big) \subset B(0, r/4), \qquad \mathcal{N}_i^\pm := \exp_{p_i}(U_i^\pm) \subset B_g(p_i, r/4) \subset \Omega_i.$$

Also, the normal-coordinate volume density is uniformly bounded below on $B(0,r)$. Thus

$$\mathrm{vol}_g(\mathcal{N}_i^\pm) = \int_{U_i^\pm} \sqrt{\det g_i(u)}\, du \ge c\,|U_i^\pm| \ge c_0 m^{-d_j}. \tag{19}$$

Consequently, every active patch indexed by $i \in \mathcal{I}$ contains two witness regions $\mathcal{N}_i^\pm$ of the natural patch-scale volume on which the rotated profile has uniformly opposite directional derivatives. It remains to transfer this directional property to $\nabla_g(\rho_j^2 a_i)$.

Fix $x \in \mathcal{N}_i^{\pm}$, write $u := \exp_{p_i}^{-1}(x) \in U_i^{\pm}$. Then $mu \in B(\pm u_0, \tau_0)$, so

$$F_i(x) := \phi_i\big(m \exp_{p_i}^{-1}(x)\big) \qquad \text{satisfies} \qquad \pm\, \partial_{e_i(p_i)}\phi_i(mu) \geq b_0. \tag{20}$$

For each $x \in \Omega_i$, let $\gamma : [0,1] \to \mathcal{M}_j$ be the unique constant-speed minimizing geodesic from p_i to x. Parallel transport $e_i(p_i)$ along γ and denote its endpoint value by $e_i(x)$. A standard Jacobi-field estimate in Proposition 27 along the radial geodesic from p_i to x gives

$$\big\|d(\exp_{p_i})_u\,(e_i(p_i)) - e_i(x)\big\|_g \leq C|u|^2 \implies \Big\|\big(d(\exp_{p_i})_u\big)^{-1} e_i(x) - e_i(p_i)\Big\|_g \leq C|u|^2.$$

Here we use the fact that $d(\exp_{p_i})_u$ is uniformly invertible for $|u| \leq r/4$. By the chain rule,

$$\langle \nabla_g F_i(x), e_i(x)\rangle_g = m\left\langle \nabla\phi_i(mu), \big(d(\exp_{p_i})_u\big)^{-1} e_i(x)\right\rangle.$$

This estimate, along with the smoothness of ϕ_i, and $|u| \leq r/4 \lesssim m^{-1}$, gives

$$\Big|\langle \nabla_g F_i(x), e_i(x)\rangle_g - m\,\partial_{e_i(p_i)}\phi_i(mu)\Big| \leq Cm|u|^2 \leq Cm^{-1}.$$

Combining with (20) and enlarging m_0 if necessary, we obtain

$$\pm\,\langle \nabla_g F_i(x), e_i(x)\rangle_g \geq \frac{b_0}{2}m, \qquad x \in \mathcal{N}_i^{\pm}. \tag{21}$$

The same chain-rule argument and the uniform boundedness of $\big(d(\exp_{p_i})_u\big)^{-1}$ give

$$\|\nabla_g F_i(x)\|_g \leq C_\phi m, \qquad x \in \mathcal{N}_i^{\pm}. \tag{22}$$

Combining (21) and (22), and controlling the discrepancy between the true normalized eigenfunction gradient and its parallel-transported approximation $e_i(x)$, we obtain

$$\begin{aligned}\pm\left\langle \nabla_g F_i(x), \frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g}\right\rangle_g &= \pm\,\langle \nabla_g F_i(x), e_i(x)\rangle_g \pm \left\langle \nabla_g F_i(x), \frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g} - e_i(x)\right\rangle_g \\ &\geq \frac{b_0}{2}m - C_\phi\, m\left\|\frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g} - e_i(x)\right\|_g.\end{aligned}$$

Indeed, the parallel-transported approximation $e_i(x)$ satisfies

$$\begin{aligned}\big\|\nabla_g f_{\ell,j}(x) - \|\nabla_g f_{\ell,j}(p_i)\|_g e_i(x)\big\|_g &\leq \int_0^1 \big\|\nabla_{\dot\gamma(t)}\nabla_g f_{\ell,j}\big\|_g\, dt \leq \|\nabla_g^2 f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)}\int_0^1 \|\dot\gamma(t)\|_g\, dt \\ &= d_g(p_i,x)\|\nabla_g^2 f_{\ell,j}\|_{L^\infty(\mathcal{M}_j)} \leq C_{\ell,j}\, m^{-1}.\end{aligned}$$

Applying the inequality

$$\left\|\frac{a}{\|a\|} - \frac{b}{\|b\|}\right\| \leq \frac{2\|a-b\|}{\min\{\|a\|,\|b\|\}},$$

in $T_x\mathcal{M}_j$ with $a = \nabla_g f_{\ell,j}(x)$ and $b = \|\nabla_g f_{\ell,j}(p_i)\|_g\, e_i(x)$, together with (18), yields

$$\left\|\frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g} - e_i(x)\right\|_g \leq C_{\ell,j}m^{-1} \qquad x \in \Omega_i.$$

Consequently, enlarging m_0 once more, we obtain, for all $x \in \mathcal{N}_i^{\pm}$,

$$\pm \left\langle \nabla_g F_i(x), \frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g} \right\rangle_g \geq \frac{b_0}{2} m - C_\phi C_{\ell,j} \geq \frac{b_0}{4} m.$$

By construction, $\rho_j^2 a_i = \rho_j F_i$, and so $\nabla_g(\rho_j^2 a_i) = \rho_j \nabla_g F_i + F_i \nabla_g \rho_j$. Taking the inner product with the normalized eigenfunction gradient and applying the uniform bounds on ϕ and $\nabla_g \rho_j$ yields, for large enough m, there exists $c_a > 0$ independent of m and i such that

$$\pm \left\langle \nabla_g(\rho_j^2 a_i)(x), \frac{\nabla_g f_{\ell,j}(x)}{\|\nabla_g f_{\ell,j}(x)\|_g} \right\rangle_g \geq \rho_{\min} \frac{b_0}{4} m - \|\phi_i\|_{L^\infty} \|\nabla_g \rho_j\|_{L^\infty(\mathcal{M}_j)} \geq c_a m$$

for all $x \in \mathcal{N}_i^{\pm}$. Multiplying by $\|\nabla_g f_{\ell,j}(x)\|_g$ and applying (18) completes the proof. ∎

Lemma 8 identifies a collection of active patches, indexed by $\mathcal{I}$, on which the local perturbations produce a sign-definite spectral effect. We next compare the eigenpairs induced by any pair of distinct configurations on their differing active set $\mathcal{D} := \{i \in \mathcal{I} : c_i^{(1)} \neq c_i^{(2)}\}$. By the Varshamov–Gilbert bound (Tsybakov, 2009, Lemma 2.9), there exists a subset of sign configurations $\mathcal{T} \subset \{-1, 1\}^{|\mathcal{I}|}$ such that $\log |\mathcal{T}| \geq c_T |\mathcal{I}| \gtrsim m^{d_j}$ and for distinct $\mathbf{c}^{(1)}, \mathbf{c}^{(2)} \in \mathcal{T}$

$$|\mathcal{D}| = \sum_{i \in \mathcal{I}} \mathbf{1}_{\{c_i^{(1)} \neq c_i^{(2)}\}} \geq c_{\mathrm{H}} |\mathcal{I}| \gtrsim m^{d_j},$$

for some constants $c_{\mathrm{T}}, c_{\mathrm{H}} > 0$ independent of m. For $b = 1, 2$, write the perturbed density

$$\rho_j^{(b)} := \rho_j^{(\mathbf{c}^{(b)})} = \rho_j \left(1 + m^{-s} \sum_{i=1}^{K} c_i^{(b)} a_i\right).$$

Since the supports of a_i are pairwise disjoint, (13) gives

$$\left\| \sum_{i=1}^{K} c_i^{(b)} a_i \right\|_{L^2(\mathcal{M}_j)}^2 = \sum_{i=1}^{K} \|a_i\|_{L^2(\mathcal{M}_j)}^2 \lesssim K\, m^{-d_j} \lesssim 1.$$

It follows that the perturbation size of densities satisfies

$$\|\rho_j^{(b)} - \rho_j\|_{L^2(\mathcal{M}_j)} = m^{-s} \left\| \rho_j \sum_{i=1}^{K} c_i^{(b)} a_i \right\|_{L^2(\mathcal{M}_j)} \lesssim m^{-s}, \qquad b = 1, 2. \tag{23}$$

Define the corresponding perturbed measure and component operator by

$$d\mu_j^{(b)} := \rho_j^{(b)}\, d\mathrm{vol}_g, \qquad \Delta_j^{(b)} f_{\ell,j} := -\frac{1}{\rho_j^{(b)}} \operatorname{div}_g \left(\left(\rho_j^{(b)}\right)^2 \nabla_g f_{\ell,j} \right).$$

Consider the linear density path

$$\rho_{j,t}^{(b)} := \rho_j + t\left(\rho_j^{(b)} - \rho_j\right), \qquad 0 \leq t \leq 1. \tag{24}$$

By the eigengap condition (6), the perturbation bound (23), and standard perturbation theory for simple isolated eigenvalues Kato (1995), the reference eigenpair admits a unique simple continuation along this path for all sufficiently large m. The quantitative eigenvalue and eigenfunction estimates along this branch are given by Proposition 26.

Let $(\lambda_{\ell,j}^{(b)}, f_{\ell,j}^{(b)})$ denote the resulting component eigenpair at $t = 1$, normalized by

$$\Delta_j^{(b)} f_{\ell,j}^{(b)} = \lambda_{\ell,j}^{(b)} f_{\ell,j}^{(b)}, \qquad \|f_{\ell,j}^{(b)}\|_{L^2(\mu_j^{(b)})} = w_j^{-1}, \qquad \lambda_\ell^{(b)} := w_j \lambda_{\ell,j}^{(b)}.$$

The feasibility of the perturbed family implies the uniform measure equivalence, so

$$\|f_{\ell,j}^{(b)}\|_{L^2(\mu_j)} \le C \|f_{\ell,j}^{(b)}\|_{L^2(\mu_j^{(b)})} = C, \qquad b = 1, 2. \tag{25}$$

We choose the signs of $f_{\ell,j}^{(1)}$ and $f_{\ell,j}^{(2)}$ according to the convention fixed throughout, so that both are aligned with the reference eigenfunction $f_{\ell,j}$. The next lemma derives the governing differential inequality for the difference $h := f_{\ell,j}^{(1)} - f_{\ell,j}^{(2)}$, establishing a strict lower bound for the forcing term induced by the local density perturbations on the witness regions.

Lemma 9 *There exist $c, C_{\ell,j} > 0$ independent of m such that for any distinct $\mathbf{c}^{(1)}, \mathbf{c}^{(2)} \in \mathcal{T}$ and each differing active patch $i \in \mathcal{D}$, there exists a witness region $\mathcal{N}_i^\star \in \{\mathcal{N}_i^+, \mathcal{N}_i^-\}$ on which the difference $h = f_{\ell,j}^{(1)} - f_{\ell,j}^{(2)}$ satisfies the pointwise inequality*

$$\operatorname{div}_g(\rho_j^2 \nabla_g h)(x) + (\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}) \rho_j(x) f_{\ell,j}^{(1)}(x) + \lambda_{\ell,j}^{(2)} \rho_j(x) h(x) \ge c\sqrt{\lambda_{\ell,j}}\, m^{1-s} - |\mathcal{R}_m(x)|, \tag{26}$$

where the remainder satisfies $\|\mathcal{R}_m\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j} \lambda_{\ell,j} m^{-s}$.

Proof For brevity, for $b = 1, 2$, set

$$A_b := \sum_{i=1}^K c_i^{(b)} a_i, \qquad A_\Delta := \frac{A_1 - A_2}{2} = \sum_{i \in \mathcal{D}} \frac{c_i^{(1)} - c_i^{(2)}}{2} a_i =: \sum_{i \in \mathcal{D}} \varsigma_i a_i.$$

Note that $\varsigma_i \in \{\pm 1\}$ for each $i \in \mathcal{D}$ and vanishes for $i \notin \mathcal{D}$. Moreover, $A_\Delta = \varsigma_i a_i$ on Ω_i. Since the support of a_i are pairwise disjoint, we write $B := A_1^2 = A_2^2 = \sum_i a_i^2$. Thus

$$\rho_j^{(b)} = \rho_j (1 + m^{-s} A_b), \qquad \big(\rho_j^{(b)}\big)^2 = \rho_j^2 \big(1 + 2m^{-s} A_b + m^{-2s} B\big).$$

For $b = 1, 2$, the perturbed eigen-equation is

$$\operatorname{div}_g\big((\rho_j^{(b)})^2 \nabla_g f_{\ell,j}^{(b)}\big) + \lambda_{\ell,j}^{(b)} \rho_j^{(b)} f_{\ell,j}^{(b)} = 0.$$

Subtracting the equation for $b = 2$ from that for $b = 1$ and separating the terms gives

$$\begin{aligned} &\operatorname{div}_g(\rho_j^2 \nabla_g h) + (\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}) \rho_j f_{\ell,j}^{(1)} + \lambda_{\ell,j}^{(2)} \rho_j h \\ &\qquad = -2m^{-s} \operatorname{div}_g \left[\rho_j^2 \left(A_1 \nabla_g f_{\ell,j}^{(1)} - A_2 \nabla_g f_{\ell,j}^{(2)}\right)\right] - m^{-2s} \operatorname{div}_g(\rho_j^2 B \nabla_g h) \\ &\qquad\quad - m^{-s} \rho_j \left(\lambda_{\ell,j}^{(1)} A_1 f_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)} A_2 f_{\ell,j}^{(2)}\right). \end{aligned}$$

The first term on the right-hand side contains the principal forcing term. Using

$$\operatorname{div}_g(\rho_j^2 A_b \nabla_g f_{\ell,j}^{(b)}) = \left\langle \nabla_g(\rho_j^2 A_b), \nabla_g f_{\ell,j}^{(b)} \right\rangle_g + \rho_j^2 A_b \Delta_g f_{\ell,j}^{(b)}$$

and

$$\begin{aligned}
&\left\langle \nabla_g(\rho_j^2 A_1), \nabla_g f_{\ell,j}^{(1)} \right\rangle_g - \left\langle \nabla_g(\rho_j^2 A_2), \nabla_g f_{\ell,j}^{(2)} \right\rangle_g \\
&= 2\left\langle \nabla_g(\rho_j^2 A_\Delta), \nabla_g f_{\ell,j} \right\rangle_g + \left\langle \nabla_g(\rho_j^2 A_1), \nabla_g(f_{\ell,j}^{(1)} - f_{\ell,j}) \right\rangle_g - \left\langle \nabla_g(\rho_j^2 A_2), \nabla_g(f_{\ell,j}^{(2)} - f_{\ell,j}) \right\rangle_g,
\end{aligned}$$

we obtain

$$\operatorname{div}_g(\rho_j^2 \nabla_g h) + (\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)})\rho_j f_{\ell,j}^{(1)} + \lambda_{\ell,j}^{(2)} \rho_j h = -4m^{-s} \left\langle \nabla_g \left(\rho_j^2 A_\Delta\right), \nabla_g f_{\ell,j} \right\rangle_g + \mathcal{R}_m,$$

where $\mathcal{R}_m = \mathcal{R}_m^{(1)} + \mathcal{R}_m^{(2)} + \mathcal{R}_m^{(3)} + \mathcal{R}_m^{(4)} + \mathcal{R}_m^{(5)}$ with

$$\begin{aligned}
&\mathcal{R}_m^{(1)} := -m^{-s}\rho_j \left(\lambda_{\ell,j}^{(1)} A_1 f_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)} A_2 f_{\ell,j}^{(2)}\right), \qquad \mathcal{R}_m^{(2)} := -2m^{-s}\rho_j^2 \left(A_1 \Delta_g f_{\ell,j}^{(1)} - A_2 \Delta_g f_{\ell,j}^{(2)}\right), \\
&\mathcal{R}_m^{(3)} := -2m^{-s}\left[\left\langle \nabla_g(\rho_j^2 A_1), \nabla_g(f_{\ell,j}^{(1)} - f_{\ell,j}) \right\rangle_g - \left\langle \nabla_g(\rho_j^2 A_2), \nabla_g(f_{\ell,j}^{(2)} - f_{\ell,j}) \right\rangle_g\right], \\
&\mathcal{R}_m^{(4)} := -m^{-2s}\left\langle \nabla_g(\rho_j^2 B), \nabla_g h \right\rangle_g, \qquad \mathcal{R}_m^{(5)} := -m^{-2s}\rho_j^2 B \Delta_g h.
\end{aligned}$$

We derive uniform L^1 bounds for these terms. Disjoint supports of a_i and (12) imply

$$\|A_b\|_{L^\infty(\mathcal{M}_j)} + \|A_\Delta\|_{L^\infty(\mathcal{M}_j)} + \|B\|_{L^\infty(\mathcal{M}_j)} \le C \tag{27}$$

and

$$\|\nabla_g(\rho_j^2 A_b)\|_{L^\infty(\mathcal{M}_j)} + \|\nabla_g(\rho_j^2 B)\|_{L^\infty(\mathcal{M}_j)} \le Cm. \tag{28}$$

Integrating the derivative bounds (88) and (89) along the linear density paths (24) gives

$$|\lambda_{\ell,j}^{(b)} - \lambda_{\ell,j}| \le C_{\ell,j}\lambda_{\ell,j} m^{-s}, \qquad \|f_{\ell,j}^{(b)} - f_{\ell,j}\|_{H^1(\mathcal{M}_j)} \le C_{\ell,j}\frac{\lambda_{\ell,j}}{\gamma_\ell} m^{-s}, \qquad b = 1, 2. \tag{29}$$

In particular, $\lambda_{\ell,j}^{(b)} \lesssim \lambda_{\ell,j}$ and the uniform elliptic estimates from (4) give

$$\|f_{\ell,j}^{(b)}\|_{L^2(\mathcal{M}_j)} + \|\Delta_g f_{\ell,j}^{(b)}\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}, \qquad b = 1, 2. \tag{30}$$

Using (27) and (30), Hölder's inequality gives

$$\|\mathcal{R}_m^{(1)}\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}\lambda_{\ell,j} m^{-s}, \qquad \|\mathcal{R}_m^{(2)}\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}\lambda_{\ell,j} m^{-s}. \tag{31}$$

For the third term, the finite perturbation estimate (29) gives

$$\|\mathcal{R}_m^{(3)}\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j} m^{-s} m \sum_{b=1}^{2} \|\nabla_g(f_{\ell,j}^{(b)} - f_{\ell,j})\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}\lambda_{\ell,j}\gamma_\ell^{-1} m^{1-2s}. \tag{32}$$

Similarly,

$$\|\nabla_g h\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}\lambda_{\ell,j}\gamma_\ell^{-1}m^{-s} \implies \|\mathcal{R}_m^{(4)}\|_{L^2(\mathcal{M}_j)} \le C_{\ell,j}\lambda_{\ell,j}\gamma_\ell^{-1}m^{1-3s}. \tag{33}$$

Finally, by (30),

$$\|\mathcal{R}_m^{(5)}\|_{L^2(\mathcal{M}_j)} \le Cm^{-2s}\left(\|\Delta_g f_{\ell,j}^{(1)}\|_{L^2(\mathcal{M}_j)} + \|\Delta_g f_{\ell,j}^{(2)}\|_{L^2(\mathcal{M}_j)}\right) \le C_\ell\lambda_{\ell,j}m^{-2s}. \tag{34}$$

Combining (31)–(34) and absorbing the fixed eigengap into $C_{\ell,j}$ gives the remainder's scale. To control the principle forcing term, we fix $i \in \mathcal{D}$. Since $A_\Delta = \varsigma_i a_i$ on Ω_i, Lemma 8 allows one to choose $\mathcal{N}_i^\star \in \{\mathcal{N}_i^+, \mathcal{N}_i^-\}$ so that

$$-4m^{-s}\left\langle \nabla_g(\rho_j^2 A_\Delta), \nabla_g f_{\ell,j}\right\rangle_g = -4m^{-s}\varsigma_i\left\langle \nabla_g(\rho_j^2 a_i), \nabla_g f_{\ell,j}\right\rangle_g \ge c\sqrt{\lambda_{\ell,j}}\, m^{1-s} \qquad \text{on } \mathcal{N}_i^\star.$$

This proves (26). Since $|\mathcal{D}| \gtrsim m^{d_j}$, the selected witness regions satisfy $\mathrm{vol}_g\left(\bigcup_{i\in\mathcal{D}}\mathcal{N}_i^\star\right) \gtrsim |\mathcal{D}|m^{-d_j} \gtrsim 1$. Thus, the local principal forcing accumulates globally at scale m^{1-s}, dominating the remainder $\mathcal{R}_m = \mathcal{O}(m^{-s})$. We therefore complete the proof.

■

Lemma 10 (Sufficient spectral separation) *Fix a differing active patch $i \in \mathcal{D}$. Then either*

$$|\lambda_\ell^{(1)} - \lambda_\ell^{(2)}| \ge c\sqrt{\lambda_\ell w_j}\, m^{1-s}, \tag{35}$$

or

$$\frac{\lambda_\ell\rho_{\max}}{m}\|f_\ell^{(1)} - f_\ell^{(2)}\|_{L^2(\mathcal{M}_j)} + w_j\|\nabla_g(f_\ell^{(1)} - f_\ell^{(2)})\|_{L^2(\mathcal{M}_j)} \ge c\sqrt{\lambda_\ell w_j}\, m^{-s}. \tag{36}$$

Proof Let $c_1 > 0$. If $|\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}| \ge c_1\sqrt{\lambda_{\ell,j}}\, m^{1-s}$, then multiplication by w_j gives the desired eigenvalue separation (35). It therefore remains to consider

$$|\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}| < c_1\sqrt{\lambda_{\ell,j}}\, m^{1-s}. \tag{37}$$

For $i \in \mathcal{D}$, choose $\mathcal{N}_i^\star \in \{\mathcal{N}_i^+, \mathcal{N}_i^-\}$ so that Equation (26) in Lemma 9 is satisfied. Define

$$\mathfrak{r} := |\mathcal{R}_m| + |\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}|\,\rho_j|f_{\ell,j}^{(1)}|.$$

Then Equation (26) implies

$$\mathrm{div}_g(\rho_j^2\nabla_g h) + \lambda_{\ell,j}^{(2)}\rho_j h + \mathfrak{r} \ge c_0\sqrt{\lambda_{\ell,j}}\, m^{1-s} \qquad \text{on } \mathcal{N}_i^\star. \tag{38}$$

Applying Lemma 28 with $V(x) := \lambda_{\ell,j}^{(2)}\rho_j(x)$, $\Lambda := \lambda_{\ell,j}^{(2)}\rho_{\max}$, and $A := c_0\sqrt{\lambda_{\ell,j}}$ yields

$$\frac{\lambda_{\ell,j}^{(2)}\rho_{\max}}{m}\int_{\mathcal{N}_i^\star}|h|\,d\mathrm{vol}_g + \int_{\mathcal{N}_i^\star}\|\nabla_g h\|_g\,d\mathrm{vol}_g \ge c\sqrt{\lambda_{\ell,j}}\, m^{-(d_j+s)} - \frac{C}{m}\int_{\mathcal{N}_i^\star}\left(|\mathcal{R}_m| + |\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}|\rho_j|f_{\ell,j}^{(1)}|\right)d\mathrm{vol}_g.$$

Since $\mathcal{N}_i^\star$ are pairwise disjoint, summing over $i \in \mathcal{D}$ then yields

$$\frac{\lambda_{\ell,j}^{(2)}\rho_{\max}}{m}\int_{\mathcal{M}_j}|h|\,d\mathrm{vol}_g+\int_{\mathcal{M}_j}\|\nabla_g h\|_g\,d\mathrm{vol}_g \\ \geq c\sqrt{\lambda_{\ell,j}}\,m^{-s}-\frac{C}{m}\int_{\mathcal{M}_j}|\mathcal{R}_m|\,d\mathrm{vol}_g-\frac{C}{m}|\lambda_{\ell,j}^{(1)}-\lambda_{\ell,j}^{(2)}|\int_{\mathcal{M}_j}|f_{\ell,j}^{(1)}|\,d\mu_j. \tag{39}$$

The remainder's scale from Lemma 9 and Hölder's inequality gives $\|\mathcal{R}_m\|_{L^1(\mathcal{M}_j)} \leq C_{\ell,j}\lambda_{\ell,j}m^{-s}$ which allows us to drop the second term on the right that is of strictly lower order. Meanwhile, the integral in the third term is of constant order according to (25) and so (37) implies that it is at most of the same order as the leading term. Choosing the constants sufficiently small allows this term to be absorbed, yielding

$$\frac{\lambda_{\ell,j}^{(2)}\rho_{\max}}{m}\|h\|_{L^1(\mathcal{M}_j)}+\|\nabla_g h\|_{L^1(\mathcal{M}_j)} \geq c\sqrt{\lambda_{\ell,j}}\,m^{-s}. \tag{40}$$

The eigenvalue perturbation estimate (88) in Proposition 26 ensures that $\lambda_{\ell,j}^{(2)} \leq C\lambda_{\ell,j}$ for all sufficiently large m. Moreover, since $\mathcal{M}_j$ has finite volume, Hölder's inequality yields

$$\|h\|_{L^1(\mathcal{M}_j)} \leq C\|h\|_{L^2(\mathcal{M}_j)}, \qquad \|\nabla_g h\|_{L^1(\mathcal{M}_j)} \leq C\|\nabla_g h\|_{L^2(\mathcal{M}_j)}.$$

By definition, $f_{\ell,j}^{(b)}$ and $f_\ell^{(b)}$ coincide on $\mathcal{M}_j$, while $f_\ell^{(b)}$ vanishes on all the other components. Therefore, multiplying by w_j and applying the relation $w_j\lambda_{\ell,j} = \lambda_\ell$ proves (36). ■

2.2.2 Upper bound on the KL divergence

We verify the KL divergence requirement (81) of Fano's method. We show that although the perturbation templates oscillate at spatial scale m^{-1}, their amplitude is only of order m^{-s}. As a consequence, the resulting densities remain close in KL divergence.

Lemma 11 (KL divergence upper bound) *For all sufficiently large m and $n_j = nw_j$*

$$\mathrm{KL}\Big((\mu^{(1)})^{\otimes n}\,\Big\|\,(\mu^{(2)})^{\otimes n}\Big) \leq Cn_j\,m^{-2s}. \tag{41}$$

Proof The feasibility of the perturbed family ensures that $\rho_j^{(2)} \geq (1/2)\cdot\rho_j$. Using the standard inequality $\mathrm{KL}(p\|q) \leq \int(p-q)^2/q\,d\mathrm{vol}_g$, we obtain

$$\mathrm{KL}\Big(\mu_j^{(1)}\,\Big\|\,\mu_j^{(2)}\Big) \leq \int_{\mathcal{M}_j}\frac{\big(\rho_j^{(1)}-\rho_j^{(2)}\big)^2}{\rho_j^{(2)}}\,d\mathrm{vol}_g \leq C_1 m^{-2s}\int_{\mathcal{M}_j}\left(\sum_{i=1}^K(c_i^{(1)}-c_i^{(2)})a_i\right)^2 d\mu_j.$$

Since the templates a_i have pairwise disjoint supports,

$$\int_{\mathcal{M}_j}\left(\sum_{i=1}^K(c_i^{(1)}-c_i^{(2)})a_i\right)^2 d\mu_j = \int_{\mathcal{M}_j}\sum_{i=1}^K(c_i^{(1)}-c_i^{(2)})^2a_i^2\,d\mu_j \leq 4\sum_{i=1}^K\int_{\mathrm{supp}(a_i)}a_i^2\,d\mu_j \\ \leq C_2\sum_{i=1}^K\mu_j(\Omega_i) \leq C_2\mu_j(\mathcal{M}_j) = C_2.$$

Combining this estimate with the additivity and joint convexity of KL divergence implies

$$\mathrm{KL}\Big((\mu^{(1)})^{\otimes n}\,\Big\|\,(\mu^{(2)})^{\otimes n}\Big) = n\,\mathrm{KL}\Big((\mu^{(1)})\,\Big\|\,\mu^{(2)}\Big) \le n\sum_{k=1}^{N} w_k\,\mathrm{KL}\Big(\mu_k^{(1)}\,\Big\|\,\mu_k^{(2)}\Big) = Cnw_j m^{-2s}.$$

■

2.3 Minimax Lower Bound via Fano's Method

We are now in a position to establish Theorem 4 and its accompanying Corollary 5.

2.3.1 Proof of Theorem 4

It suffices to establish the lower bound over the restricted family constructed above, in which the manifold geometry, mixture weights, and component densities outside $\mathcal{M}_j$ are held fixed. Since this family is contained in the full parameter space $\{(\mathcal{M},\rho) : \mathcal{M}\in\mathbf{M}, \rho\in\mathcal{P}^{(s)}\}$, any minimax lower bound for the restricted problem applies directly to the full parameter space.

Recall that the codebook $\mathcal{T}$ satisfies $\log|\mathcal{T}| \gtrsim m^{d_j}$. For two distinct sign configurations $\mathbf{c}^{(1)},\mathbf{c}^{(2)}\in\mathcal{T}$ and for sufficiently large n_j, Lemma 11 gives the KL divergence upper bound

$$\frac{\max\limits_{\mathbf{c}\neq\mathbf{c}'}\mathrm{KL}\Big(\big(\mu^{(1)}\big)^{\otimes n}\,\Big\|\,\big(\mu^{(2)}\big)^{\otimes n}\Big)+\log 2}{\log|\mathcal{T}|} \le C_1 n_j\, m^{-(d_j+2s)} + C_2 m^{-d_j} \le \frac{1}{2}.$$

Lemma 10 implies that corresponding perturbed eigenpairs are separated at scale m^{-s}. In particular, the left side of (36) is bounded above by a constant multiple of the $H^1(\mathcal{M}_j)$ distance for $m\ge 1$. Consequently

$$d_{\mathrm{eig}}\Big(\theta(\rho^{(1)}),\theta(\rho^{(2)})\Big) \ge c_{\ell,j}\sqrt{\lambda_\ell w_j}\,m^{-s}.$$

Choose $m = \Big\lceil A n_j^{1/(d_j+2s)}\Big\rceil$ with $A>0$ sufficiently large. Then Fano's method yields

$$\inf_{\widehat{\theta}}\max_{\mathbf{c}\in\mathcal{T}}\mathbb{E}_{\left(\mu^{(\mathbf{c})}\right)^{\otimes n}}\Big[d_{\mathrm{eig}}\Big(\theta(\rho^{(\mathbf{c})}),\widehat{\theta}(\mathcal{X}_n)\Big)\Big] \ge c_{\ell,j}\sqrt{\lambda_\ell w_j}\,n_j^{-s/(d_j+2s)}.$$

Because the restricted family is contained in the full parameter space, this proves

$$\inf_{(\widehat{\lambda}_\ell,\widehat{f}_\ell)}\sup_{\mathcal{M}\in\mathbf{M},\,\rho\in\mathcal{P}^{(s)}}\mathbb{E}_{\mathcal{X}_n\sim\mu_\rho^{\otimes n}}\Big[|\lambda_\ell-\widehat{\lambda}_\ell| + \|f_\ell-\widehat{f}_\ell\|_{H^1(\mathcal{M}_j)}\Big] \ge c_{\ell,j}\sqrt{\lambda_\ell w_j}\,n_j^{-s/(d_j+2s)}.$$

Repeating the argument for every admissible supporting component establishes (8).

2.3.2 Proof of Corollary 5

We use the same finite family, Hamming-separated codebook $\mathcal{T}$, and KL bound as in the proof of Theorem 4. It remains only to strengthen the pairwise eigenfunction separation from the H^1 scale m^{-s} to the L^2 scale $m^{-(s+1)}$. Since $h = f_{\ell,j}^{(1)} - f_{\ell,j}^{(2)}$ captures the full difference $f_\ell^{(1)} - f_\ell^{(2)}$ by the assumption, we use h to denote this difference interchangeably.

If the eigenvalue alternative (35) from Lemma 10 holds, then

$$|\lambda_\ell^{(1)} - \lambda_\ell^{(2)}| \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{1-s} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{-(s+1)}.$$

Otherwise, if the first term on the left side of (36) is at least half of its right side, then

$$\|h\|_{L^2(\mathcal{M}_j)} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{1-s} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{-(s+1)},$$

with all the fixed factors absorbed into $c_{\ell,j}$. Otherwise, $\|\nabla_g h\|_{L^2(\mathcal{M}_j)} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{-s}$. Since $\mathcal{M}_j$ is closed, integration by parts and Cauchy–Schwarz yield

$$\|\nabla_g h\|^2_{L^2(\mathcal{M}_j)} = -\int_{\mathcal{M}_j} h\, \Delta_g h \, d\mathrm{vol}_g \leq \|h\|_{L^2(\mathcal{M}_j)} \|\Delta_g h\|_{L^2(\mathcal{M}_j)}.$$

It remains to upper bound $\|\Delta_g h\|_{L^2}$. Expanding the component eigen-equations yields

$$\Delta_g f_{\ell,j}^{(b)} = \frac{\lambda_{\ell,j}^{(b)}}{\rho_j^{(b)}} f_{\ell,j}^{(b)} - 2\left\langle \nabla_g \log \rho_j^{(b)}, \nabla_g f_{\ell,j}^{(b)} \right\rangle_g, \qquad b = 1, 2.$$

Subtracting $b = 1$ from $b = 2$ and by the triangle inequality, $\|\Delta_g h\|_{L^2(\mathcal{M}_j)}$ is controlled by

$$|\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}| \|f_{\ell,j}^{(1)}\|_{L^2(\mathcal{M}_j)} + \lambda_{\ell,j}^{(2)} \|h\|_{L^2(\mathcal{M}_j)} + \lambda_{\ell,j}^{(2)} \left\| \frac{1}{\rho_j^{(1)}} - \frac{1}{\rho_j^{(2)}} \right\|_{L^\infty(\mathcal{M}_j)} \|f_{\ell,j}^{(2)}\|_{L^2(\mathcal{M}_j)}$$
$$+ \|\nabla_g \log \rho_j^{(1)}\|_{L^\infty(\mathcal{M}_j)} \|\nabla_g h\|_{L^2(\mathcal{M}_j)} + \|\nabla_g \log \rho_j^{(1)} - \nabla_g \log \rho_j^{(2)}\|_{L^\infty(\mathcal{M}_j)} \|\nabla_g f_{\ell,j}^{(2)}\|_{L^2(\mathcal{M}_j)}.$$

The perturbed densities (14), with the templates bounds (12) and disjoint supports, imply

$$\|\rho_j^{(1)} - \rho_j^{(2)}\|_{L^\infty(\mathcal{M}_j)} \leq C m^{-s}, \qquad \|\nabla_g \rho_j^{(1)} - \nabla_g \rho_j^{(2)}\|_{L^\infty(\mathcal{M}_j)} \leq C m^{1-s}.$$

Since the perturbed densities are uniformly bounded away from zero,

$$\left\| \frac{1}{\rho_j^{(1)}} - \frac{1}{\rho_j^{(2)}} \right\|_{L^\infty(\mathcal{M}_j)} \leq C m^{-s}, \qquad \|\nabla_g \log \rho_j^{(1)} - \nabla_g \log \rho_j^{(2)}\|_{L^\infty(\mathcal{M}_j)} \leq C m^{1-s}. \tag{42}$$

Moreover, integrating the eigenpair perturbation bounds (88) and (89) along the corresponding density paths gives

$$|\lambda_{\ell,j}^{(1)} - \lambda_{\ell,j}^{(2)}| \leq C_{\ell,j} m^{-s}, \qquad \|h\|_{H^1(\mathcal{M}_j)} \leq C_{\ell,j} m^{-s}. \tag{43}$$

Equations (42), (43), and the uniform H^1 bounds on $f_{\ell,j}^{(1)}, f_{\ell,j}^{(2)}$ give $\|\Delta_g h\|_{L^2(\mathcal{M}_j)} \leq C_{\ell,j} m^{1-s}$. Consequently,

$$\|h\|_{L^2(\mathcal{M}_j)} \geq \frac{\|\nabla_g h\|^2_{L^2(\mathcal{M}_j)}}{\|\Delta_g h\|_{L^2(\mathcal{M}_j)}} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{-(s+1)},$$

where any fixed factors are absorbed into $c_{\ell,j}$. We therefore have established

$$|\lambda_\ell^{(1)} - \lambda_\ell^{(2)}| + \|f_\ell^{(1)} - f_\ell^{(2)}\|_{L^2(\mathcal{M}_j)} \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, m^{-(s+1)}. \tag{44}$$

Choosing m as in the proof of Theorem 4, then Fano's method gives

$$\inf_{(\widehat{\lambda}_\ell, \widehat{f}_\ell)} \sup_{\mathcal{M}\in\mathbf{M},\, \rho\in\mathcal{P}^{(s)}} \mathbb{E}_{\mathcal{X}_n \sim \mu_\rho^{\otimes n}} \left[|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{L^2(\mathcal{M}_j)} \right] \geq c_{\ell,j}\sqrt{\lambda_\ell w_j}\, n_j^{-(s+1)/(d_j+2s)}.$$

Taking the maximum over all admissible supporting components completes the proof.

2.3.3 Proof of Theorem 6

It suffices to estimate the density on $\mathcal{M}_j$. Choosing $h_j \asymp n_j^{-1/(2s+d_j)}$ in Lemma 23 gives

$$\mathbb{E}\|\widehat{\rho}_j - \rho_j\|_{L^2(\mathcal{M}_j)} \le C n_j^{-s/(d_j+2s)}.$$

By Proposition 26, the simple eigenpair is stable under L^2-perturbations of the density:

$$|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M})} \le C_{\ell,j}\|\widehat{\rho}_j - \rho_j\|_{L^2(\mathcal{M}_j)}.$$

Taking expectations and combining the two estimates yields

$$\mathbb{E}_{\mathcal{X}_n\sim\rho}\left[|\lambda_\ell - \widehat{\lambda}_\ell| + \|f_\ell - \widehat{f}_\ell\|_{H^1(\mathcal{M})}\right] \le C_{\ell,j} n_j^{-s/(d_j+2s)}.$$

Taking the worst case over all admissible supporting components proves the claim.

3 Spectral Convergence of the Graph Laplacian

We establish an upper bound for the convergence of the graph Laplacian $\mathcal{L}$ to the Laplace-Beltrami operator $\Delta_{\mathcal{M}}$, thereby proving Theorem 7. In contrast to the oracle setting in the construction of Theorem 6, manifold components are not known a priori nor required to be separated in the ambient space. The graph Laplacian then must recover the intrinsic geometry of each manifold component while suppressing interactions across distinct components. To address this challenge, we employ the annular proximity graph with angle constraints introduced in Trillos et al. (2023); see Appendix C.1 for a high-level summary of this construction. Our analysis improves upon the best known spectral convergence result in that work

$$|\hat{\lambda}_\ell - \sigma_\eta\lambda_\ell| \lesssim \mathcal{O}(n^{-\frac{1}{3d-1}}), \qquad \|\hat{f}_\ell - f_\ell\|_{L^2(\mathcal{M})} \lesssim \mathcal{O}(n^{-\frac{1}{6d-2}})$$

and achieves the sharper convergence rate stated in Theorem 7. The analytical framework developed here is conditionally based on the high-probability events in Theorems 30 and 31, which simultaneously promote strong within-manifold connectivity while maintaining sparse between-manifold connections. These underlying probabilities are then made explicit and rigorously aggregated in the final proof in Section 3.3.

We focus on the manifold components with maximal intrinsic dimension $d := \max_j d_j$ which governs the bottleneck of the overall convergence rate. For some inner and outer annulus radii $0 < \epsilon_- < \epsilon_+ < 1$ to be chosen appropriately, define the Euclidean and geodesic annulus kernels respectively by

$$K_\epsilon^E(x,y) := \eta_{\epsilon_+}(\|x-y\|) - \eta_{\epsilon_-}(\|x-y\|), \qquad K_\epsilon^g(x,y) := \eta_{\epsilon_+}(d_g(x,y)) - \eta_{\epsilon_-}(d_g(x,y)).$$

When working in normal coordinates via the change of variables $y = \exp_x(\epsilon_+ v)$, with $\|v\| \le 1$, we further define the kernel in normal coordinates by

$$K_\epsilon^{\mathcal{N}}(v) := \eta_1(\|v\|) - \eta_{\epsilon_-/\epsilon_+}(\|v\|).$$

Setting the normalization factor $\vartheta_\varepsilon := (\epsilon_+^{d+2} - \epsilon_-^{d+2})^{-1}$, the graph Laplacian is given by

$$\mathcal{L}f_\ell(x) := \frac{\vartheta_\varepsilon}{n}\sum_{y\in\mathcal{X}_n} K_\epsilon^E(x,y)\left(f_\ell(x) - f_\ell(y)\right).$$

For $x \in \mathcal{X}_{n,j}$, the graph Laplacian decomposes into the within-manifold operator

$$\begin{aligned}\mathcal{L}^{\mathrm{wth}} f_\ell(x) &= \sum_{y \in \mathcal{X}_{n,j}} \frac{w_j}{n_j(\epsilon_+^{d+2} - \epsilon_-^{d+2})} \left[\eta_{\epsilon_+}(\|x-y\|) - \eta_{\epsilon_-}(\|x-y\|)\right](f_\ell(x) - f_\ell(y)) \\ &= \frac{\vartheta_\varepsilon}{n} \sum_{y \in \mathcal{X}_{n,j}} K_\epsilon^E(x,y)\,(f_\ell(x) - f_\ell(y))\,,\end{aligned} \tag{45}$$

and the between-manifold operator

$$\mathcal{L}^{\mathrm{btw}} f_\ell(x) = \frac{\vartheta_\varepsilon}{n} \sum_{k \neq j,\, y \in \mathcal{X}_{n,k}} K_\epsilon^E(x,y)\,(f_\ell(x) - f_\ell(y))\,. \tag{46}$$

Symmetry of K_ϵ^E identifies the quadratic energy $U := \langle f_\ell, \mathcal{L} f_\ell \rangle_{L^2(\mathcal{X}_n)} = U_{\mathrm{wth}} + U_{\mathrm{btw}}$ with

$$U_{\mathrm{wth}} := \langle f_\ell, \mathcal{L}^{\mathrm{wth}} f_\ell \rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{j=1}^{N} \sum_{x,y \in \mathcal{X}_{n,j}} K_\epsilon^E(x,y)(f_\ell(x) - f_\ell(y))^2, \tag{47}$$

and

$$U_{\mathrm{btw}} := \langle f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{j=1}^{N} \sum_{x \in \mathcal{X}_{n,j}} \sum_{k \neq j,\, y \in \mathcal{X}_{n,k}} K_\epsilon^E(x,y)(f_\ell(x) - f_\ell(y))^2. \tag{48}$$

We use $(\widehat{\lambda}_\ell, \widehat{f}_\ell)$ to denote an eigenpair of $\mathcal{L}$ with normalization $\|\widehat{f}_\ell\|^2_{L^2(\mathcal{X}_n)} = \frac{1}{n}\sum \widehat{f}_\ell(x)^2 = 1$. The graph-induced discrete H^1-norm, along with the corresponding seminorm, is given by

$$\|g\|^2_{H^1(\mathcal{X}_n)} := \|g\|^2_{L^2(\mathcal{X}_n)} + |g|^2_{H^1(\mathcal{X}_n)}, \qquad |g|^2_{H^1(\mathcal{X}_n)} := \frac{\vartheta_\varepsilon}{n^2} \sum_{x,y \in \mathcal{X}_n} K_\epsilon^E(x,y)\big(g(x) - g(y)\big)^2.$$

3.1 Preliminary Eigenpair Convergence

Our first step is to derive an error decomposition that separates the within-manifold approximation error from the between-manifold interaction. By the self-adjointness of $\mathcal{L}$,

$$\begin{aligned}\widehat{\lambda}_\ell \langle \widehat{f}_\ell, f_\ell \rangle_{L^2(\mathcal{X}_n)} &= \langle \widehat{f}_\ell, \mathcal{L} f_\ell \rangle_{\mathcal{L}^2(\mathcal{X}_n)} = \langle \widehat{f}_\ell, \sigma_\eta \Delta_{\mathcal{M}} f_\ell \rangle_{L^2(\mathcal{X}_n)} + \langle \widehat{f}_\ell, (\mathcal{L} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)} \\ &= \sigma_\eta \lambda_\ell \langle \widehat{f}_\ell, f_\ell \rangle_{L^2(\mathcal{X}_n)} + \langle \widehat{f}_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)} + \langle \widehat{f}_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)}.\end{aligned}$$

Adding and subtracting f_ℓ and moving the first term on the second line to the left gives

$$(\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell) \langle \widehat{f}_\ell, f_\ell \rangle_{L^2(\mathcal{X}_n)} = \langle \widehat{f}_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)} + \langle \widehat{f}_\ell - f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)} + \langle f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)}.$$

By Theorem 32, $\langle \widehat{f}_\ell, f_\ell \rangle_{L^2(\mathcal{X}_n)} \geq c_0$ with probability at least $1 - Cn\exp(-cn\epsilon_+^d)$ for some $0 < c_0 < 1$ independent of n. We can then isolate the eigenvalue discrepancy

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \leq C_\ell \big(\big|\langle \widehat{f}_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)}\big| + \big|\langle \widehat{f}_\ell - f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)}\big| + \big|\langle f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell \rangle_{L^2(\mathcal{X}_n)}\big| \big)$$

for some constant $C_\ell > 1$. The following Lemma 12 controls each term on the right-hand side and gives a preliminary high probability upper bound on the eigenvalue convergence rate. We will improve this rate in Section 3.2.

Lemma 12 (Preliminary eigenvalue convergence) *There exist some constant $C_\ell > 0$, depending only on the geometric, the density-class, the intersection parameters and the index ℓ such that with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$, one has*

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \le C_\ell \sqrt{\lambda_\ell} B_\ell \epsilon_+^2 \log \epsilon_+^{-1}. \tag{49}$$

Proof The proof simply follows by combining Propositions 34 through 36 in Appendix 3. Given $h : \mathcal{X}_n \to \mathbb{R}$, write $h_j := h|_{\mathcal{X}_{n,j}}$ for its restriction to the sample from $\mathcal{M}_j$. For $\varphi : \mathcal{X}_{n,j} \to \mathbb{R}$, define the componentwise discrete H^1-seminorm by

$$|\varphi|^2_{H^1(\mathcal{X}_{n,j})} := \frac{\vartheta_\varepsilon}{n_j^2} \sum_{x,y \in \mathcal{X}_{n,j}} K_\epsilon^g(x,y)\big(\varphi(x) - \varphi(y)\big)^2,$$

and its dual H^{-1}-seminorm by

$$\|h_j\|_{H^{-1}(\mathcal{X}_{n,j})} := \sup\left\{ \frac{1}{n_j} \sum_{x \in \mathcal{X}_{n,j}} h_j(x)\varphi(x) : \frac{1}{n_j} \sum_{x \in \mathcal{X}_{n,j}} \varphi(x) = 0,\ |\varphi|_{H^1(\mathcal{X}_{n,j})} \le 1 \right\}.$$

It follows that the first contribution can be controlled as

$$\big|\langle \widehat{f}_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)}\big| \le \|\widehat{f}_\ell\|_{H^1(\mathcal{X}_n)} \|(\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell\|_{H^{-1}(\mathcal{X}_n)}.$$

Note that the right side of the quadratic energy is exactly $\frac{1}{2}|g|^2_{H^1(\mathcal{X}_n)}$ for any $g : \mathcal{X}_n \to \mathbb{R}$:

$$\langle \mathcal{L} g, g \rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{x,y \in \mathcal{X}_n} K_\epsilon^E(x,y)\big(g(x) - g(y)\big)^2 = \frac{1}{2}|g|^2_{H^1(\mathcal{X}_n)}.$$

Applying to $\widehat{f}_\ell$ and using $\mathcal{L}\widehat{f}_\ell = \widehat{\lambda}_\ell \widehat{f}_\ell$ yields $|\widehat{f}_\ell|^2_{H^1(\mathcal{X}_n)} = 2\widehat{\lambda}_\ell \|\widehat{f}_\ell\|^2_{L^2(\mathcal{X}_n)} = 2\widehat{\lambda}_\ell$. Therefore

$$\|\widehat{f}_\ell\|^2_{H^1(\mathcal{X}_n)} = 1 + 2\widehat{\lambda}_\ell \quad \Longrightarrow \quad \|\widehat{f}_\ell\|_{H^1(\mathcal{X}_n)} \le C_\ell\Big(1 + \sqrt{\widehat{\lambda}_\ell}\Big).$$

Combining with Proposition 34 establishes, with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,

$$\big|\langle \widehat{f}_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)}\big| \le C_\ell \sqrt{\lambda_\ell} B_\ell \epsilon_+^2 \log \epsilon_+^{-1}.$$

For the second contribution, assume for the moment that $\|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} < \infty$. We establish the exact convergence rate in Lemma 13 below. Combining with Proposition 35 gives

$$\|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} \|\mathcal{L}^{\mathrm{btw}} f_\ell\|_{H^{-1}(\mathcal{X}_n)} \le C_\ell B_\ell \epsilon_+^2,$$

with probability at least $1 - Cn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$. For the last term, Proposition 36 gives

$$\big|\langle f_l, \mathcal{L}^{\mathrm{btw}} f_l \rangle_{L^2(\mathcal{X}_n)}\big| \le C_\ell B_\ell^2 \epsilon_+^4.$$

with probability at least $1 - Cn\exp\big(-cn\epsilon_+^{d+4}\big)$. All pieces together complete the proof. ■

The high-probability upper bound on the eigenvalues' convergence allows us to bound the eigenfunctions convergence. This is established in the next Lemma 13.

Lemma 13 (Eigenfunction convergence) *Let $(\widehat{\lambda}_\ell, \widehat{f}_\ell)$ be an $L^2(\mathcal{X}_n)$-normalized eigenpair of the graph Laplacian $\mathcal{L}$. Given the preliminary eigenvalue convergence established in Lemma 12, there exists some constant $C_\ell > 0$, depending only on the geometric, the density-class, the intersection parameters, the index ℓ, and the eigengap γ_ℓ, such that with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$, one has*

$$\|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} \le C_\ell B_\ell \epsilon_+^2 \log \epsilon_+^{-1}.$$

Here B_ℓ denotes the regularity constant defined in (9).

Proof For notational convenience, set $h_\ell := \widehat{f}_\ell - f_\ell$. Since $\mathcal{L}1 = 0$, self-adjointness gives

$$\widehat{\lambda}_\ell \langle \widehat{f}_\ell, 1\rangle_{L^2(\mathcal{X}_n)} = \langle \mathcal{L}\widehat{f}_\ell, 1\rangle_{L^2(\mathcal{X}_n)} = \langle \widehat{f}_\ell, \mathcal{L}1\rangle_{L^2(\mathcal{X}_n)} = 0.$$

Thus $\widehat{f}_\ell$ has empirical mean zero. For the population eigenfunction, define

$$\bar{f}_\ell := \langle f_\ell, 1\rangle_{L^2(\mathcal{X}_n)}, \qquad \delta_{\ell,n} := \Big|\|f_\ell\|^2_{L^2(\mathcal{X}_n)} - 1\Big|, \qquad f_\ell^\circ := f_\ell - \bar{f}_\ell, \qquad h_\ell^\circ := \widehat{f}_\ell - f_\ell^\circ.$$

Then h_ℓ° has empirical mean zero and $h_\ell^\circ = h_\ell + \bar{f}_\ell$. Since the discrete H^1-seminorm is invariant under the addition of constants, $|h_\ell^\circ|_{H^1(\mathcal{X}_n)} = |h_\ell|_{H^1(\mathcal{X}_n)}$. Moreover, by orthogonality to the constants,

$$\|h_\ell\|^2_{L^2(\mathcal{X}_n)} = \|h_\ell^\circ\|^2_{L^2(\mathcal{X}_n)} + |\bar{f}_\ell|^2. \tag{50}$$

Thus, the discrete H^1-norm is given by

$$\|h_\ell\|^2_{H^1(\mathcal{X}_n)} = \|h_\ell\|^2_{L^2(\mathcal{X}_n)} + |h_\ell|^2_{H^1(\mathcal{X}_n)} = \|h_\ell^\circ\|^2_{L^2(\mathcal{X}_n)} + |\bar{f}_\ell|^2 + |h_\ell|^2_{H^1(\mathcal{X}_n)}.$$

Since $\Delta_{\mathcal{M}} f_\ell^\circ = \Delta_{\mathcal{M}} f_\ell = \lambda_\ell f_\ell = \lambda_\ell f_\ell^\circ + \lambda_\ell \bar{f}_\ell$, the centered error satisfies

$$(\mathcal{L} - \widehat{\lambda}_\ell I) h_\ell^\circ = g_\ell^\circ - \sigma_\eta \lambda_\ell \bar{f}_\ell, \tag{51}$$

where

$$g_\ell^\circ := -(\mathcal{L} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell^\circ - (\sigma_\eta \lambda_\ell - \widehat{\lambda}_\ell) f_\ell^\circ.$$

The final term in (51) is constant and hence vanishes when paired with h_ℓ°, as well as with every positive-eigenvalue eigenvector of $\mathcal{L}$. Therefore,

$$\langle \mathcal{L} h_\ell^\circ, h_\ell^\circ\rangle_{L^2(\mathcal{X}_n)} - \widehat{\lambda}_\ell \|h_\ell^\circ\|^2_{L^2(\mathcal{X}_n)} = \langle g_\ell^\circ, h_\ell^\circ\rangle_{L^2(\mathcal{X}_n)}. \tag{52}$$

By the symmetry of K_ε^E,

$$\langle \mathcal{L} h_\ell^\circ, h_\ell^\circ\rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{x,y\in\mathcal{X}_n} K_\varepsilon^E(x,y)\big(h_\ell^\circ(x) - h_\ell^\circ(y)\big)^2 = \frac{1}{2}|h_\ell^\circ|^2_{H^1(\mathcal{X}_n)}.$$

Let $S := \operatorname{span}\{\widehat{f}_\ell\}$ and let $P_S^\perp$ denote the orthogonal projection onto $S^\perp$. Let $\{(\widehat{\lambda}_k, \widehat{f}_k)\}_{k=1}^n$ be an $L^2(\mathcal{X}_n)$-orthonormal eigenbasis of $\mathcal{L}$, and write

$$P_S^\perp h_\ell^\circ = \sum_{k\neq\ell \text{ s.t. } \widehat{\lambda}_k > 0} a_k \widehat{f}_k, \qquad a_k := \langle h_\ell^\circ, \widehat{f}_k\rangle_{L^2(\mathcal{X}_n)}, \qquad b_k := \langle g_\ell^\circ, \widehat{f}_k\rangle_{L^2(\mathcal{X}_n)}.$$

The zero mode does not appear because h_ℓ° is empirically mean zero. Define the empirical eigengap by $\widehat{\gamma}_\ell := \min_{k\neq\ell} |\widehat{\lambda}_k - \widehat{\lambda}_\ell|$. Taking the inner product of (51) with $\widehat{f}_k$ gives

$$(\widehat{\lambda}_k - \widehat{\lambda}_\ell)a_k = \langle g_\ell^\circ, \widehat{f}_k\rangle_{L^2(\mathcal{X}_n)} = b_k, \qquad k \neq \ell, \quad \widehat{\lambda}_k > 0.$$

The Parseval identity gives $\|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 = \sum_{k\neq\ell} |a_k|^2$. By the Cauchy-Schwarz inequality,

$$\begin{aligned}\widehat{\gamma}_\ell\|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 &= \sum_{k\neq\ell,\,\widehat{\lambda}_k>0} \widehat{\gamma}_\ell|a_k|^2 \le \sum_{k\neq\ell,\,\widehat{\lambda}_k>0} |\widehat{\lambda}_k - \widehat{\lambda}_\ell||a_k|^2 = \sum_{k\neq\ell,\,\widehat{\lambda}_k>0} |b_k||a_k| \\ &\le \left(\sum_{k\neq\ell,\,\widehat{\lambda}_k>0} \frac{|b_k|^2}{\widehat{\lambda}_k}\right)^{\frac{1}{2}} \left(\sum_{k\neq\ell,\,\widehat{\lambda}_k>0} \widehat{\lambda}_k|a_k|^2\right)^{\frac{1}{2}} = C\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)}|h_\ell^\circ|_{H^1(\mathcal{X}_n)}.\end{aligned} \tag{53}$$

On the other hand, choose the sign of $\widehat{f}_\ell$ such that $\zeta_{\ell,n} := \langle f_\ell, \widehat{f}_\ell\rangle_{L^2(\mathcal{X}_n)} = \langle f_\ell^\circ, \widehat{f}_\ell\rangle_{L^2(\mathcal{X}_n)} \ge 0$. Since $\|\widehat{f}_\ell\|_{L^2(\mathcal{X}_n)} = 1$,

$$\|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 = \|f_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 - \zeta_{\ell,n}^2, \qquad \|h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 = (1-\zeta_{\ell,n})^2 + \|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2.$$

Moreover, $0 \le \zeta_{\ell,n} \le \|f_\ell^\circ\|_{L^2(\mathcal{X}_n)}$, and hence

$$(1-\zeta_{\ell,n})^2 \le 2(1-\|f_\ell^\circ\|_{L^2(\mathcal{X}_n)})^2 + 2(\|f_\ell^\circ\|_{L^2(\mathcal{X}_n)} - \zeta_{\ell,n})^2 \le 2(1-\|f_\ell^\circ\|_{L^2(\mathcal{X}_n)})^2 + 2\|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2.$$

Since

$$|1 - \|f_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2| \le \delta_{\ell,n} + |\bar{f}_\ell|^2, \qquad |1 - \|f_\ell^\circ\|_{L^2(\mathcal{X}_n)}| \le |1 - \|f_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2|,$$

it follows that

$$\|h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 \le 3\|P_S^\perp h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 + C\delta_{\ell,n}^2 + C|\bar{f}_\ell|^4. \tag{54}$$

On the preliminary spectral-convergence event from Lemma 12, for n sufficiently large, the upper bound in (49) converges to zero. Since $\gamma_\ell > 0$ is fixed, we therefore have

$$|\widehat{\lambda}_k - \sigma_\eta\lambda_k| \le \frac{\sigma_\eta\gamma_\ell}{4}$$

for every index k entering the definition of $\widehat{\gamma}_\ell$. Consequently,

$$|\widehat{\lambda}_k - \widehat{\lambda}_\ell| \ge \sigma_\eta|\lambda_k - \lambda_\ell| - |\widehat{\lambda}_k - \sigma_\eta\lambda_k| - |\widehat{\lambda}_\ell - \sigma_\eta\lambda_\ell| \ge \sigma_\eta\gamma_\ell - \frac{\sigma_\eta\gamma_\ell}{2} = \frac{\sigma_\eta\gamma_\ell}{2}.$$

Taking the minimum over $k \neq \ell$ and increasing n further if necessary,

$$\widehat{\gamma}_\ell \ge \frac{\sigma_\eta\gamma_\ell}{2}, \qquad \frac{\widehat{\lambda}_\ell}{\widehat{\gamma}_\ell} \le \frac{\sigma_\eta\lambda_\ell + |\widehat{\lambda}_\ell - \sigma_\eta\lambda_\ell|}{\widehat{\gamma}_\ell} \le \frac{2\sigma_\eta\lambda_\ell}{\frac{\sigma_\eta\gamma_\ell}{2}} \le 4\frac{\lambda_\ell}{\gamma_\ell}.$$

Combining (53) and (54) yields

$$\widehat{\lambda}_\ell\|h_\ell^\circ\|_{L^2(\mathcal{X}_n)}^2 \le C\frac{\lambda_\ell}{\gamma_\ell}\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)}|h_\ell^\circ|_{H^1(\mathcal{X}_n)} + C_\ell(\delta_{\ell,n}^2 + |\bar{f}_\ell|^4). \tag{55}$$

Substituting (55) into (52) gives

$$\frac{1}{2}|h_\ell^\circ|_{H^1(\mathcal{X}_n)}^2 \le C\frac{\lambda_\ell}{\gamma_\ell}\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)}|h_\ell^\circ|_{H^1(\mathcal{X}_n)} + C_\ell\big(\delta_{\ell,n}^2 + |\bar{f}_\ell|^4\big).$$

The elementary implication $x^2 \le ax + b^2 \Rightarrow x \le a + b$ therefore yields

$$|h_\ell^\circ|_{H^1(\mathcal{X}_n)} \le C\frac{\lambda_\ell}{\gamma_\ell}\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} + C_\ell\big(\delta_{\ell,n} + |\bar{f}_\ell|^2\big). \tag{56}$$

Equations (53), (54), and (56) similarly imply

$$\|h_\ell^\circ\|_{L^2(\mathcal{X}_n)} \le C_{\ell,\gamma}\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} + C_\ell\big(\delta_{\ell,n} + |\bar{f}_\ell|^2\big).$$

Together with (50) and $|h_\ell|_{H^1} = |h_\ell^\circ|_{H^1}$, this gives

$$\|h_\ell\|_{H^1(\mathcal{X}_n)} \le C\frac{\lambda_\ell}{\gamma_\ell}\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} + C_\ell\big(\delta_{\ell,n} + |\bar{f}_\ell|\big). \tag{57}$$

Since both $\mathcal{L}$ and $\Delta_\mathcal{M}$ annihilate constants, $(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell^\circ = (\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell$. Therefore,

$$\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} \le \|(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)} + |\widehat{\lambda}_\ell - \sigma_\eta\lambda_\ell|\|f_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)}.$$

By the discrete Poincaré inequality and $\|f_\ell^\circ\|_{L^2(\mathcal{X}_n)} \le 1 + \delta_{\ell,n}$, $\|f_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} \le C$. Hence (49)

$$\|g_\ell^\circ\|_{H^{-1}(\mathcal{X}_n)} \le C\left(\|(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)} + |\widehat{\lambda}_\ell - \sigma_\eta\lambda_\ell|\right).$$

It remains to bound $|\bar{f}_\ell|$ and $\delta_{\ell,n}$. Since population eigenfunction satisfies $\int_\mathcal{M} f_\ell d\mu = 0$, Bernstein's inequality gives, for any $t > 0$, with probability at least $1 - 2\exp(-ct^2)$,

$$|\bar{f}_\ell| \le C\left(\frac{t}{\sqrt{n}} + \frac{\|f_\ell\|_{L^\infty(\mathcal{M})}t^2}{n}\right).$$

Setting $t = \sqrt{n}\epsilon_+^2$ gives, with probability at least $1 - 2\exp(-cn\epsilon_+^4)$,

$$|\bar{f}_\ell| \le C_\ell(\epsilon_+^2 + \epsilon_+^4) \le C_\ell\epsilon_+^2.$$

Applying Bernstein's inequality once more gives, with probability at least $1 - 2\exp(-ct^2)$,

$$\delta_{\ell,n} = \left|\|f_\ell\|_{L^2(\mathcal{X}_n)}^2 - 1\right| \le C\left(\frac{\|f_\ell\|_{L^\infty(\mathcal{M})}t}{\sqrt{n}} + \frac{\|f_\ell\|_{L^\infty(\mathcal{M})}^2 t^2}{n}\right).$$

Setting $t = \sqrt{n}\epsilon_+^2$ gives, with probability at least $1 - 2\exp(-cn\epsilon_+^4)$, $\delta_{\ell,n} \le C_\ell\epsilon_+^2$. Therefore, both remainder terms can be absorbed into the eigenvalue convergence (49).

To complete the proof, it suffices to control $\|(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)}$. Decompose it by

$$\|(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)} \le \|(\mathcal{L}^{\text{wth}} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)} + \|\mathcal{L}^{\text{btw}}f_\ell\|_{H^{-1}(\mathcal{X}_n)}.$$

Proposition 34 and 35 together yield, with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,

$$\|(\mathcal{L} - \sigma_\eta\Delta_\mathcal{M})f_\ell\|_{H^{-1}(\mathcal{X}_n)} \le C_\ell B_\ell\epsilon_+^2\log\epsilon_+^{-1}.$$

Combining the eigenvalue estimates from Lemma 12 completes the proof. ∎

Remark 14 *We can now revisit the proof of Lemma 12. The second term is now well controlled without imposing the assumption of the eigenfunction convergence. By Lemma 13, with probability at least* $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,

$$\left|\langle \widehat{f}_\ell - f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| \le \|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)}\|\mathcal{L}^{\mathrm{btw}} f_\ell\|_{H^{-1}(\mathcal{X}_n)} \le C_\ell B_\ell \epsilon_+^4 \log \epsilon_+^{-1},$$

which is strictly smaller in ϵ_+. *The rest of the proof remains unchanged.*

3.2 Improved Eigenpair Convergence

In this section, we improve the eigenpair estimate from Section 3.1. The analysis takes the same procedure but with one more decomposition

$$\begin{aligned}|\hat{\lambda}_\ell - \sigma_\eta\lambda_\ell| \le C_\ell\big(&\left|\langle \hat{f}_\ell - f_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta\Delta_{\mathcal{M}})f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| + \left|\langle f_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta\Delta_{\mathcal{M}})f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| \\ &+ \left|\langle \hat{f}_\ell - f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| + \left|\langle f_\ell, \mathcal{L}^{\mathrm{btw}} f_\ell\rangle_{L^2(\mathcal{X}_n)}\right|\big).\end{aligned} \tag{58}$$

All terms on the right except the second has been controlled in Section 3.1. We begin with bounding the discrepancy between the expected within-manifold graph energy and its continuum counterpart:

$$\mathrm{Bias}_{\mathrm{wth}} := \left|\mathbb{E}\langle f_\ell, (\sigma_\eta\Delta_{\mathcal{M}} - \mathcal{L}^{\mathrm{wth}})f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| = \left|\sigma_\eta\mathbb{E}\langle f_\ell, \Delta_{\mathcal{M}} f_\ell\rangle_{L^2(\mathcal{X}_n)} - \mathbb{E}\langle f_\ell, \mathcal{L}^{\mathrm{wth}} f_\ell\rangle_{L^2(\mathcal{X}_n)}\right|.$$

By the definition of $\Delta_{\mathcal{M}}$, the continuum term admits the decomposition:

$$\begin{aligned}\sigma_\eta\mathbb{E}\langle f_\ell, \Delta_{\mathcal{M}} f_\ell\rangle_{L^2(\mathcal{X}_n)} &= -\sigma_\eta\sum_{j=1}^N w_j\int_{\mathcal{M}_j} f_\ell(w_j\Delta_{\mathcal{M}_j} f_\ell)\rho_j\,\mathrm{dvol}_g \\ &= -\sigma_\eta\sum_{j=1}^N w_j^2\int_{\mathcal{M}_j} f_\ell\langle\nabla_g\rho_j, \nabla_g f_\ell\rangle\rho_j\,\mathrm{dvol}_g - \frac{\sigma_\eta}{2}\sum_{j=1}^N w_j^2\int_{\mathcal{M}_j} f_\ell\rho_j^2\Delta_{\mathcal{M}_j} f_\ell\,\mathrm{dvol}_g.\end{aligned} \tag{59}$$

We replace the ambient Euclidean kernel K_ϵ^E by its geodesic counterpart K_ϵ^g. The discrepancy introduced by this replacement will be collected into a geometric remainder and estimated separately. For $x, y \in \mathcal{X}_{n,j}$, expand $f_\ell(y)$ along the geodesic from x to y,

$$f_\ell(y) - f_\ell(x) = \langle\nabla_g f_\ell(x), \log_x(y)\rangle_g + \frac{1}{2}\langle\nabla_g^2 f_\ell(x)\log_x(y), \log_x(y)\rangle_g + \mathcal{R}_f(x,y), \tag{60}$$

where

$$\mathcal{R}_f(x,y) = \int_0^1 \frac{(1-t)^2}{2}\nabla_g^3 f_\ell(\gamma_{xy}(t))[\log_x(y), \log_x(y), \log_x(y)]\,dt$$

denotes the exact third-order Taylor remainder satisfying $|\mathcal{R}_f(x,y)| \le C\,d_g(x,y)^3$. Here γ_{xy} denotes the geodesic from x to y. The quadratic form then admits the decomposition

$$U_{\mathrm{wth}} = \langle f_\ell, \mathcal{L}^{\mathrm{wth}} f_\ell\rangle_{L^2(\mathcal{X}_n)} = U_{\mathrm{drift}} + U_{\mathrm{diff}} + U_{\mathrm{rem}} + U_{\mathrm{geo}}. \tag{61}$$

The two leading contribution, and the two remainders are defined respectively as

$$U_{\mathrm{drift}} := \left\langle f_\ell, \mathcal{L}^{\mathrm{drift}} f_\ell\right\rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{n^2}\sum_{j=1}^N\sum_{x,y\in\mathcal{X}_{n,j}} K_\epsilon^g(x,y)f_\ell(x)\langle\nabla_g f_\ell(x), \log_x(y)\rangle_g. \tag{62}$$

$$U_{\mathrm{diff}} := \left\langle f_\ell, \mathcal{L}^{\mathrm{diff}} f_\ell \right\rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{j=1}^{N} \sum_{x,y \in \mathcal{X}_{n,j}} K_\epsilon^g(x,y) f_\ell(x) \langle \nabla_g^2 f_\ell(x) \log_x(y), \log_x(y) \rangle_g. \tag{63}$$

$$U_{\mathrm{rem}} := \left\langle f_\ell, \mathcal{L}^{\mathrm{rem}} f_\ell \right\rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{n^2} \sum_{j=1}^{N} \sum_{x,y \in \mathcal{X}_{n,j}} K_\epsilon^g(x,y) f_\ell(x) \mathcal{R}_f(x,y). \tag{64}$$

$$U_{\mathrm{geo}} := \langle f_\ell, \mathcal{L}^{\mathrm{geo}} f_\ell \rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{n^2} \sum_{j=1}^{N} \sum_{x,y \in \mathcal{X}_{n,j}} \big(K_\epsilon^E(x,y) - K_\epsilon^g(x,y)\big) f_\ell(x)(f_\ell(y) - f_\ell(x)). \tag{65}$$

The expected drift contribution with respect to the sampling measure is

$$\mathbb{E} U_{\mathrm{wth}} = \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} \int_{\mathcal{M}_j} K_\epsilon^g(x,y) f_\ell(x) \langle \nabla_g f_\ell(x), \log_x(y) \rangle_g \rho_j(x) \rho_j(y) d\mathrm{vol}_g(y) d\mathrm{vol}_g(x).$$

Passing to normal coordinates with $v \in B(0,1) \subset T_x\mathcal{M}_j$ and $y = \exp_x(\epsilon_+ v)$, one has

$$\mathbb{E}\, U_{\mathrm{drift}} = \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x) \rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) \langle \nabla_g f_\ell(x), \epsilon_+ v \rangle_g \rho_j(\exp_x(\epsilon_+ v)) J_x(\epsilon_+ v) \epsilon_+^d dv d\mathrm{vol}_g. \tag{66}$$

The remaining terms have analogous normal-coordinate representations

$$\mathbb{E}\, U_{\mathrm{diff}} = \frac{\vartheta_\varepsilon}{2} \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x) \rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) \langle \nabla_g^2 f_\ell(x)\, \epsilon_+ v, \epsilon_+ v \rangle_g\, \rho_j(\exp_x(\epsilon_+ v)) J_x(\epsilon_+ v) \epsilon_+^d dv d\mathrm{vol}_g, \tag{67}$$

$$\mathbb{E}\, U_{\mathrm{rem}} = \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x) \rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) \mathcal{R}_f(x, \exp_x(\epsilon_+ v))\, \rho_j(\exp_x(\epsilon_+ v)) J_x(\epsilon_+ v) \epsilon_+^d dv d\mathrm{vol}_g, \tag{68}$$

$$\begin{aligned} \mathbb{E}\, U_{\mathrm{rem}} = \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x) \rho_j(x) \times \int_{B(0,1)} & [K_\epsilon^E(x, \exp_x(\epsilon_+ v)) - K_\epsilon^g(x, \exp_x(\epsilon_+ v))] \\ & (f_\ell(\exp_x(\epsilon_+ v)) - f_\ell(x))\, \rho_j(\exp_x(\epsilon_+ v)) J_x(\epsilon_+ v) \epsilon_+^d d\mathrm{vol}_g. \end{aligned} \tag{69}$$

Throughout these expressions, the density is expanded in normal coordinates according to

$$\rho_j(\exp_x(\epsilon_+ v)) = \rho_j(x) + \epsilon_+ \langle \nabla_g \rho_j(x), v \rangle_g + \mathcal{R}_\rho(x,v), \qquad |\mathcal{R}_\rho(x,v)| \le C \epsilon_+^2 \|v\|^2.$$

The Jacobian determinant of the exponential map is denoted by $J_x(\epsilon_+ v)$ which satisfies

$$J_x(\epsilon_+ v) = 1 + q_x(\epsilon_+ v) + \mathcal{R}_J(x,v), \qquad |\mathcal{R}_J(x,v)| \le C \epsilon_+^3 \|v\|^3,$$

where $q_x(\cdot)$ is a quadratic polynomial whose coefficients depend smoothly on the local curvature. Multiplying the two expansions yields

$$\rho_j(\exp_x(\epsilon_+ v)) J_x(\epsilon_+ v) = \rho_j(x) + \epsilon_+ \langle \nabla_g \rho_j(x), v \rangle_g + Q_{\rho,J}(x, v; \epsilon_+) + T_{\rho,J}(x, v; \epsilon_+) + \mathcal{O}(\epsilon_+^4 \|v\|^4), \tag{70}$$

where

$$Q_{\rho,J}(x,v;\epsilon_+) := \rho_j(x)q_x(\epsilon_+ v) + \mathcal{R}_\rho(x,v) \tag{71}$$

collects quadratic-order contribution, and

$$T_{\rho,J}(x,v;\epsilon_+) = \rho_j(x)\mathcal{R}_J(x,v) + \epsilon_+\langle \nabla_g \rho_j(x), v\rangle_g q_x(\epsilon_+ v) \tag{72}$$

collects cubic-order contribution. Uniformly over admissible manifolds and density classes

$$|Q_{\rho,J}(x,v;\epsilon_+)| \le C\epsilon_+^2\|v\|^2, \qquad |T_{\rho,J}(x,v;\epsilon_+)| \le C\epsilon_+^3\|v\|^3. \tag{73}$$

The remainder of the proof identifies the leading-order contributions and controls all higher-order errors. The drift term (66) recovers the first continuum contribution in the Laplace-Beltrami decomposition (59), which is established in Lemma 15. Similarly, the diffusion term (67) recovers the second continuum contribution in (59), as shown in Lemma 16. The Taylor remainder (68) and the geometric correction (69) are shown to be of higher-order errors $\mathcal{O}(\epsilon_+^2)$ in Lemmas 17 and 18, respectively. Combining these estimates yield Proposition 19, establishing the bias bound for the within-manifold graph Laplacian and its convergence to the tensorized Laplace–Beltrami operator.

Lemma 15 (Drift contribution) *The expected drift contribution in (66) satisfies*

$$\left| \mathbb{E}\big\langle f_\ell, \mathcal{L}^{\text{drift}} f_\ell \big\rangle_{L^2(\mathcal{X}_n)} - \sigma_\eta \sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell \rho_j \langle \nabla_g \rho_j, \nabla_g f_\ell \rangle d\text{vol}_g \right| \le C_\ell B_\ell^2 \epsilon_+^2$$

where B_ℓ is defined in (9) and C_ℓ depends only on the geometric and density regularity, the index ℓ, the kernel profile η, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights.

Proof Substituting the expansion (70) into (66), we decompose the drift contribution according to the order in ϵ_+. The zeroth-order term is

$$\vartheta_\varepsilon \sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j^2(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \, \epsilon_+^d dv d\text{vol}_g(x).$$

Since $K_\epsilon^{\mathcal{N}}$ is radial, it is even in v. On the other hand, $\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g$ is odd in v. Therefore, the inner integral is odd and so vanishes over the symmetric ball $B(0,1)$.

For the first-order term, the definition of $K_\epsilon^{\mathcal{N}}(v)$, σ_η, and the change of variable yield

$$\begin{aligned}
&\int_{B_0(1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \langle \nabla_g \rho_j(x), \epsilon_+ v\rangle_g \epsilon_+^d dv \\
&= \epsilon_+^{d+2} \frac{\langle \nabla_g f_\ell(x), \nabla_g \rho_j(x)\rangle_g}{d} \left[\int_{B(0,1)} \eta_1(\|v\|)\|v\|^2 dv - \int_{B(0,\frac{\epsilon_-}{\epsilon_+})} \eta_1\left(\frac{\epsilon_+}{\epsilon_-}\|v\|\right)\|v\|^2 dv \right] \\
&= \epsilon_+^{d+2} \frac{\langle \nabla_g f_\ell(x), \nabla_g \rho_j(x)\rangle_g}{d} \left[1 - \left(\frac{\epsilon_-}{\epsilon_+}\right)^{d+2} \right] \int_{B(0,1)} \eta_1(\|v\|)\|v\|^2 dv \\
&= \sigma_\eta \langle \nabla_g f_\ell(x), \nabla_g \rho_j(x)\rangle_g (\epsilon_+^{d+2} - \epsilon_-^{d+2}).
\end{aligned}$$

Therefore, using the definition of ϑ_ε

$$\vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \langle \nabla_g \rho_j(x), \epsilon_+ v\rangle_g \epsilon_+^d dv d\mathrm{vol}_g(x)$$
$$= \sigma_\eta \sum_{j=1}^{K} w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\langle \nabla_g f_\ell(x), \nabla_g \rho_j(x)\rangle_g \rho_j(x) d\mathrm{vol}_g(x).$$

Hence, the first-order drift contribution exactly matches the first continuum term in (59).

It remains to control the higher-order contributions. We begin with the quadratic term $Q_{\rho,J}$. The part involving q_x vanishes by symmetry: $q_x(\epsilon_+ v)$ and $K_\epsilon^{\mathcal{N}}$ are even in v, while $\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g$ is odd. Hence the corresponding inner integral is zero. The remaining part of $Q_{\rho,J}$ involves the density Taylor remainder $\mathcal{R}_\rho$. Since we only assume $\rho_j \in C^2$, we use the integral remainder formula

$$\mathcal{R}_\rho(x,v) = \epsilon_+^2 \int_0^1 (1-t)\nabla_g^2 \rho_j(\exp_x(t\epsilon_+ v)) \left[d(\exp_x)_{t\epsilon_+ v}(v), d(\exp_x)_{t\epsilon_+ v}(v)\right] dt.$$

For each $t \in [0,1]$, define the Hessian factor

$$\mathcal{D}_t(x,v) := \nabla_g^2 \rho_j(\exp_x(t\epsilon_+ v)) \left[d(\exp_x)_{t\epsilon_+ v}(v), d(\exp_x)_{t\epsilon_+ v}(v)\right].$$

The corresponding contribution is

$$\mathcal{I}_t := \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \epsilon_+^{d+2} \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \, \mathcal{D}_t(x,v)\, dv\, d\mathrm{vol}_g(x).$$

A direct estimate would only give a $\mathcal{O}(\epsilon_+)$ bound. To recover the desired $\mathcal{O}(\epsilon_+^2)$ rate, we use the geodesic symmetrization argument from Section C.3, which yields $|\mathcal{I}_t| \le C B_\ell^2 \epsilon_+^2$ uniformly for all $t \in [0,1]$. Consequently,

$$\left|\vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell \rho_j \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell, \epsilon_+ v\rangle_g \mathcal{R}_\rho(x,v)\epsilon_+^d dv d\mathrm{vol}_g\right| = \left|\int_0^1 (1-t)\mathcal{I}_t dt\right| \le C_\ell B_\ell^2 \epsilon_+^2.$$

Combining this with the symmetry cancellation of the q_x-term shows that the full quadratic contribution $Q_{\rho,J}$ contributes only an $\mathcal{O}(B_\ell^2 \epsilon_+^2)$ error.

Finally, we control the cubic contribution $T_{\rho,J}$. Since $|\mathcal{R}_J(x,v)| \le C\epsilon_+^3 \|v\|^3$, we have

$$\left|\vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \rho_j \mathcal{R}_J(x,v)\epsilon_+^d \, dv \, d\mathrm{vol}_g(x)\right|$$
$$\le C_\ell \vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} |f_\ell(x)||\rho_j(x)|^2 |\nabla_g f_\ell(x)| d\mathrm{vol}_g(x) \int_{B(0,1)} |K_\epsilon^{\mathcal{N}}(v)| \|v\|^4 dv$$
$$\le C_\ell \vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^{N} w_j^2 \|f_\ell\|_{L^2(\mathcal{M}_j)} \|\nabla_g f_\ell\|_{L^2(\mathcal{M}_j)} \le C_\ell B_\ell^2 \epsilon_+^2.$$

Similarly, since $q_x(\epsilon_+ v) = \mathcal{O}(\epsilon_+^2\|v\|^2)$, the mixed cubic term satisfies

$$\begin{aligned}
&\left|\vartheta_\varepsilon \sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g f_\ell(x), \epsilon_+ v\rangle_g \langle \nabla_g \rho_j, \epsilon_+ v\rangle_g q_x(\epsilon_+ v)\epsilon_+^d dv d\mathrm{vol}_g(x)\right| \\
\le\ & C_\ell \vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} |f_\ell(x)||\rho_j(x)||\nabla_g f_\ell(x)||\nabla_g \rho_j(x)| d\mathrm{vol}_g(x) \int_{B(0,1)} |K_\epsilon^{\mathcal{N}}(v)|\|v\|^4 dv \\
\le\ & C_\ell \vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^N w_j^2 \|f_\ell\|_{L^2(\mathcal{M}_j)} \|\nabla_g f_\ell\|_{L^2(\mathcal{M}_j)} \le C_\ell B_\ell^2 \epsilon_+^2.
\end{aligned}$$

In both estimates, the moment $\int_{B(0,1)} |K_\epsilon^{\mathcal{N}}(v)|\|v\|^4\, dv$ is bounded by a constant depending only on the kernel profile η and the fixed ratio ϵ_-/ϵ_+. The fourth-order product remainder in (70) is even smaller. Hence all contributions beyond the first-order drift term are $\mathcal{O}(B_\ell^2\epsilon_+^2)$. Combining the zeroth-order cancellation, the exact identification of the first-order contribution, and the quadratic and cubic-remainder estimate completes the proof.■

Lemma 16 (Diffusion contribution) *The expected diffusion contribution* (67) *satisfies*

$$\left|\mathbb{E}\big\langle f_\ell, \mathcal{L}^{\mathrm{diff}} f_\ell\big\rangle_{L^2(\mathcal{X}_n)} - \frac{\sigma_\eta}{2}\sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j^2(x)\Delta_{\mathcal{M}_j} f_\ell(x) \mathrm{dvol}_g(x)\right| \le C_\ell B_\ell^2 \epsilon_+^2,$$

where B_ℓ is defined in (9) *and C_ℓ depends only on the geometric and density regularity, the index ℓ, the kernel profile η, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights.*

Proof The proof follows the same decomposition strategy as in Lemma 15, but is substantially simpler. Unlike the drift contribution, the diffusion integrand already contains the quadratic factor $\langle \nabla_g^2 f_\ell(x)\epsilon_+ v, \epsilon_+ v\rangle_g$. Thus, it suffices to track the terms through $Q_{\rho,J}$ in the expansion (70); the cubic term $T_{\rho,J}$ and the remainders are of higher order $\mathcal{O}(\epsilon_+^3)$.

First, consider the zeroth-order term arising from $\rho_j(x)$. By isotropy of the radial kernel,

$$\begin{aligned}
&\frac{\vartheta_\varepsilon}{2}\sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g^2 f_\ell(x)\,\epsilon_+ v, \epsilon_+ v\rangle_g\, \rho_j(x)\epsilon_+^d dv d\mathrm{vol}_g(x) \\
&\qquad = \frac{\sigma_\eta}{2}\sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j^2(x)\Delta_{\mathcal{M}_j} f_\ell(x) d\mathrm{vol}_g(x).
\end{aligned}$$

Thus, the leading diffusion contribution exactly matches the second continuum term in (59).

Next, the integrand involving the first-order density term $\epsilon_+\langle \nabla_g \rho_j(x), v\rangle$ contains

$$\langle \nabla_g^2 f_\ell(x)\epsilon_+ v, \epsilon_+ v\rangle_g \langle \nabla_g \rho_j(x), \epsilon_+ v\rangle_g$$

which is cubic and odd in v. Since $K_\epsilon^{\mathcal{N}}$ is radial and even, the integral over $B(0,1)$ vanishes.

It remains to estimate the quadratic remainder $Q_{\rho,J}$. Substituting the bound (73) yields

$$\begin{aligned}
&\left|\frac{\vartheta_\varepsilon}{2}\sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g^2 f_\ell(x)\,\epsilon_+ v, \epsilon_+ v\rangle_g\, Q_{\rho,J}(x,v;\epsilon_+)\epsilon_+^d\, dv\, d\mathrm{vol}_g(x)\right| \\
&\qquad \le C_\ell \epsilon_+^2 \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} |f_\ell(x)||\rho_j(x)|\|\nabla_g^2 f_\ell(x)\| d\mathrm{vol}_g(x) \int_{B(0,1)} |K_\epsilon^{\mathcal{N}}(v)|\|v\|^4 dv \\
&\qquad \le C_\ell \epsilon_+^2 \sum_{j=1}^{N} w_j^2 \|f_\ell\|_{L^2(\mathcal{M}_j)} \|\nabla_g^2 f_\ell\|_{L^2(\mathcal{M}_j)} \le C_\ell B_\ell^2 \epsilon_+^2.
\end{aligned}$$

Combining the leading-order identification, the cancellation of the first-order term, and the estimate for the quadratic remainder completes the proof. ■

Lemma 17 (Taylor remainder) *The higher-order Taylor remainder term* (69) *satisfies*

$$\left|\mathbb{E}\big\langle f_\ell, \mathcal{L}^{\mathrm{rem}} f_\ell\big\rangle_{L^2(\mathcal{X}_n)}\right| \le C_\ell B_\ell^2 \epsilon_+^2,$$

where B_ℓ is defined in (9) *and C_ℓ depends only on the geometric and density regularity, the kernel profile η, the index ℓ, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights.*

Proof We follow the same decomposition strategy as in the proofs of Lemmas 15 and 16. The key simplification here is that the Taylor remainder is already third order in ϵ_+. It suffices to analyze the expansion (70) only through the first-order term; $Q_{\rho,J}$, $T_{\rho,J}$, and the remaining higher-order terms automatically generate errors of order $\mathcal{O}(\epsilon_+^2)$ or smaller.

Due to the symmetry of the unit ball, the resulting integrand involving the zeroth-order contribution $\rho_j(x)$ vanishes identically since it is odd in v. The first nontrivial contribution arises from the first-order density term $\epsilon_+\langle \nabla_g \rho_j(x), v\rangle_g$. Substituting this term into the expected Taylor remainder gives

$$\begin{aligned}
\int_0^1 \frac{(1-t)^2}{2} \mathcal{II}_t dt = \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \\
\times \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\mathcal{R}_f(x, \exp_x(\epsilon_+ v))\langle \nabla_g \rho_j(x), \epsilon_+ v\rangle_g \epsilon_+^d dv d\mathrm{vol}_g(x),
\end{aligned}$$

where

$$\mathcal{II}_t := \vartheta_\varepsilon \sum_{j=1}^{N} w_j^2 \epsilon_+^{d+4} \int_{\mathcal{M}_j} f_\ell(x)\rho_j(x) \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v)\langle \nabla_g \rho_j(x), v\rangle_g \mathcal{T}_t(x,v) dm(x,v),$$

with

$$\mathcal{T}_t(x,v) := \nabla_g^3 f_\ell(\exp_x(t\epsilon_+ v))[d(\exp_x)_{t\epsilon_+ v}(v), d(\exp_x)_{t\epsilon_+ v}(v), d(\exp_x)_{t\epsilon_+ v}(v)].$$

Arguing exactly as in the proof of the estimate for $\mathcal{I}_t$, we obtain

$$|\mathcal{II}_t| \le C\vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^{N} w_j^2 \left(\|f_\ell\|_{L^2(\mathcal{M}_j)} \|\nabla_g^3 f_\ell\|_{L^\infty(\mathcal{M}_j)} + \|\nabla_g f_\ell\|_{L^2(\mathcal{M}_j)} \|\nabla_g^3 f_\ell\|_{L^2(\mathcal{M}_j)} \right) \le C B_\ell^2 \epsilon_+^2,$$

uniformly in $t \in [0,1]$. Consequently,

$$\left| \int_0^1 \frac{(1-t)^2}{2} \mathcal{II}_t dt \right| \le C_\ell B_\ell^2 \epsilon_+^2.$$

Finally, the quadratic term $Q_{\rho,J}$ and cubic term $T_{\rho,J}$ in (70) contribute strictly higher-order factors and are absorbed into the same $\mathcal{O}(B_\ell^2 \epsilon_+^2)$ bound. ∎

Lemma 18 (Geometric correction) *The geometric correction term in (69) satisfies*

$$\left| \mathbb{E} \langle f_\ell, \mathcal{L}^{\mathrm{geo}} f_\ell \rangle_{L^2(\mathcal{X}_n)} \right| \le C_\ell B_\ell^2 \epsilon_+^2, \tag{74}$$

where B_ℓ is defined in (9) and C_ℓ depends only on the geometric and density regularity, the index ℓ, the kernel profile η, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights.

Proof We begin by expanding the geometric kernel discrepancy

$$K_\epsilon^E(x,y) - K_\epsilon^g(x,y) = \left[\eta_{\epsilon_+}(\|x-y\|) - \eta_{\epsilon_+}(d_g(x-y))\right] - \left[\eta_{\epsilon_-}(\|x-y\|) - \eta_{\epsilon_-}(d_g(x-y))\right].$$

By the distance comparison estimate from Belkin and Niyogi (2005),

$$\|x-y\| - d_g(x,y) = \mathcal{O}(d_g(x,y)^3) = \mathcal{O}(\epsilon_+^3 \|v\|^3).$$

By the mean-value theorem, together with the Lipschitz continuity of the kernel profile,

$$K_\epsilon^E(x, \exp_x(\epsilon_+ v)) - K_\epsilon^g(x, \exp_x(\epsilon_+ v)) = \epsilon_+^2 \mathcal{S}(x,v) + \mathcal{O}(\epsilon_+^3 \|v\|^4), \tag{75}$$

where $|\mathcal{S}(x,v)| \le C\|v\|^3$. Substituting this expansion together with (70) and the Taylor expansion of f_ℓ (60), the geometric correction is governed by the products of $\epsilon_+^2 \mathcal{S}(x,v)$ with

$$[\rho_j(x) + \epsilon_+ \langle \nabla_g \rho_j(x), v \rangle_g + Q_{\rho,J}(x,v;\epsilon_+)] \left[\epsilon_+ \langle \nabla_g f_\ell(x), v \rangle_g + \frac{\epsilon_+^2}{2} \langle \nabla_g^2 f_\ell(x) v, v \rangle_g + \mathcal{O}(\epsilon_+^3 \|v\|^3). \right]$$

Since the kernel discrepancy already contributes a factor of order $\mathcal{O}(\epsilon_+^2 \|v\|^3)$, it suffices to retain only terms up to second order in the above product. We first consider the contribution

$$\epsilon_+^2 \mathcal{S}(x,v) \rho_j(x) \epsilon_+ \langle \nabla_g f_\ell(x), v \rangle_g.$$

Since both K_ϵ^E and K_ϵ^g are radial kernels, $\mathcal{S}(x,v)$ is even in v, whereas $\langle \nabla_g f_\ell(x), v \rangle_g$ is odd. Thus, the integrand is odd and its integral over $B(0,1)$ vanishes.

The first nontrivial term arises from

$$\epsilon_+^2 \left(\langle \nabla_g f_\ell(x), v \rangle_g \langle \nabla_g \rho_j(x), v \rangle_g + \frac{1}{2} \rho_j(x) \langle \nabla_g^2 f_\ell(x) v, v \rangle_g \right).$$

The boundedness of the kernel moment $\int_{B(0,1)} |\mathcal{S}(x,v)|\|v\|^2 dv$ gives

$$|\mathbb{E}\langle f_\ell, \mathcal{L}^{\mathrm{geo}} f_\ell\rangle_{L^2(\mathcal{X}_n)}| \le C\vartheta_\varepsilon \epsilon_+^{d+4} \sum_{j=1}^N w_j^2 \int_{\mathcal{M}_j} \left(|f_\ell(x)||\nabla_g f_\ell(x)| + |f_\ell(x)||\nabla_g^2 f_\ell(x)|\right) \rho_j d\mathrm{vol}_g(x).$$

Applying Cauchy–Schwarz and the definition of B_ℓ yields (74). All remaining terms contain at least one additional factor of ϵ_+ and therefore contribute only $\mathcal{O}(B_\ell^2 \epsilon_+^3)$. ■

Combining Lemmas 15 through 18 yields the bias estimate of the graph Laplacian.

Proposition 19 (Bias of the graph Laplacian) *The discrepancy between the expected within-manifold graph energy and its continuum counterpart*

$$\left|\mathbb{E}\langle f_\ell, (\sigma_\eta \Delta_{\mathcal{M}} - \mathcal{L}^{\mathrm{wth}}) f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| \le C_\ell B_\ell^2 \epsilon_+^2, \tag{76}$$

where B_ℓ is defined in (9) and C depends only on the geometric and density regularity, the kernel profile η, the index ℓ, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights.

Proposition 20 (Concentration of the within-manifold graph Laplacian) *Let B_ℓ be defined in (9). For some constant C_ℓ depending only on the geometric and density regularity, the kernel profile η, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights,*

$$\mathbb{P}\left[\left|\langle f_\ell, (\sigma_\eta \Delta_{\mathcal{M}} - \mathcal{L}^{\mathrm{wth}}) f_\ell\rangle_{L^2(\mathcal{X}_n)}\right| \ge C_\ell B_\ell^2 \epsilon_+^2\right] \le CN \exp\left(-cn\epsilon_+^{d+4}\right) \tag{77}$$

Proof Given Proposition 19, it suffices to control the fluctuation of the within-manifold graph Laplacian around its expectation and the desired concentration estimate then follows from Bernstein's inequality for U-statistic stated in Lemma 25. To this end, we consider the decomposition introduced in (61) and estimate the second-moment bound B_Ψ and the uniform bound C_Ψ in (84) and (85). We treat each contribution separately and combine the resulting bounds via a union bound.

Recall from Section C.3 that the symmetrized representation of U_{drift} is

$$U_{\mathrm{drift}} = \frac{1}{n^2} \sum_{j=1}^N \sum_{x,y \in \mathcal{X}_{n,j}} K_\epsilon^g(x,y) \left(\frac{\vartheta_\varepsilon}{2}\left[f_\ell(x)\langle \nabla_g f_\ell(x), \log_x(y)\rangle_g + f_\ell(y)\langle \nabla_g f_\ell(y), \log_y(x)\rangle_g\right]\right).$$

Let $(\bar{x}, \bar{v})$ be the geodesic-flow counterpart of (x,v) and define

$$\Psi_{\mathrm{drift}}(x, \epsilon_+ v) := \frac{\vartheta_\varepsilon \epsilon_+^{d+1}}{2}[f_\ell(x)\langle \nabla_g f_\ell(x), v\rangle_g + f_\ell(\bar{x})\langle \nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g].$$

The cancellation estimate in Section C.3 has

$$|f_\ell(x)\langle \nabla_g f_\ell(x), v\rangle_g + f_\ell(\bar{x})\langle \nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g| \le C B_\ell^2 \|v\|^2 \epsilon_+.$$

Substituting this estimate into the definition of Ψ_{drift}, we obtain

$$\left|\int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) \Psi_{\mathrm{drift}}(x, \exp_x(\epsilon_+ v)) dv\right| \le C_\ell \vartheta_\varepsilon \epsilon_+^{d+2} B_\ell^2 \int_{B(0,1)} \|v\|^2 dv \le C_\ell B_\ell^2,$$

where we used $\vartheta_\varepsilon \asymp \epsilon_+^{-(d+2)}$ and the finiteness of the kernel moment $\int_{B(0,1)} \|v\|^2 dv$. Consequently, the second-moment bound satisfies

$$B_{\Psi_{\text{drift}}} := \max_{j\in[N]} \int_{\mathcal{M}_j} \left(\int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) \Psi_{\text{drift}}(x, \exp_x(\epsilon_+ v)) dv \right)^2 d\text{vol}_g(x) \le C_\ell B_\ell^4.$$

For the uniform bound, using $|\log_x(y)| \le \epsilon_+$ and the fixed mixture weights, we obtain

$$C_{\Psi_{\text{drift}}} := \max_{j\in[N]} \sup_{x\in\mathcal{M}_j} \sup_{\|v\|\le 1} \left| \Psi_{\text{drift}}(x, \exp_x(\epsilon_+ v)) \right| \le \max_{j\in[N]} \frac{C_\ell}{\epsilon_+} \|f_\ell\|_{L^\infty(\mathcal{M}_j)} \|\nabla_g f_\ell\|_{L^\infty(\mathcal{M}_j)} \le C_\ell B_\ell^2 \epsilon_+^{-1}.$$

We consider the quadratic contribution and the Taylor remainder simultaneously. Define

$$[\Psi_{\text{diff}} + \Psi_{\text{rem}}](x, \exp_x(\epsilon_+ v)) := \vartheta_\varepsilon \epsilon_+^{d+2} f_\ell(x) \left[\frac{1}{2} \langle \nabla_g^2 f_\ell(x) v, v \rangle_g + \epsilon_+ \langle \nabla_g^3 f_\ell(x) v, v, v \rangle_g \right].$$

Using the finiteness of the kernel moment and the definition of B_ℓ,

$$\left| \int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) [\Psi_{\text{diff}} + \Psi_{\text{rem}}](x, \exp_x(\epsilon_+ v)) dv \right| \le C \vartheta_\varepsilon \epsilon_+^{d+2} B_\ell^2 \int_{B(0,1)} \|v\|^2 + \epsilon_+ \|v\|^3 dv \le C B_\ell^2 (1 + \epsilon_+).$$

Using the inequality $(a+b)^2 \le 2a^2 + 2b^2$, we obtain the second-moment bound as

$$B_{\Psi_{\text{diff}} + \Psi_{\text{rem}}} := \max_{j\in[N]} \int_{\mathcal{M}_j} \left(\int_{B(0,1)} K_\epsilon^{\mathcal{N}}(v) [\Psi_{\text{diff}} + \Psi_{\text{rem}}](x, \exp_x(\epsilon_+ v)) dv \right)^2 d\text{vol}_g(x) \le C B_\ell^4 (1 + \epsilon_+^2).$$

Analogous to the estimate of the drift uniform bound, we have

$$C_{\Psi_{\text{diff}} + \Psi_{\text{rem}}} \le \max_{j\in[N]} C \left(\|f_\ell\|_{L^\infty(\mathcal{M}_j)} \|\nabla_g^2 f_\ell\|_{L^\infty(\mathcal{M}_j)} + \epsilon_+ \|f_\ell\|_{L^\infty(\mathcal{M}_j)} \|\nabla_g^3 f_\ell\|_{L^\infty(\mathcal{M}_j)} \right) \le C B_\ell^2 (1 + \epsilon_+).$$

Finally, for the geometric correction, recall that the kernel discrepancy admits the expansion (75), which implicitly contributes a factor of order ϵ_+^2. Consequently, we define

$$\Psi_{\text{geo}}(x, \exp_x(\epsilon_+ v)) := \vartheta_\varepsilon \epsilon_+^{d+2} f_\ell(x) [\epsilon_+ \langle \nabla_g f_\ell(x), v \rangle_g + \mathcal{O}(\epsilon_+^2 v^2)].$$

Retaining the leading order term is sufficient to hit the target order. Proceeding exactly as above, the second-moment and the uniform bound satisfy

$$B_{\Psi_{\text{geo}}} \le C B_\ell^4 \epsilon_+^2, \qquad C_{\Psi_{\text{geo}}} \le C B_\ell^2 \epsilon_+.$$

All together and using $\Psi = \Psi_{\text{drift}} + \Psi_{\text{diff}} + \Psi_{\text{rem}} + \Psi_{\text{geo}}$ yields

$$B_\Psi \le C_\ell B_\ell^4, \quad \text{and} \quad C_\Psi \le C_\ell B_\ell^2 (\epsilon_+^{-1} + 1 + \epsilon_+).$$

Bernstein's inequality in Lemma 25 with $t = c\epsilon_+^2 B_\ell^2$ and $U = \langle f_\ell, \mathcal{L}^{\text{wth}} f_\ell \rangle_{L^2(\mathcal{M}_j)}$ gives

$$\mathbb{P}[|U - \mathbb{E}[U]| > C_\ell \epsilon_+^2 B_\ell^2] \le C \exp\left(- \frac{n \epsilon_+^d (\epsilon_+^2 B_\ell^2)^2}{B_\ell^4 + \epsilon_+^2 B_\ell^2 B_\ell^2 (\epsilon_+^{-1} + 1 + \epsilon_+)} \right) \le C \exp\left(-cn\epsilon_+^{d+4} \right).$$

Taking a union bound over all N sub-manifolds yields the desired probability bound. ■

Lemma 21 (Improved eigenvalue convergence) *Let B_ℓ be the eigenfunction regularity defined in (9). For some constant C_ℓ depending on the geometric and density regularity, the index ℓ, the eigengap γ_ℓ, the kernel profile η, the fixed annulus ratio ϵ_-/ϵ_+, and the mixture weights, with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$, one has*

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \le C_\ell B_\ell \epsilon_+^2, \tag{78}$$

Proof The proof proceeds by controlling each contribution in the decomposition in (58). By Lemma 13 and Proposition 34, with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,

$$\left|\langle \hat{f}_\ell - f_\ell, (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell \rangle_{L^2(\mathcal{X}_n)}\right| \le C_\ell B_\ell^2 \epsilon_+^4 (\log \epsilon_+^{-1})^2.$$

By Proposition 20, with probability at least $1 - CN\exp(-cn\epsilon_+^{d+4})$,

$$\left|\langle f_\ell, (\sigma_\eta \Delta_{\mathcal{M}} - \mathcal{L}^{\mathrm{wth}}) f_\ell \rangle_{L^2(\mathcal{X}_n)}\right| \le C_\ell B_\ell^2 \epsilon_+^2.$$

The other two terms are bounded similarly as in the proof of Lemma 12. Hence, the second and third term dominates the convergence, which concludes the proof. ■

3.3 Proof of Theorem 7

The proof follows directly from Lemmas 13 and 21. Specifically, with probability at least $1 - CNn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \le C_\ell B_\ell \epsilon_+^2, \qquad \|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} \le C_\ell B_\ell \epsilon_+^2 \log \epsilon_+^{-1}.$$

This high-probability event is conditioned on Theorems 30 and 31. Under the choice of $r = \epsilon_+^3$ in our construction, the two theorems together require $\epsilon_+ \gg C(\log n/n)^{\frac{1}{5d-2}}$ so that both theorems hold simultaneously with probability at least $1 - Cn^{-2}$. Under this choice, for every $d \ge 2$, $Nn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4}) = o(n^{-2})$. Substituting this back to the eigenpair estimate yields, for all $d \ge 2$, with probability at least $1 - Cn^{-2}$,

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \le C_\ell B_\ell \Big(\frac{\log n}{n}\Big)^{\frac{1}{2.5d-1}}, \qquad \|\widehat{f}_\ell - f_\ell\|_{H^1(\mathcal{X}_n)} \le C_\ell B_\ell \Big(\frac{\log n}{n}\Big)^{\frac{1}{2.5d-1}} \log n.$$

This completes the proof.

4 Experiments

To numerically verify Theorems 4, 6, and 7, we construct synthetic data on two connected, unit-volume flat tori for each $d \in \{1, 2, 3\}$, parameterized by $\mathbb{T}^d := \mathbb{R}^d/\mathbb{Z}^d$ and identified with the periodic coordinate domain $[0, 1)^d$. Define the unit-speed, unit-length circle $\gamma(t) := \frac{1}{2\pi}\big(\cos(2\pi t) - 1,\ \sin(2\pi t)\big)$. For $d = 1$, the embeddings into $\mathbb{R}^4$ are given by

$$F_1^{(1)}(u) = (\gamma(u), 0_2), \qquad F_2^{(1)}(u) = (0_2, \gamma(u)).$$

r	φ_r	a_{1r}	a_{2r}	$\boldsymbol{k}_{1r}^{(2)}$	$\boldsymbol{k}_{2r}^{(2)}$	$\boldsymbol{k}_{1r}^{(3)}$	$\boldsymbol{k}_{2r}^{(3)}$
1	cos	0.19429357	0.21530560	$(2,0)$	$(2,0)$	$(2,0,0)$	$(2,0,0)$
2	cos	0.07146951	0.16111551	$(0,2)$	$(0,2)$	$(0,2,0)$	$(0,2,0)$
3	cos	0.035	0.040	$(1,1)$	$(1,-1)$	$(1,1,0)$	$(1,-1,0)$
4	sin	0.025	0.020	$(2,1)$	$(1,2)$	$(2,1,0)$	$(1,2,0)$
5	cos	0.11797347	0.07914921	$(4,0)$	$(4,0)$	$(0,0,2)$	$(0,0,2)$
6	cos	0.05728538	0.10603882	$(0,4)$	$(0,4)$	$(0,4,0)$	$(0,4,0)$
7	cos	0.03634769	0.02649998	$(2,2)$	$(2,-2)$	$(1,1,1)$	$(1,1,1)$
8	sin	0.015	0.018	$(4,1)$	$(1,4)$	$(1,1,2)$	$(1,1,2)$

Table 1: Density amplitudes and frequencies used in the numerical experiments. Coefficients are displayed to eight decimal places; the computations use their unrounded values.

For $d \in \{2,3\}$, let $\Gamma_d(u) := (\gamma(u_2), \dots, \gamma(u_d))$, and define the embeddings into $\mathbb{R}^{4d-2}$ by

$$F_1^{(d)}(u) = \big(\gamma(u_1), \Gamma_d(u), 0_{2(d-1)}\big), \qquad F_2^{(d)}(u) = \big(\gamma(u_1), 0_{2(d-1)}, \Gamma_d(u)\big).$$

Setting $\mathcal{M}_j^{(d)} := F_j^{(d)}(\mathbb{T}^d)$, these embeddings induce the standard flat metric satisfying

$$(DF_j^{(d)})^{\mathsf{T}} DF_j^{(d)} = I_d, \qquad \mathrm{vol}_g(\mathcal{M}_j^{(d)}) = 1.$$

The two components intersect at the origin when $d = 1$, whereas for $d \in \{2,3\}$, they intersect along the common unit-length circle

$$\mathcal{M}_1^{(d)} \cap \mathcal{M}_2^{(d)} = \big\{(\gamma(t), 0_{2(d-1)}, 0_{2(d-1)}) : t \in \mathbb{T}^1\big\},$$

with their transverse tangent directions mutually orthogonal.

Our nonuniform periodic sampling density on each component is given by

$$\rho_j^{(d)}\big(F_j^{(d)}(u)\big) = 1 + \sum_{r=1}^{8} a_{jr}\, \varphi_r\Big(2\pi \boldsymbol{k}_{jr}^{(d)} \cdot u\Big), \qquad j \in \{1,2\},$$

where $\varphi_r(\theta)$ equals $\cos\theta$ for $r \in \{1,2,3,5,6,7\}$ and $\sin\theta$ for $r \in \{4,8\}$. The amplitudes are shared across all d, while the frequencies are specified in Table 1. For $d = 1$, we set

$$\boldsymbol{k}_{jr}^{(1)} = \boldsymbol{k}_{jr,1}^{(2)} + 4\boldsymbol{k}_{jr,2}^{(2)},$$

implying that the one-dimensional chart density is precisely the restriction of the two-dimensional chart density to $u \mapsto (u, 4u \bmod 1)$. All frequencies are nonzero, so the densities are properly normalized. Moreover, $\sum_{r=1}^{8} |a_{1r}| \approx 0.552370$ and $\sum_{r=1}^{8} |a_{2r}| \approx 0.666109$, which imply the dimension-independent bounds $0.3338 \le \rho_j^{(d)} \le 1.6662$. The densities are C^∞ and hence naturally satisfy the density smoothness assumptions with $s = 2$. See Figure 1 for a visualization for this density construction.

Assigning equal weights to both components, we sample from the mixture distribution

$$\mu^{(d)} = \frac{1}{2}\mu_1^{(d)} + \frac{1}{2}\mu_2^{(d)}, \qquad d\mu_j^{(d)} = \rho_j^{(d)}\, d\,\mathrm{vol}_g\,.$$

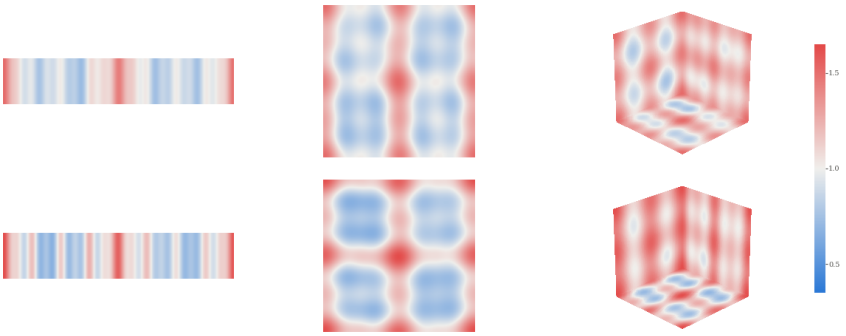


Figure 1: The nonuniform sampling densities $\rho_j^{(d)}$ on the unit-volume flat tori for $d \in \{1, 2, 3\}$ and $j \in \{1, 2\}$. Each column displays the densities for the two components at dimension d.

Specifically, for each observation, we independently draw a component index $Z_i \in \{1, 2\}$ with $\mathbb{P}(Z_i = j) = 1/2$, sample the periodic coordinate U_i according to the chart density $\rho_{Z_i}^{(d)} \circ F_{Z_i}^{(d)}$, and set $X_i = F_{Z_i}^{(d)}(U_i)$. For the PBC coordinates, samples are generated via rejection sampling from the uniform distribution on $[0, 1)^d$, utilizing the envelope constant $B_j = 1 + \sum_{r=1}^{8} |a_{jr}|$ and the acceptance probability $\rho_j^{(d)}\big(F_j^{(d)}(u)\big)/B_j$. The oracle KDE and PBC experiments share all identical component manifolds, densities, and mixture weights.

We first verify Theorems 4 and 6 using oracle KDE with periodic Gaussian kernel

$$\widetilde{\rho}_{j,h}(x) = \frac{1}{n_j} \sum_{i:Z_i=j} K_h(x - X_i), \qquad K_h(x) = \sum_{z \in \mathbb{Z}^2} \frac{1}{2\pi h^2} \exp\left(-\frac{|x + z|^2}{2h^2}\right).$$

The component labels and intrinsic coordinates are known a priori. Our experiment uses its equivalent discrete Fourier form with prescribed bandwidth

$$K_{h_n}(x) = \sum_{k \in \mathbb{Z}^2} e^{-2\pi^2 h_n^2 |\boldsymbol{\xi}|^2} e^{2\pi i \boldsymbol{\xi} \cdot x}, \qquad h_n = 0.2\,(n/2)^{-1/(d+2s)}.$$

Let $N = 256$. The samples are first binned onto an $N \times N$ grid of the torus. Let C_j denote the grid counts of component j and n_j its sample size. The smoothed density is computed in the Fourier domain as

$$\widetilde{\rho}_j = \frac{N^2}{n_j} \operatorname{IFFT}\Big[\operatorname{FFT}(C_j)(\boldsymbol{\xi})\, e^{-2\pi^2 h_n^2 |\boldsymbol{\xi}|^2}\, \mathbf{1}_{\{|\boldsymbol{\xi}|_\infty \le 2m_n\}}\Big], \qquad m_n = \left\lceil \frac{2.5}{2\pi h_n} \right\rceil + 3,$$

where $\boldsymbol{\xi} \in \mathbb{Z}^2$, the factor $e^{-2\pi^2 h_n^2 |\boldsymbol{\xi}|^2}$ is the Fourier multiplier of a Gaussian kernel of bandwidth h_n, and the indicator truncates the spectrum at $|\boldsymbol{\xi}|_\infty \le 2m_n$. To keep the estimate positive and normalized on the grid, we set

$$\widehat{\rho}_{j,ab} = \frac{\max\{\widetilde{\rho}_{j,ab},\, 0.05\}}{N^{-2} \sum_{c,d} \max\{\widetilde{\rho}_{j,cd},\, 0.05\}}, \qquad a, b = 1, \ldots, N,$$

so $\widehat{\rho}_j$ has unit mean on the grid. In all experiments, the lower bound 0.05 was never active.

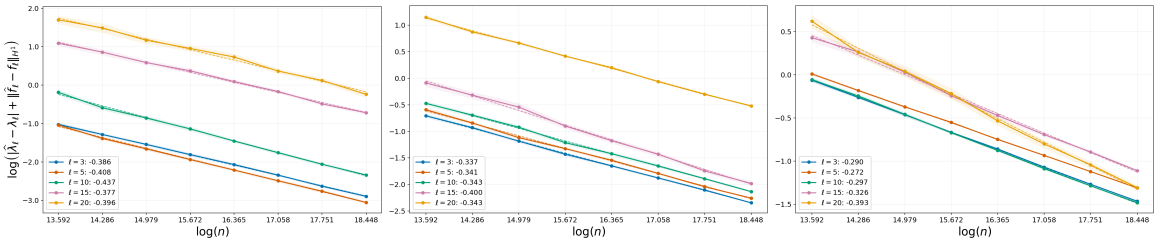


Figure 2: Oracle KDE spectral convergence rate for $d \in \{1, 2, 3\}$.

The leading eigenpairs are therefore obtained from the problems

$$-\frac{1}{2\widehat{\rho}_{j,h}}\,\mathrm{div}\big(\widehat{\rho}_{j,h}^{2}\nabla \widehat{f}_{\ell,j}\big) = \widehat{\lambda}_{\ell}\,\widehat{f}_{\ell,j}, \qquad j = 1, 2,$$

whose weak form is

$$\frac{1}{2}\int_{\mathbb{T}^2} \widehat{\rho}_{j,h}^{2}\,\nabla \widehat{f}_{\ell,j}\cdot\nabla \overline{v}\,dx = \widehat{\lambda}_{\ell}\int_{\mathbb{T}^2} \widehat{\rho}_{j,h}\,\widehat{f}_{\ell,j}\,\overline{v}\,dx \quad \text{for all } v \in H^1(\mathbb{T}^2).$$

The problem is solved by a Fourier–Galerkin method in $V_{m_n} = \mathrm{span}\big\{e^{2\pi i\,\boldsymbol{\xi}\cdot x} : |\boldsymbol{\xi}|_\infty \le m_n\big\}$. The spectra of the two components are then merged and sorted in increasing order to give $\widehat{\lambda}_1 \le \widehat{\lambda}_2 \le \cdots$. The reference spectrum $\lambda_1 \le \lambda_2 \le \cdots$ is computed in the same way from the true densities ρ_j, using a Fourier truncation at $|\boldsymbol{\xi}|_\infty \le 18$. Both the population and the estimated eigenfunctions are normalized using corresponding densities ρ_j and $\widehat{\rho}_{j,h}$. Experimental results for $d \in \{1, 2, 3\}$, $\ell \in \{3, 5, 10, 15, 20\}$ and $n = \{0.8, 1.6, 3.2, 6.4, 12.8, 25.6, 51.2, 102.8\} \times 10^6$, are given in the log-log plot in Figure 2. The dashed lines are least-squares fits, and the shaded bands show the standard deviation. All the errors are averaged over 50 independently generated data sets.

For $s = 2$, Theorems 4 and 6 give the rate $n^{-s/(2s+d)}$, that is, $n^{-0.400}$, $n^{-0.333}$ and $n^{-0.286}$ for $d = 1, 2, 3$. The fitted rates are in close agreement with these exponents: 13 of the 15 pairs (d, ℓ) lie within 0.05 of the theoretical exponent, and averaging over ℓ gives 0.401, 0.353 and 0.316 for $d = 1, 2, 3$, against 0.400, 0.333 and 0.286. In particular, the convergence slows down as the dimension grows, as the theory demonstrates.

Two further features are worth noting. First, the largest deviations occur for large ℓ, namely $\ell = 15$ for $d = 2$ (0.400) and $\ell = 20$ for $d = 3$ (0.393). There the error curves are less regular at small n and the fitted slopes are steeper than the theoretical one, which suggests that the higher eigenpairs are resolved only once n is sufficiently large. Second, a slope slightly steeper than the theoretical exponent does not contradict the lower bound: the minimax rate describes the worst case over the smoothness class, whereas the experiments use a single C^∞ density. Finally, the error level generally increases with ℓ while the slopes remain close to each other, so ℓ appears to affect the constant rather than the rate.

d	C_ϵ	β	c_ϵ	C_r	p	α
1	0.028	1/3	0.6	0.50	3	$3.5\epsilon_+$
2	0.11	1/8	0.78	0.60	3	$3.5\epsilon_+$
3	0.18	1/13	0.8	0.72	3	$3.8\epsilon_+$

Table 2: Parameters used in the PBC experiments.

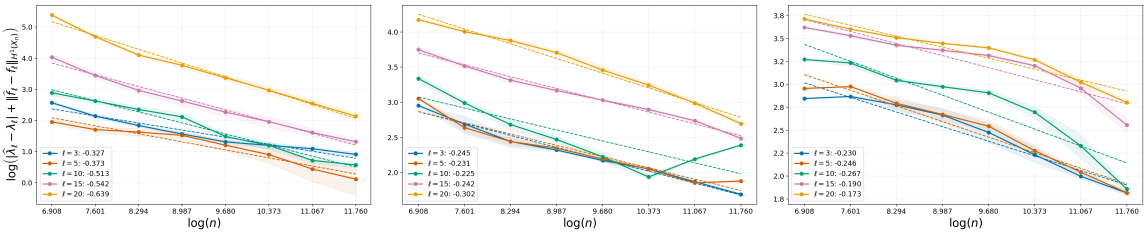


Figure 3: Realistic spectral convergence rate for $d \in \{1, 2, 3\}$.

Next, we test the convergence rate in Theorem 7 where a practical graph based method is applied; the latent component labels are used only to generate the samples and to evaluate the results. We restrict to $n \in \{0.1, 0.2, 0.4, 0.8, 1.6, 3.2, 6.4, 12.8\} \times 10^4$. Given the sample size n, the graph scales are set by the schedule

$$t_n = \left[\frac{(\log n)/n}{(\log n_0)/n_0}\right]^\beta, \qquad \epsilon_+ = C_\epsilon\, t_n, \qquad \epsilon_- = c_\epsilon\, \epsilon_+, \qquad r = C_r\, C_\epsilon\, t_n^p, \tag{79}$$

where n_0 is a reference sample size, with $n_0 = 2000$ for $d = 1, 2$ and $n_0 = 4000$ for $d = 3$. The values used for each d are listed in Table 2. Figure 3 shows the experimental results for a practical spectral convergence rate on a log–log scale; as before, all errors are averaged over 50 independently generated data sets. The fitted rates are $n^{-0.479}$ for $d = 1$, $n^{-0.249}$ for $d = 2$, and $n^{-0.222}$ for $d = 3$. Compared with Figure 2, the rates for $d = 2$ and $d = 3$ are slower than those of the oracle KDE, whereas the rate for $d = 1$ is faster. Recall that Theorem 7 guarantees, for $d \geq 2$, a convergence rate at least as fast as $n^{-1/(2.5d-1)}$, that is, $n^{-1/4}$ for $d = 2$ and $n^{-2/13} \approx n^{-0.154}$ for $d = 3$. The experimental results are consistent with this bound: the observed exponent for $d = 2$ essentially matches the predicted one, and that for $d = 3$ slightly exceeds it. This suggests that the rate established in Theorem 7 may not be sharp and could potentially be improved. Moreover, the theory makes no statement for $d = 1$, where we observe a rate of $n^{-0.479}$. We view both observations as directions for future work: tightening the upper bound and extending the analysis to $d = 1$.

The errors in Figure 3 fluctuate considerably, which may be due to two factors. First, because of the limited scalability of the implementation, the sample sizes are relatively small compared with those used for the oracle KDE. Second, there is no general guidance for tuning the parameters. We fix β and p according to the theory so that the parameters have the scale prescribed by the graph construction, but the results still depend on the prefactors C_ϵ, c_ϵ, C_r and α. A more reliable way to identify the critical local scale of the

approximation graph would likely yield more accurate results. Overall, practical methods still trail the theory, and principled parameter-selection rules are needed.

5 Conclusion & Future Direction

In this work, we establish the minimax lower bound for the spectral estimation of the tensorized Laplace–Beltrami operator in multi-manifold clustering. We further show that this lower bound is sharp by constructing a matching oracle estimator when the manifold components are known. Beyond this information-theoretic characterization, we establish an improved spectral convergence guarantee for the graph Laplacian to the tensorized Laplace–Beltrami operator in the more practical setting where manifold components are unknown. Our analysis yields sharper convergence rates for both eigenvalues and eigenfunctions than previously established results, advancing the theoretical understanding of spectral estimation in multi-manifold learning.

Several important questions remain open. A central challenge is to close the gap between the minimax rate achieved by the oracle estimator and the convergence rate established for the practical graph Laplacian. Whether this gap can be eliminated through sharper analysis or improved graph constructions, or instead reflects an intrinsic statistical cost of unknown manifold membership, remains to be investigated. Another important direction is to extend the minimax analysis beyond simple eigenpairs supported on individual manifold components to more general eigenspaces spanning multiple components, particularly when the manifolds intersect. Such extensions would provide a more complete characterization of the statistical limits of multi-manifold spectral estimation and further clarify the relationship between information-theoretic optimality and achievable algorithmic guarantees.

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Appendix A. Auxiliary Statistical Tools

A.1 Fano's Method

We briefly review Fano's method, adapting it to our specific notation and problem setting; see Wainwright (2019) for a comprehensive treatment.

Proposition 22 (Fano's method; cf. Wainwright (2019)) *Let $\mathcal{P}_M = \{\rho^{(1)}, \dots, \rho^{(M)}\} \subset \mathcal{P}^{(s)}$, and let $\mu^{(a)}$ be the sampling measure induced by $\rho^{(a)}$. Suppose that, for some $\alpha, \beta > 0$,*

$$d_{\mathrm{eig}}(\theta(\rho^{(a)}), \theta(\rho^{(b)})) \geq 2\alpha \qquad \text{for all } a \neq b, \tag{80}$$

$$\mathrm{KL}(\mu^{(a)} || \mu^{(b)}) \leq \beta. \qquad \text{for all } a \neq b. \tag{81}$$

Then the minimax risk satisfies

$$\mathfrak{M}_n\big(\theta(\mathcal{P}_M); d_{\mathrm{eig}}\big) \geq \ \alpha \left(1 - \frac{n\beta + \log 2}{\log M}\right).$$

Here $\mathrm{KL}(\cdot \,\|\, \cdot)$ denotes the Kullback–Leibler (KL) divergence between probability measures

$$\mathrm{KL}(\mu^{(a)} || \mu^{(b)}) = \int_{\mathcal{M}} \log\left(\frac{d\mu^{(a)}}{d\mu^{(b)}}(x)\right) d\mu^{(a)}(x),$$

provided $\mu^{(a)}$ is absolutely continuous with respect to $\mu^{(b)}$. For n i.i.d. samples,

$$\mathrm{KL}((\mu^{(a)})^{\otimes n} \| (\mu^{(b)})^{\otimes n}) = n\,\mathrm{KL}(\mu^{(a)} \| \mu^{(b)}).$$

A.2 Kernel Density Estimation on Manifolds

Kernel Density Estimation (KDE) is a non-parametric statistical method that applies kernel smoothing for probability density estimation. For a fixed point x, classical KDE in a d-dimensional Euclidean space has its kernel density estimator

$$\hat{\rho}_h(x) = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right),$$

with $K(\cdot)$ the kernel function and $h > 0$ the smoothing bandwidth. The following result is a higher-order extension of Pelletier (2005, Theorem 3.1), which establishes the case $s = 2$.

Lemma 23 (KDE upper bound on a known manifold) *Let $\mathcal{M}$ be a known compact d-dimensional Riemannian manifold without boundary satisfying Assumption 1, and let $X_1, \dots, X_n$ be independently sampled from $\rho\,\mathrm{dvol}_g$, where $\rho \in \mathcal{P}^{(s)}_{\mathcal{M}}$ for an integer $s \geq 2$. Then, for all sufficiently small $h > 0$, there exists a KDE estimator $\widehat{\rho}_h \in \mathcal{P}^{(s)}_{\mathcal{M}}$ such that*

$$\mathbb{E}\|\widehat{\rho}_h - \rho\|^2_{L^2(\mathcal{M})} \leq C\left(\frac{1}{nh^d} + h^{2s}\right).$$

Consequently, choosing $h \asymp n^{-1/(2s+d)}$ gives

$$\mathbb{E}\|\widehat{\rho}_h - \rho\|_{L^2(\mathcal{M})} \leq C n^{-s/(2s+d)}.$$

Here C depends only on the geometric and density-class parameters, and the kernel.

Proof Choose a compactly supported radial order-s kernel $\mathsf{K}_s : \mathbb{R}^d \to \mathbb{R}$ satisfying

$$\int_{\mathbb{R}^d} \mathsf{K}_s(z)\,dz = 1, \qquad \int_{\mathbb{R}^d} z^\alpha \mathsf{K}_s(z)\,dz = 0, \quad 1 \le |\alpha| \le s-1,$$

and

$$\int_{\mathbb{R}^d} \|z\|^s |\mathsf{K}_s(z)|\,dz < \infty, \qquad \int_{\mathbb{R}^d} \mathsf{K}_s(z)^2\,dz < \infty.$$

The radiality makes the kernel independent of the choice of orthonormal coordinates on each tangent space. For $x \in \mathcal{M}$, let $J_x(u)$ denote the normal-coordinate volume density

$$d\mathrm{vol}_g(y) = d\mathrm{vol}_g(\exp_x u) = J_x(u)\,du, \qquad u \in T_x\mathcal{M}.$$

Choose $h_0 > 0$ below the injectivity radius under the geometric assumptions so that the exponential map is a diffeomorphism on $B(0,h_0) \subset T_x\mathcal{M}$, uniformly in x,

$$0 < c_J \le J_x(u) \le C_J \qquad \text{for } x \in \mathcal{M}, \quad \|u\| \le h_0.$$

For $0 < h \le h_0$, define the volume-corrected kernel

$$\mathsf{K}_{s,h}(x,y) := \begin{cases} \dfrac{1}{h^d J_x(\log_x y)} \mathsf{K}_s\!\left(\dfrac{\log_x y}{h}\right), & d_g(x,y) < h, \\ 0, & d_g(x,y) \ge h, \end{cases}$$

and the preliminary estimator $\widetilde{\rho}_h(x) := \frac{1}{n}\sum_{i=1}^n \mathsf{K}_{s,h}(x, X_i)$. We proceed by making the bias–variance decomposition

$$\mathbb{E}\|\widetilde{\rho}_h - \rho\|^2_{L^2(\mathcal{M})} = \int_{\mathcal{M}} \left(\mathbb{E}\widetilde{\rho}_h(x) - \rho(x)\right)^2 d\mathrm{vol}_g(x) + \int_{\mathcal{M}} \mathrm{Var}\big(\widetilde{\rho}_h(x)\big)\, d\mathrm{vol}_g(x).$$

We first control its bias. Change of variables with $u = \log_x(y) = hz$ for $z \in B(0,1)$ implies

$$\mathbb{E}\widetilde{\rho}_h(x) = \int_{B_g(x,h)} \frac{\mathsf{K}_s\big(h^{-1}\log_x y\big)}{h^d J_x(\log_x y)} \rho(y)\, d\mathrm{vol}_g(y).$$

Using the normal-coordinate change of variables, the volume correction cancels the Jacobian:

$$\mathbb{E}\widetilde{\rho}_h(x) = \int_{B(0,h)} \frac{1}{h^d} \mathsf{K}_s\Big(\frac{u}{h}\Big)\, \rho(\exp_x u)\, du = \int_{B(0,1)} \mathsf{K}_s(z) \rho(\exp_x(hz))\, dz.$$

The covariant Taylor expansion of ρ along $t \mapsto \exp_x(thz)$ gives

$$\rho(\exp_x(hz)) = \rho(x) + \sum_{\nu=1}^{s-1} \frac{h^\nu}{\nu!} (\nabla_g^\nu \rho)_x[z^{\otimes \nu}] + R_{s,x}(hz),$$

where the density-class assumption implies $|R_{s,x}(hz)| \le Ch^s\|z\|^s$ uniformly in x. The order-s moment conditions eliminate all terms of order one through $s-1$. Therefore,

$$\mathbb{E}\widetilde{\rho}_h(x) - \rho(x) = \int_{B(0,1)} \mathsf{K}_s(z) R_{s,x}(hz)\, dz,$$

and hence $\sup_{x\in\mathcal{M}} |\mathbb{E}\widetilde{\rho}_h(x) - \rho(x)| \le Ch^s$. Since $\mathcal{M}$ has uniformly bounded volume,

$$\int_{\mathcal{M}} |\mathbb{E}\widetilde{\rho}_h(x) - \rho(x)|^2 \, d\mathrm{vol}_g(x) \le Ch^{2s}. \tag{82}$$

We next estimate the variance. Independence gives

$$\mathrm{Var}\big(\widetilde{\rho}_h(x)\big) = \frac{1}{n}\,\mathrm{Var}\big(\mathsf{K}_{s,h}(x, X_1)\big) \le \frac{1}{n}\mathbb{E}\big[\mathsf{K}_{s,h}(x, X_1)^2\big].$$

Applying the same normal-coordinate change of variables,

$$\mathrm{Var}\big(\widetilde{\rho}_h(x)\big) \le \frac{1}{nh^{2d}} \int_{B(0,h)} \mathsf{K}_s\Big(\frac{u}{h}\Big)^2 \frac{\rho(\exp_x u)}{J_x(u)}\, du = \frac{1}{nh^d} \int_{B(0,1)} \mathsf{K}_s(z)^2 \frac{\rho(\exp_x(hz))}{J_x(hz)}\, dz \le \frac{C}{nh^d},$$

where the last inequality uses the uniform bounds on ρ, J_x, K_s. Integrating over $\mathcal{M}$ yields

$$\int_{\mathcal{M}} \mathrm{Var}\big(\widetilde{\rho}_h(x)\big)\, d\mathrm{vol}_g(x) \le \frac{C}{nh^d}. \tag{83}$$

Combining (82) and (83) with the bias–variance decomposition gives

$$\mathbb{E}\|\widetilde{\rho}_h - \rho\|^2_{L^2(\mathcal{M})} \le C\left(h^{2s} + \frac{1}{nh^d}\right).$$

For $s > 2$, the order-s kernel may be signed, so $\widetilde{\rho}_h$ need not itself belong to the admissible density class. Choose an approximate L^2-projection $\widehat{\rho}_h \in \mathcal{P}^{(s)}_{\mathcal{M}}$ satisfying

$$\|\widehat{\rho}_h - \widetilde{\rho}_h\|_{L^2(\mathcal{M})} \le \inf_{\varrho\in\mathcal{P}^{(s)}_{\mathcal{M}}} \|\varrho - \widetilde{\rho}_h\|_{L^2(\mathcal{M})} + n^{-1}.$$

Since $\rho \in \mathcal{P}^{(s)}_{\mathcal{M}}$, $\|\widehat{\rho}_h - \rho\|_{L^2(\mathcal{M})} \le 2\|\widetilde{\rho}_h - \rho\|_{L^2(\mathcal{M})} + n^{-1}$. It follows that

$$\mathbb{E}\|\widehat{\rho}_h - \rho\|^2_{L^2(\mathcal{M})} \le C\left(h^{2s} + \frac{1}{nh^d}\right).$$

Balancing the two terms gives $h \asymp n^{-1/(2s+d)}$, which leads to the first claimed identity. Jensen's inequality then yields the second. ∎

A.3 Bernstein Concentration Inequality for the U-statistic

We first recall the classic Bernstein's inequality for i.i.d. real-valued random variables.

Lemma 24 (Bernstein's inequality) *Let $X_1, X_2, \ldots, X_n$ be independent, identically distributed (i.i.d.) real-valued random variables for which $|X_i| \le M$ and $Var(X_i) \le \sigma^2$. Then*

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n X_i - \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^n X_i\right]\right| \ge t\right] \le 2\exp\left(-\frac{cnt^2}{\sigma^2 + Mt}\right), \quad \forall t > 0.$$

There is a Bernstein-type concentration inequality for the U (unbiased)-statistic

$$U = \frac{1}{n}\sum_{j=1}^{N} w_j \sum_{y\in\mathcal{M}_j} \eta_{\epsilon_+}(d(x,y))\mathcal{K}(x,y)$$

for a kernel function $\mathcal{K}: \mathcal{M}_j \times \mathcal{M}_j \to \mathbb{R}$, which we state next lemma.

Lemma 25 (Bernstein's inequality for the U-statistic) *Given a function $\Psi: \mathcal{M}_j \times \mathcal{M}_j \to \mathbb{R}$ which satisfies*

$$\max_{j\in[N]} \int_{\mathcal{M}_j} \left(\int_{B_1(0)\subseteq T_x\mathcal{M}_j} \eta(\|v\|)\Psi(x, \exp_x(\epsilon_+ v))\mathrm{d}v \right)^2 d\mathrm{vol}_g(x) \le B_\Psi < \infty, \tag{84}$$

and

$$\max_{j\in[N]} \sup_{x\in\mathcal{M}_j} \sup_{v\in B_1(0)\subseteq T_x\mathcal{M}_j} |\Psi(x, \exp_x(\epsilon_+ v))| \le C_\Psi < \infty. \tag{85}$$

Then the corresponding U-statistic has the Bernstein's concentration bound

$$\mathbb{P}\left[|U - \mathbb{E}(U)| > t\right] \le 2\exp\left(-\frac{cn\epsilon_+^d t^2}{B_\Psi + tC_\Psi}\right).$$

Appendix B. Technical Proofs of Section 2

B.1 Construction of Perturbation Templates

We verify the support, normalization, and derivative properties stated in (12). Since $\phi_i \in C_c^\infty(B(0,1/4))$, the rescaled profile $\phi_i(m\,\exp_{p_i}^{-1}(x))$ vanishes near $\partial B_g(p_i, r/4)$. Its extension by zero is therefore regular up to the order permitted by ρ_j. Moreover,

$$\mathrm{supp}(\widetilde{a}_i) \subset B_g(p_i, r/4) \subset B_g(p_i, r/2) \subset \Omega_i.$$

Because the cells $\{\Omega_i\}_{i=1}^K$ are pairwise disjoint up to their boundaries, the raw templates $\{\widetilde{a}_i\}_{i=1}^K$ have pairwise disjoint supports. To establish the derivative scaling, the chain rule in normal coordinates gives

$$\|\nabla_g^\nu \widetilde{a}_i\|_{L^\infty(\mathcal{M}_j)} \le C m^\nu, \qquad \nu = 0, \dots, s, \tag{86}$$

where the factor ρ_j^{-1} from differentiation is uniformly bounded by Assumption 3.

It remains to impose the exact weighted mean zero condition. For $u = \exp_{p_i}^{-1}(x) \in B(0, r/4)$, write $J_i(u) = \sqrt{\det g_i(u)}$ for the normal-coordinate volume density. Then

$$\int_{\mathcal{M}_j} \widetilde{a}_i \, d\mu_j = \int_{B(0,r/4)} \phi_i(mu) J_i(u)\, du = \int_{B(0,r/4)} \phi_i(mu)\big(J_i(u) - 1\big)\, du.$$

The second equality follows from $\int_{\mathbb{R}^{d_j}} \phi_i = 0$. Since $|J_i(u) - 1| \le C|u|^2$ uniformly in i, we can bound the error by applying the change of variables $v = mu$. This yields $du = m^{-d_j} dv$

and $|u|^2 = m^{-2}|v|^2$, which gives

$$\left|\int_{\mathcal{M}_j} \widetilde{a}_i \, d\mu_j\right| \le C \int_{\mathbb{R}^{d_j}} |\phi_i(mu)| \, |u|^2 \, du = Cm^{-(d_j+2)} \int_{B(0,1/4)} |\phi_i(v)| \, |v|^2 \, dv \le C' m^{-(d_j+2)}. \tag{87}$$

Choose a nonnegative function $\chi_i \in C_c^\infty\Big(B_g(p_i, r/2) \setminus \overline{B_g(p_i, r/4)}\Big)$ such that

$$\int_{\mathcal{M}_j} \chi_i \, d\mu_j \asymp m^{-d_j}, \qquad \|\nabla_g^\nu \chi_i\|_{L^\infty(\mathcal{M}_j)} \le Cm^\nu, \quad \nu = 0, \dots, s.$$

Such functions are obtained by rescaling a fixed annular bump in normal coordinates. Define

$$b_i := \frac{\int_{\mathcal{M}_j} \widetilde{a}_i \, d\mu_j}{\int_{\mathcal{M}_j} \chi_i \, d\mu_j}, \quad a_i := \widetilde{a}_i - b_i \chi_i \quad \Longrightarrow \quad \int_{\mathcal{M}_j} a_i \, d\mu_j = 0, \qquad |b_i| \le Cm^{-2}.$$

Consequently, $\|\nabla_g^\nu (b_i \chi_i)\|_{L^\infty(\mathcal{M}_j)} \le Cm^{\nu-2}$ for $\nu = 0, \dots, s$. Combining with (86) proves

$$\operatorname{supp}(a_i) \subset B_g(p_i, r/2) \subset \Omega_i, \qquad \|\nabla_g^\nu a_i\|_{L^\infty(\mathcal{M}_j)} \le Cm^\nu, \quad \nu = 0, \dots, s.$$

B.2 Eigenpair stability under small density perturbation

We introduce ODE-like quantities analogous to the static ones in the main text for analysis. Let $\rho(t)$ be a C^1 curve in the admissible density class on a compact manifold $\mathcal{M}$, and define

$$d\mu_t = \rho(t) \, d\mathrm{vol}_g, \qquad \Delta_{\rho(t)} = -\frac{1}{\rho(t)} \operatorname{div}_g\big(\rho(t)^2 \nabla_g(\cdot)\big).$$

Suppose that $\lambda_\ell(t)$ is a simple and isolated eigenvalue with eigengap

$$\gamma_\ell := \inf_{|t|<t_*} \min_{k \neq \ell} |\lambda_k(t) - \lambda_\ell(t)| > 0.$$

By standard perturbation theory for self-adjoint operators with compact resolvent Kato (1995), the eigenpair admits a C^1 continuation, normalized by $\|f_\ell(t)\|_{L^2(\mu_t)} = 1$. We write

$$\rho = \rho(0), \quad \dot\rho = \partial_t \rho(t)|_{t=0}, \quad \lambda_\ell = \lambda_\ell(0), \quad \dot\lambda_\ell = \partial_t \lambda_\ell(t)|_{t=0}, \quad f_\ell = f_\ell(0), \quad \dot f_\ell = \partial_t f_\ell(t)|_{t=0}.$$

Proposition 26 (Infinitesimal stability of a simple eigenpair) *There exists a constant $C_\ell < \infty$, depending only on the fixed geometric and density-class parameters, the eigenvalue index ℓ, and the eigengap γ_ℓ, such that*

$$|\dot\lambda_\ell| \le C_\ell \, \lambda_\ell \, \|\dot\rho\|_{L^2(\mathcal{M})}, \tag{88}$$

and

$$\|\dot f_\ell\|_{H^1(\mathcal{M})} = \|\dot f_\ell\|_{L^2(\mathcal{M})} + \|\nabla_g \dot f_\ell\|_{L^2(\mathcal{M})} \ \le \ C_\ell \frac{\lambda_\ell}{\gamma_\ell} \, \|\dot\rho\|_{L^2(\mathcal{M})}. \tag{89}$$

Proof Introduce the parameter-dependent bilinear forms

$$a_t(u,v) := \int_{\mathcal{M}} \rho(t)^2 \langle \nabla_g u, \nabla_g v \rangle_g \, d\mathrm{vol}_g, \qquad b_t(u,v) := \int_{\mathcal{M}} \rho(t) uv \, d\mathrm{vol}_g.$$

The eigenvalue equation and normalization condition then take the weak form

$$a_t(f_\ell(t), v) = \lambda_\ell(t) b_t(f_\ell(t), v), \qquad b_t(f_\ell(t), f_\ell(t)) = 1. \tag{90}$$

Differentiating the first identity with respect to t at $t = 0$ and testing with $v = f_\ell$ yields

$$\dot{\lambda}_\ell = 2 \int_{\mathcal{M}} \rho\dot{\rho} \|\nabla_g f_\ell\|_g^2 \, d\mathrm{vol}_g - \lambda_\ell \int_{\mathcal{M}} \dot{\rho} f_\ell^2 \, d\mathrm{vol}_g.$$

where the terms involving $\dot{f}_\ell$ cancel out due to the unperturbed eigenvalue equation $a_0(\dot{f}_\ell, f_\ell) = \lambda_\ell b_0(\dot{f}_\ell, f_\ell)$. Appplying the eigenfunction bounds from (5), Hölder's inequality, and the finite volume of $\mathcal{M}$, we obtain

$$|\dot{\lambda}_\ell| \le C_\ell \Big(\|\nabla_g f_\ell\|_{L^\infty(\mathcal{M})}^2 + \lambda_\ell \|f_\ell\|_{L^\infty(\mathcal{M})}^2 \Big) \|\dot{\rho}\|_{L^2(\mathcal{M})} \le C_\ell \lambda_\ell \|\dot{\rho}\|_{L^2(\mathcal{M})}.$$

Now set the unperturbed bilinear form $B(u,v) := a_0(u,v) - \lambda_\ell b_0(u,v)$. Differentiating the weak eigenvalue equation (90) at $t = 0$ for an arbitrary $v \in H^1(\mathcal{M})$ yields the variational equation $B(\dot{f}_\ell, v) = F(v)$, where the forcing functional $F \in H^{-1}(\mathcal{M})$ is defined by

$$F(v) := -2 \int_{\mathcal{M}} \rho\dot{\rho} \langle \nabla_g f_\ell, \nabla_g v \rangle_g \, d\mathrm{vol}_g + \dot{\lambda}_\ell b_0(f_\ell, v) + \lambda_\ell \int_{\mathcal{M}} \dot{\rho} f_\ell v \, d\mathrm{vol}_g.$$

Applying Hölder's inequality alongside the uniform bounds for f_ℓ and the estimate (88), we readily bound the operator norm of F:

$$\|F\|_{H^{-1}(\mathcal{M})} = \sup_{\|v\|_{H^1}=1} |F(v)| \le C_\ell \lambda_\ell \|\dot{\rho}\|_{L^2(\mathcal{M})}. \tag{91}$$

To estimate $\dot{f}_\ell$, we introduce the b_0-orthogonal decomposition

$$\dot{f}_\ell = \alpha f_\ell + \dot{f}_{\ell,\perp}, \qquad \text{where} \quad b_0(\dot{f}_{\ell,\perp}, f_\ell) = 0.$$

Differentiating the normalization identity (90) at $t = 0$ fixes the projection coefficient α:

$$2b_0(\dot{f}_\ell, f_\ell) + \int_{\mathcal{M}} \dot{\rho} f_\ell^2 \, d\mathrm{vol}_g = 0 \implies 2\alpha + \int_{\mathcal{M}} \dot{\rho} f_\ell^2 \, d\mathrm{vol}_g = 0,$$

which immediately implies $|\alpha| \le C_\ell \|\dot{\rho}\|_{L^2(\mathcal{M})}$. Since f_ℓ lies in the kernel of B, we have $B(\dot{f}_{\ell,\perp}, v) = B(\dot{f}_\ell, v) - \alpha B(f_\ell, v) = F(v)$ for all v. We now restrict this problem to the invariant subspace $V = \{u \in H^1(\mathcal{M}) : b_0(u, f_\ell) = 0\}$. Since λ_ℓ is simple, the spectral gap condition, combined with standard elliptic regularity, ensures that B satisfies the resolvent stability estimate on V. Consequently,

$$\|\dot{f}_{\ell,\perp}\|_{H^1(\mathcal{M})} \le C_\ell \gamma_\ell^{-1} \|F\|_{H^{-1}(\mathcal{M})} \le C_\ell \frac{\lambda_\ell}{\gamma_\ell} \|\dot{\rho}\|_{L^2(\mathcal{M})}.$$

Finally, summing the orthogonal components yields (89):

$$\|\dot{f}_\ell\|_{H^1(\mathcal{M})} \le |\alpha| \|f_\ell\|_{H^1(\mathcal{M})} + \|\dot{f}_{\ell,\perp}\|_{H^1(\mathcal{M})} \le C_\ell \frac{\lambda_\ell}{\gamma_\ell} \|\dot{\rho}\|_{L^2(\mathcal{M})}.$$

■

B.3 Local Coercivity and Geometric Estimates

Proposition 27 (Uniform Jacobi-field comparison) *Let $e_i(x) \in T_x\mathcal{M}_j$ denote the parallel transport of a unit vector $e_i(p_i) \in T_{p_i}\mathcal{M}_j$ along the radial geodesic $\gamma(t) := \exp_{p_i}(tu)$. Then there exist constants $C_{\exp} > 0$, independent of the patch index i, such that for $u \in T_{p_i}\mathcal{M}_j$ with $|u| \le r/4$, the differential of the exponential map satisfies*

$$\big\| d(\exp_{p_i})_u\big(e_i(p_i)\big) - e_i(x)\big\|_g \le C_{\exp}|u|^2.$$

Proof Consider the variation through radial geodesics and its associated variation vector field along γ,

$$\Gamma(s,t) := \exp_{p_i}\big(t(u + se_i(p_i))\big), \qquad \text{with} \qquad J(t) := \frac{\partial}{\partial s}\Gamma(s,t)\Big|_{s=0}.$$

By definition, $J(t)$ is a Jacobi field satisfying the initial conditions $J(0) = 0$, $\nabla_{\dot\gamma(0)}J(0) = e_i(p_i)$. Moreover, evaluating at $t = 1$ yields $J(1) = d(\exp_{p_i})_u\big(e_i(p_i)\big)$. Along the geodesic $\gamma(t)$, the Jacobi field satisfies the second-order linear ODE:

$$\nabla^2_{\dot\gamma(t)}J(t) + R(J(t),\dot\gamma(t))\dot\gamma(t) = 0.$$

The uniform curvature bound implies that there exists a constant C_0 such that

$$\big\|R\big(J(t),\dot\gamma(t)\big)\dot\gamma(t)\big\|_g \le C_0\|J(t)\|_g|u|^2.$$

Let $E(t)$ be the parallel vector field along γ determined by $E(0) = e_i(p_i)$, so that $E(1) = e_i(x)$. The flat comparison field $tE(t)$ shares the same initial conditions as $J(t)$, namely

$$tE(t)\big|_{t=0} = 0, \qquad \nabla_{\dot\gamma(0)}(tE(t)) = e_i(p_i).$$

Define the error field $K(t) := J(t) - tE(t)$. Since $E(t)$ is parallel ($\nabla_{\dot\gamma}E \equiv 0$), we have $\nabla^2_{\dot\gamma(t)}K(t) = \nabla^2_{\dot\gamma(t)}J(t)$. The Jacobi equation then yields

$$\nabla^2_{\dot\gamma(t)}K(t) = -R\big(J(t),\dot\gamma(t)\big)\dot\gamma(t),$$

with initial conditions $K(0) = 0$ and $\nabla_{\dot\gamma(0)}K(0) = 0$. Integrating this second derivative twice with respect to $t \in [0,1]$ yields

$$K(1) = \int_0^1\int_0^\tau \nabla^2_{\dot\gamma(s)}K(s)\,ds\,d\tau.$$

By standard Jacobi field estimates, $\|J(t)\|_g$ is uniformly bounded for $t \in [0,1]$. Taking the norm of the double integral gives, for some uniform constant $C_{\exp}$,

$$\|K(1)\|_g \le \int_0^1\int_0^\tau C_0\|J(s)\|_g|u|^2\,ds\,d\tau \le C_{\exp}|u|^2.$$

Since $K(1) = J(1) - E(1) = d(\exp_{p_i})_u\big(e_i(p_i)\big) - e_i(x)$, the claim is proved. ∎

Lemma 28 (Local L^1 coercivity) *Fix an active patch $i \in \mathcal{D}$ and let $h \in C^2(\mathcal{N}_i^\star)$, $\mathfrak{r} \in L^1(\mathcal{N}_i^\star)$, and $A > 0$. Suppose that*

$$\operatorname{div}_g(\rho_j^2 \nabla_g h) + V(x)h + \mathfrak{r} \geq A\, m^{1-s} \qquad \text{on } \mathcal{N}_i^\star, \tag{92}$$

where $0 \leq V(x) \leq \Lambda$ on $\mathcal{N}_i^\star$. Then there exist constants $c, C > 0$, depending only on the density-class and geometric bounds and on τ_0, such that

$$\frac{\Lambda}{m} \int_{\mathcal{N}_i^\star} |h| \, d\mathrm{vol}_g + \int_{\mathcal{N}_i^\star} \|\nabla_g h\|_g \, d\mathrm{vol}_g \geq cA\, m^{-(d_j+s)} - \frac{C}{m} \int_{\mathcal{N}_i^\star} |\mathfrak{r}| \, d\mathrm{vol}_g. \tag{93}$$

Proof Let $z_i^* \in \{\pm u_0/m\}$ denote the center of $\mathcal{N}_i^*$. Choose a fixed cut-off function $\chi_0 \in C_c^\infty(B(0,\tau_0))$ satisfying $0 \leq \chi_0 \leq 1$ and $\chi_0 \geq \frac{1}{2}$ on $B(0,\tau_0/2)$. Define $\chi \in C_c^\infty(\mathcal{N}_i^\star)$ by

$$\chi(x) := \chi_0\big(m\big(\psi_i^{-1}(x) - z_i^\star\big)\big), \qquad x \in \mathcal{N}_i^\star.$$

By construction, χ satisfies $0 \leq \chi \leq 1$, $\|\nabla_g \chi\|_g \leq Cm$, and is supported in $\mathcal{N}_i^\star$ with $\chi \geq \frac{1}{2}$ on $\psi_i\left(B\left(z_i^\star, \frac{\tau_0}{2m}\right)\right)$. The uniform bi-Lipschitz property of normal coordinates at scale m^{-1} implies

$$\int_{\mathcal{N}_i^\star} \chi \, d\mathrm{vol}_g \geq \frac{1}{2} \operatorname{vol}_g\left(\psi_i\left(B\left(z_i^\star, \tfrac{\tau_0}{2m}\right)\right)\right) \geq c_0 m^{-d_j}.$$

Multiplying (92) by χ and integrating over $\mathcal{N}_i^\star$ yields

$$\int_{\mathcal{N}_i^\star} \chi \left[\operatorname{div}_g(\rho_j^2 \nabla_g h) + Vh + \mathfrak{r}\right] d\mathrm{vol}_g \geq A\, m^{1-s} \int_{\mathcal{N}_i^\star} \chi \, d\mathrm{vol}_g \geq c_0 A\, m^{-(d_j+s-1)}. \tag{94}$$

On the other hand, since $0 \leq V \leq \Lambda$ and $0 \leq \chi \leq 1$,

$$\int_{\mathcal{N}_i^\star} \chi \left[\operatorname{div}_g(\rho_j^2 \nabla_g h) + Vh + \mathfrak{r}\right] d\mathrm{vol}_g \leq \int_{\mathcal{N}_i^\star} \chi \operatorname{div}_g(\rho_j^2 \nabla_g h) \, d\mathrm{vol}_g + \Lambda \int_{\mathcal{N}_i^\star} |h| \, d\mathrm{vol}_g + \int_{\mathcal{N}_i^\star} |\mathfrak{r}| \, d\mathrm{vol}_g. \tag{95}$$

Because χ is compactly supported in $\mathcal{N}_i^\star$, integration by parts gives

$$\int_{\mathcal{N}_i^\star} \chi \operatorname{div}_g(\rho_j^2 \nabla_g h) \, d\mathrm{vol}_g = -\int_{\mathcal{N}_i^\star} \rho_j^2 \langle \nabla_g \chi, \nabla_g h \rangle_g \, d\mathrm{vol}_g.$$

Using $\rho_j \leq \rho_{\max}$ and $\|\nabla_g \chi\|_g \leq Cm$, we obtain

$$\left| \int_{\mathcal{N}_i^\star} \chi \operatorname{div}_g(\rho_j^2 \nabla_g h) \, d\mathrm{vol}_g \right| \leq Cm \int_{\mathcal{N}_i^\star} \|\nabla_g h\|_g \, d\mathrm{vol}_g.$$

Combining (94) and (95) therefore gives

$$c_0 A\, m^{-(d_j+s-1)} \leq Cm \int_{\mathcal{N}_i^\star} \|\nabla_g h\|_g \, d\mathrm{vol}_g + \Lambda \int_{\mathcal{N}_i^\star} |h| \, d\mathrm{vol}_g + \int_{\mathcal{N}_i^\star} |\mathfrak{r}| \, d\mathrm{vol}_g.$$

Dividing by m and adjusting the constants proves (93). ∎

Appendix C. Technical Proofs of Section 3

C.1 Annular proximity graph with angle constraints

To make the paper self-contained, we give a high-level summary of the annular proximity graph with angle constraints introduced in Trillos et al. (2023) and the related theorems.

Definition 29 *Let $\alpha \in (0, \pi/2)$ and $r > 0$. We say x_{i_1} and x_{i_m} is (α, r)-connected if there exists a path between the two points such that for all $j = 1, 2 \ldots, m-1$,*

$$\angle(x_{i_m} - x_{i_1}, x_{i_{j+1}} - x_{i_j}) < \alpha, \qquad \|x_{i_{j+1}} - x_{i_j}\| < r.$$

Given $0 < \epsilon_- < C\epsilon_+$, construct a graph with weights $w(x, y) = 1$ only if $\epsilon_- < \|x - y\| < \epsilon_+$ and x, y is (α, r)-connected; otherwise $w(x, y) = 0$. Here $C > 0$ is a constant strictly less than 1. The resulting graph is called the annular proximity graph with angle constraints.

Several theorems are established following this graph construction. The first concerns strong within-manifold connectivity, while the second justifies the sparsity of between-manifold connections. Together, these yield the eigenpair convergence claimed in that work. Finally, a supplementary lemma characterizes the exact conditions for pairwise path connectivity.

Theorem 30 (Theorem 14; Trillos et al. (2023)) *Let $\alpha \in [0, \pi/4)$ and $r \le C\epsilon_+$. Then for $j \in [N]$, there exists $C_j > 0$, depending on the geometric parameters of $\mathcal{M}_j$, such that any points $x, y \in \mathcal{X}_{n,j}$ satisfying $\|x - y\| < \epsilon_+$ is (α, r)-connected with probability at least*

$$1 - C_{\mathrm{wth}}(n), \qquad \text{with} \qquad C_{\mathrm{wth}}(n) := \frac{C_j n^2 \epsilon_+}{r} \exp\left(-C_j n_j r^{d_j} (\tan\alpha)^{d_j - 1}\right).$$

Theorem 31 (Theorem 15; Trillos et al. (2023)) *Suppose $\alpha < \arcsin(C_{j,k} r/\epsilon_+)$. Choose $\epsilon_- \ge c\epsilon_+$ for some $0 < c < 1$. Then with probability at least*

$$1 - C_{j,k} n \exp\left(-C_{j,k} n \max\{q^{d_j - d_{jk}}, q^{d_k - d_{jk}}, \epsilon_+^{d_j}, \epsilon_+^{d_k}\}\right),$$

the number of connections $\mathcal{N}_{jk}$ between $\mathcal{M}_j$ and $\mathcal{M}_k$ satisfies

$$\mathcal{N}_{jk} \le C_{j,k} n^2 \max\{q^{d_j - d_{jk}} \epsilon_+^{d_k}, q^{d_k - d_{jk}} \epsilon_+^{d_j}\}, \qquad q := \left(\frac{r^2}{\epsilon_+} + r\epsilon_+\right).$$

Theorem 32 (Theorem 10; Trillos et al. (2023)) *Suppose the conditions for Theorems 30 and 31 hold. Then with probability at least $1 - Cn\exp(-cn\epsilon_+^d) - C_{\mathrm{wth}}(n)$,*

$$|\widehat{\lambda}_\ell - \sigma_\eta \lambda_\ell| \le C_\ell \lambda_\ell, \qquad \|\widehat{f}_\ell - f_\ell\|_{L^2(\mathcal{X}_n)} \le C_\ell.$$

Lemma 33 (Lemma 20; Trillos et al. (2023)) *Suppose $x \in \mathcal{X}_{n,j}$ and $y \in \mathcal{X}_{n,k}$ satisfy $\epsilon_- < \|x - y\| < \epsilon_+$ with $\epsilon_- \ge c\epsilon_+$ for some $0 < c < 1$. Let $\alpha > 0$ satisfy*

$$\sin(\alpha) < Cr/\epsilon_+.$$

Assume, in addition, that $\mathrm{dist}(x, \mathcal{M}_k) > Cq$ and $\mathrm{dist}(y, \mathcal{M}_j) > r$. Then there does not exist an (α, r)-constrained path connecting x and y.

Our construction throughout the section chooses $r = \epsilon_+^3$ so that $q \asymp \epsilon_+^4$.

C.2 Within-manifold and Between-manifold Laplacian Estimates

The analysis in this section relies on the multiscale Poincaré inequality of Trillos et al. (2026). For completeness, we record the relevant quantities in our notation.

For each j, fix a cover of $\mathcal{M}_j$ by $N_{\nu,j}$ radius-2ν balls and choose the multiscale depth m_j so that $3^{-m_j} \asymp \epsilon_+$. For $p \in [m_j]$, define the level-p cell size by $s_{p,j} := 3^{p-m_j}$. Thus $s_{1,j} \asymp \epsilon_+$, $s_{m_j,j} = 1$, and $m_j \asymp \log(\epsilon_+^{-1})$. For each chart a and level p, let $\Upsilon_{p,a,j}^{m_j}$ denote the cell-index set, $\square_{p,a,j,\upsilon}^{m_j} \subset \mathcal{M}_j$ the geometric cell indexed by υ, and $Q_{p,a,j,\upsilon}^{m_j} := \square_{p,a,j,\upsilon}^{m_j} \cap \mathcal{X}_{n,j}$ its sample set. On the relevant high-probability occupancy event, uniformly over a, p, υ,

$$|\Upsilon_{p,a,j}^{m_j}| \asymp s_{p,j}^{-d_j}, \qquad |Q_{p,a,j,\upsilon}^{m_j}| \asymp n_j s_{p,j}^{d_j}.$$

For $\xi : \mathcal{X}_{n,j} \to \mathbb{R}$, define its root-mean-square cell average at chart a and level p by

$$T_{p,a,j}(\xi) := \left(\frac{1}{|\Upsilon_{p,a,j}^{m_j}|} \sum_{\upsilon \in \Upsilon_{p,a,j}^{m_j}} \left| \frac{1}{|Q_{p,a,j,\upsilon}^{m_j}|} \sum_{x \in Q_{p,a,j,\upsilon}^{m_j}} \xi(x) \right|^2 \right)^{1/2}. \tag{96}$$

See Trillos et al. (2026) for the precise multiscale construction.

Applying Proposition 3.6 of Trillos et al. (2026) componentwise and weighting the resulting estimates, the global negative Sobolev seminorm satisfies

$$\|h\|_{H^{-1}(\mathcal{X}_n)} \le C \sum_{j=1}^{N} w_j \left[\epsilon_+ \|h_j\|_{L^2(\mathcal{X}_{n,j})} + \sum_{a=1}^{N_{\nu,j}} \sum_{p=1}^{m_j} s_{p,j} T_{p,a,j}(h_j) \right] \tag{97}$$

with probability at least $1 - C\epsilon_+^{-d} \exp(-cn\epsilon_+^d)$. Our first result in this subsection controls the within-manifold consistency error in this weighted negative Sobolev seminorm.

Proposition 34 (Consistency of the within-manifold operator) *There exists some constant $C_\ell > 0$, depending only on the index ℓ, the geometric and density class parameters, and the kernel profile η, such that with probability at least $1 - CNn\epsilon_+^{-d} \exp(-cn\epsilon_+^{d+4})$,*

$$\|(\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell\|_{H^{-1}(\mathcal{X}_n)} \le C_\ell (B_\ell + 1) \epsilon_+^2 \log \epsilon_+^{-1}.$$

Here B_ℓ denotes the regularity constant defined in (9).

Proof Set $h := (\mathcal{L}^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}}) f_\ell$. It follows that $h_j = h|_{\mathcal{X}_{n,j}} = (\mathcal{L}_j^{\mathrm{wth}} - \sigma_\eta \Delta_{\mathcal{M}_j}) f_{\ell,j}$ on $\mathcal{X}_{n,j}$. Applying Lemma 8(3) of Li et al. (2025) componentwise with the rescaling $t \mapsto t/\sqrt{\epsilon_+}$ gives, with probability at least $1 - C\epsilon_+^{-d_j} \exp(-cn_j \epsilon_+^{d_j} t^2)$,

$$\epsilon_+ \|h_j\|_{L^2(\mathcal{X}_{n,j})} \le C_{\ell,j} (\epsilon_+^2 + t),$$

for some constant $C_{\ell,j}$ depending only on the index ℓ and the geometric and density-class parameters. Choosing $t \asymp \epsilon_+^2$, weighting the resulting estimates, and taking a union bound over the N fixed number of components yields, with probability at least $1 - CN\epsilon_+^{-d} \exp(-cn\epsilon_+^{d+4})$,

$$\sum_{j=1}^{N} w_j \epsilon_+ \|h_j\|_{L^2(\mathcal{X}_{n,j})} \le C_\ell \epsilon_+^2.$$

For the multiscale cell-average, fix j, a, p, $\upsilon \in \Upsilon_{p,a,j}^{m_j}$, and define the normalized cell indicator

$$\psi_{p,a,j,\upsilon}(x) := \frac{\mathbf{1}_{\square_{p,a,j,\upsilon}^{m_j}}(x)}{\mu_j(\square_{p,a,j,\upsilon}^{m_j})}.$$

The corresponding componentwise form of Proposition 3.10 of Trillos et al. (2026) gives

$$\left| \frac{1}{n_j} \sum_{x \in \mathcal{X}_{n,j}} \psi_{p,a,j,\upsilon}(x) h_j(x) \right| \le C_{\ell,j}(B_\ell + 1)\epsilon_+^2 s_{p,j}^{-1}$$

for each fixed cell, except on an event of probability at most $Cn_j \exp(-cn_j\epsilon_+^{d_j+4})$. Here $C_{\ell,j}$ depends on the geometric and density class parameters, and the kernel profile η. Moreover,

$$\frac{1}{|Q_{p,a,j,\upsilon}^{m_j}|} \sum_{x \in Q_{p,a,j,\upsilon}^{m_j}} h_j(x) = \frac{n_j \mu_j(\square_{p,a,j,\upsilon}^{m_j})}{|Q_{p,a,j,\upsilon}^{m_j}|} \times \frac{1}{n_j} \sum_{x \in \mathcal{X}_{n,j}} \psi_{p,a,j,\upsilon}(x) h_j(x).$$

On the occupancy event, $|Q_{p,a,j,\upsilon}^{m_j}| \asymp n_j \mu_j(\square_{p,a,j,\upsilon}^{m_j})$, and hence

$$\left| \frac{1}{|Q_{p,a,j,\upsilon}^{m_j}|} \sum_{x \in Q_{p,a,j,\upsilon}^{m_j}} h_j(x) \right| \le C_{\ell,j}(B_\ell + 1)\epsilon_+^2 s_{p,j}^{-1}.$$

It follows from (96) that

$$s_{p,j} T_{p,a,j}(h_j) \le s_{p,j} \left(\frac{1}{|\Upsilon_{p,a,j}^{m_j}|} \sum_{\upsilon \in \Upsilon_{p,a,j}^{m_j}} C_{\ell,j}(B_\ell + 1)^2 \epsilon_+^4 s_{p,j}^{-2} \right)^{1/2} \le C_{\ell,j}(B_\ell + 1)\epsilon_+^2.$$

Since $m_j \asymp \log(\epsilon_+^{-1})$ and the number of charts is uniformly bounded, summing over all components, charts, and scales gives, with probability at $1 - Cn \exp(-cn\epsilon_+^{d+4})$,

$$\sum_{j=1}^{N} w_j \sum_{a=1}^{N_{\nu,j}} \sum_{p=1}^{m_j} s_{p,j} T_{p,a,j}(h_j) \le C_\ell(B_\ell + 1)\epsilon_+^2 \log \epsilon_+^{-1}.$$

Combining the preceding two estimates with the multiscale Poincaré inequality proves the asserted seminorm bound on the simultaneous high-probability event.

■

We next control the contribution from the between-manifold edges. Fix two intersecting components $\mathcal{M}_j$ and $\mathcal{M}_k$, both of maximal intrinsic dimension $d_j = d_k = d$. Let

$$d_{jk} := \dim(\mathcal{M}_j \cap \mathcal{M}_k) \le d - 1, \qquad c_{jk} := d - d_{jk} \ge 1,$$

so that c_{jk} is the codimension of the intersection in either component. The argument uses the same tubular-neighborhood counting twice: globally for the L^2 estimate and cellwise in the multiscale Poincaré inequality for the H^1-estimate.

Proposition 35 (Between-manifold Laplacian bound) *There exists some constant $C_\ell > 0$, depending only on the geometric, the density-class, and the intersection parameters such that for all $d \geq 1$, with probability at least $1 - Cn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4})$,*

$$\|\mathcal{L}^{\mathrm{btw}} f_\ell\|_{L^2(\mathcal{X}_n)} \leq C_\ell B_\ell \epsilon_+ \tag{98}$$

and

$$\|\mathcal{L}^{\mathrm{btw}} f_\ell\|_{H^{-1}(\mathcal{X}_n)} \leq C_\ell B_\ell \epsilon_+^2. \tag{99}$$

Here B_ℓ denotes the regularity constant defined in (9).

Proof Since the number of components is fixed, it suffices to work with one ordered pair $j \neq k$ and then sum the resulting estimates. For tube widths $\alpha, \beta \leq \epsilon_+$, define

$$\mathsf{N}_{jk}(\alpha) := \sum_{x \in \mathcal{X}_{n,j}} \mathbf{1}_{\{\mathrm{dist}(x, \mathcal{M}_k) \leq \alpha\}}, \qquad \mathsf{D}_{jk}(x;\beta) := \sum_{y \in \mathcal{X}_{n,k}} \mathbf{1}_{\{K_\epsilon^E(x,y) \neq 0,\, \mathrm{dist}(y, \mathcal{M}_j) \leq \beta\}}.$$

Thus, $\mathsf{N}_{jk}(\alpha)$ counts candidate j-endpoints in the α-tube of $\mathcal{M}_{jk} := \mathcal{M}_j \cap \mathcal{M}_k$, whereas $\mathsf{D}_{jk}(x;\beta)$ counts the admissible k-neighbors lying within the ambient ϵ_+-neighborhood of a fixed x and the β-tube of $\mathcal{M}_{jk}$. Tubular coordinates around $\mathcal{M}_{jk}$ give

$$\mathrm{vol}_{\mathcal{M}_j}\{x \in \mathcal{X}_{n,j} : \mathrm{dist}(x, \mathcal{M}_k) \leq \alpha\} \leq C_{jk}\alpha^{c_{jk}},$$

and

$$\mathrm{vol}_{\mathcal{M}_k}\{y \in \mathcal{X}_{n,k} : \|x - y\| \leq \epsilon_+,\ \mathrm{dist}(y, \mathcal{M}_j) \leq \beta\} \leq C_{jk}\epsilon_+^{d-c_{jk}}\beta^{c_{jk}}.$$

Furthermore, for a level-p cell of side length $s := s_{p,j}$, the corresponding local estimate is

$$\mathrm{vol}_{\mathcal{M}_j}\left(\square_{p,a,j,v}^{m_j} \cap \{x \in \mathcal{X}_{n,j} : \mathrm{dist}(x, \mathcal{M}_k) \leq \alpha\}\right) \leq C_{jk} s^{d-c_{jk}} \alpha^{c_{jk}}.$$

Since $\alpha, \beta \leq \epsilon_+$,

$$\alpha^{c_{jk}} \leq \epsilon_+^{c_{jk}-1}\alpha, \qquad \epsilon_+^{d-c_{jk}}\beta^{c_{jk}} \leq \epsilon_+^{d-1}\beta.$$

Bernstein's inequality and a union bound over the sample points and the $O(\epsilon_+^{-d})$ multiscale cells therefore give an event $\mathcal{E}_{\mathrm{btw}}$ satisfying

$$\mathbb{P}(\mathcal{E}_{\mathrm{btw}}) \geq 1 - Cn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+3}) \geq 1 - Cn\epsilon_+^{-d}\exp(-cn\epsilon_+^{d+4}),$$

on which, simultaneously for the relevant choices $\alpha, \beta \in \{q, r, \epsilon_+\}$,

$$\mathsf{N}_{jk}(\alpha) \leq C_{jk} n \epsilon_+^{c_{jk}-1}\alpha, \qquad \mathsf{D}_{jk}(x;\beta) \leq C_{jk} n \epsilon_+^{d-1}\beta. \tag{100}$$

We work on this event throughout the remainder of the proof.

We first prove (98). By the kernel support and global $C_\ell B_\ell$-Lipschitz continuity of f_ℓ,

$$|f_{\ell,j}(x) - f_{\ell,k}(y)| \leq C_\ell B_\ell \epsilon_+ \qquad \text{whenever } K_\epsilon^E(x,y) \neq 0. \tag{101}$$

Using $w_j/n_j = 1/n$, the boundedness of the kernel profile, and Cauchy-Schwarz inequality,

$$\|\mathcal{L}^{\mathrm{btw}} f_\ell\|_{L^2(\mathcal{X}_n)}^2 \leq C_\ell \frac{\vartheta_\varepsilon^2 B_\ell^2 \epsilon_+^2}{n^3} \sum_{j=1}^N \sum_{k \neq j} \sum_{x \in \mathcal{X}_{n,j}} \mathsf{D}_{jk}(x;\epsilon_+)^2. \tag{102}$$

For upper tube widths (α, β), define the pairwise squared-neighbor count by

$$\mathfrak{C}_{jk}^{(2)}(\alpha,\beta) := \sum_{x\in\mathcal{X}_{n,j},\,\mathrm{dist}(x,\mathcal{M}_k)\le\alpha} \mathsf{D}_{jk}(x;\beta)^2 \le \mathsf{N}_{jk}(\alpha)\sup_x \mathsf{D}_{jk}(x;\beta)^2.$$

Lemma 33 restricts every retained j–k edge to one of the three configurations summarized in Table 3. Under the choice $r = \epsilon_+^3$, Case (3) dominates and so (102) yields (98).

Case	$\mathrm{dist}(x,\mathcal{M}_k)$	$\mathrm{dist}(y,\mathcal{M}_j)$	$\mathsf{N}_{jk}(\alpha)$	$\sup_x \mathsf{D}_{jk}(x;\beta)$	$\mathfrak{C}_{jk}^{(2)}(\alpha,\beta)$
(1)	$(r,\epsilon_+]$	$(0,q]$	$C_{jk}n\epsilon_+^{c_{jk}}$	$C_{jk}n\epsilon_+^{d+3}$	$C_{jk}n^3\epsilon_+^{2d+c+6} \le C_{jk}n^3\epsilon_+^{2d+7}$
(2)	$(q,r]$	$(0,r]$	$C_{jk}n\epsilon_+^{c_{jk}+2}$	$C_{jk}n\epsilon_+^{d+2}$	$C_{jk}n^3\epsilon_+^{2d+c+6} \le C_{jk}n^3\epsilon_+^{2d+7}$
(3)	$(0,q]$	$(0,\epsilon_+]$	$C_{jk}n\epsilon_+^{c_{jk}+3}$	$C_{jk}n\epsilon_+^{d}$	$C_{jk}n^3\epsilon_+^{2d+c_{jk}+3} \le C_{jk}n^3\epsilon_+^{2d+4}$

Table 3: Global squared-neighbor counts for an ordered pair (j,k). The last column counts j-endpoints weighted by the square of their admissible k-neighbor counts.

We next prove (99). Set $h := \mathcal{L}^{\mathrm{btw}} f_\ell$ and define the ordered-pair contribution by

$$h_{jk}(x) := \frac{\vartheta_\varepsilon}{n}\sum_{k\neq j,\, y\in\mathcal{X}_{n,k}} K_\epsilon^E(x,y)\big(f_{\ell,j}(x) - f_{\ell,k}(y)\big), \qquad h_j = h|_{\mathcal{X}_{n,j}} = \sum_{k\neq j} h_{jk}.$$

Recall the componentwise multiscale Poincaré estimate (97). By Cauchy–Schwarz and (98),

$$\sum_{j=1}^N w_j\epsilon_+\|h_j\|_{L^2(\mathcal{X}_{n,j})} \le \epsilon_+\|h\|_{L^2(\mathcal{X}_n)} \le C_\ell B_\ell \epsilon_+^2. \tag{103}$$

It remains to control the multiscale term. Fix a, p, υ, and abbreviate the cell side length by $s := s_{p,j}$ and $Q := Q_{p,a,j,\upsilon}^{m_j}$. Define the local endpoint count by

$$\mathsf{N}_{jk,Q}(\alpha) := \sum_{x\in Q} \mathbf{1}_{\{\mathrm{dist}(x,\mathcal{M}_k)\le\alpha\}}.$$

The local tubular estimate and the same concentration event give

$$\mathsf{N}_{jk,Q}(\alpha) \le C_{jk} n s^{d-c_{jk}} \epsilon_+^{c_{jk}-1}\alpha.$$

In parallel with the global squared count, define the cellwise ordinary connection count by

$$\mathfrak{C}_{jk,Q}^{(1)}(\alpha,\beta) := \sum_{x\in Q,\,\mathrm{dist}(x,\mathcal{M}_k)\le\alpha} \mathsf{D}_{jk}(x;\beta).$$

The resulting estimates are summarized in Table 4. Cases (1) and (3) dominate, so the number of retained j–k edges with their j-endpoint in Q is at most $C_{jk}n^2 s^{d-c_{jk}}\epsilon_+^{d+c_{jk}+3}$. Since $|Q| \asymp n_j s^d \asymp n s^d$, the Lipschitz continuity (101) and $\vartheta_\varepsilon \asymp \epsilon_+^{-d-2}$ imply

$$\begin{aligned}\left|\frac{1}{|Q|}\sum_{x\in Q} h_{jk}(x)\right| &\le \frac{\vartheta_\varepsilon}{n|Q|}\sum_{x\in Q}\sum_{y\in\mathcal{X}_{n,k}} \big|K_\epsilon^E(x,y)\big|\,|f_{\ell,j}(x) - f_{\ell,k}(y)| \le \frac{C_\ell B_\ell \vartheta_\varepsilon \epsilon_+}{n|Q|}\mathfrak{C}_{jk,Q}^{(1)} \\ &\le \frac{C_\ell B_\ell \epsilon_+^{-d-2}\epsilon_+}{n(ns^d)}\left(C_{jk}n^2 s^{d-c_{jk}}\epsilon_+^{d+c_{jk}+3}\right) = C_\ell B_\ell \epsilon_+^{c_{jk}+2} s^{-c_{jk}}.\end{aligned} \tag{104}$$

Case	$\operatorname{dist}(x,\mathcal{M}_k)$	$\operatorname{dist}(y,\mathcal{M}_j)$	$\mathsf{N}_{jk,Q}(\alpha)$	$\sup_x \mathsf{D}_{jk}(x;\beta)$	$\mathfrak{C}^{(1)}_{jk,Q}(\alpha,\beta)$
(1)	$(r,\epsilon_+]$	$(0,q]$	$C_{jk}ns^{d-c_{jk}}\epsilon_+^{c_{jk}}$	$C_{jk}n\epsilon_+^{d+3}$	$C_{jk}n^2s^{d-c_{jk}}\epsilon_+^{d+c_{jk}+3}$
(2)	$(q,r]$	$(0,r]$	$C_{jk}ns^{d-c_{jk}}\epsilon_+^{c_{jk}+2}$	$C_{jk}n\epsilon_+^{d+2}$	$C_{jk}n^2s^{d-c_{jk}}\epsilon_+^{d+c_{jk}+4}$
(3)	$(0,q]$	$(0,\epsilon_+]$	$C_{jk}ns^{d-c_{jk}}\epsilon_+^{c_{jk}+3}$	$C_{jk}n\epsilon_+^{d}$	$C_{jk}n^2s^{d-c_{jk}}\epsilon_+^{d+c_{jk}+3}$

Table 4: Cellwise ordinary connection counts for an ordered pair (j,k). The last column counts ordered retained edges (x,y) with $x \in Q$.

Define the active-cell index set for (j,k) at level (p,a) by

$$\mathcal{A}_{p,a,jk} := \left\{ \upsilon \in \Upsilon^{m_j}_{p,a,j} : Q^{m_j}_{p,a,j,\upsilon} \text{ contains the } \mathcal{M}_j\text{-endpoint of a retained } j\text{–}k \text{ edge} \right\}.$$

Only cells intersecting the $O(\epsilon_+)$-tube around $\mathcal{M}_{jk}$ are active. Since $s \gtrsim \epsilon_+$,

$$|\mathcal{A}_{p,a,jk}| \le C_{jk}s^{-(d-c_{jk})}, \qquad \frac{|\mathcal{A}_{p,a,jk}|}{|\Upsilon^{m_j}_{p,a,j}|} \le C_{jk}s^{c_{jk}}.$$

Combining this active-cell proportion with (104) and the triangle inequality gives

$$T_{p,a,j}(h_j) \le \sum_{k\neq j} T_{p,a,j}(h_{jk}) \le C_\ell B_\ell \epsilon_+^{c_{jk}+2} s^{-c_{jk}/2}.$$

Since $s = 3^{p-m_j}$ and $3^{-m_j} \asymp \epsilon_+$, a geometric series gives the codimension-c_{jk} scale factor

$$\sum_{p=1}^{m_j} s^{1-c_{jk}/2} \le C \begin{cases} 1, & c_{jk}=1, \\ \log(\epsilon_+^{-1}), & c_{jk}=2, \\ \epsilon_+^{1-c_{jk}/2}, & c_{jk}>2. \end{cases}$$

It follows that the multiscale contribution of each ordered pair is bounded by

$$\sum_{p=1}^{m_j} sT_{p,a,j}(h_j) = C_\ell B_\ell \epsilon_+^{c_{jk}+2} \sum_{p=1}^{m_j} s^{1-c_{jk}/2} \le C_\ell B_\ell \begin{cases} \epsilon_+^3, & c_{jk}=1, \\ \epsilon_+^4 \log(\epsilon_+^{-1}), & c_{jk}=2, \\ \epsilon_+^{c_{jk}/2+3}, & c_{jk}>2. \end{cases}$$

Because the numbers of charts and components are fixed, summing over a,j,k yields

$$\sum_{j=1}^{N} w_j \sum_{a=1}^{N_{\nu,j}} \sum_{p=1}^{m_j} sT_{p,a,j}(h_j) \le C_\ell B_\ell \epsilon_+^3,$$

for all $d \ge 1$. Combining this estimate with (103) proves (99).

■

Proposition 36 (Between-manifold Dirichlet energy bound) *Choose ϵ_+ such that $n\epsilon_+^{5d-2} \to \infty$. Then there exists some constant $C_\ell > 0$, depending only on the geometric, the density-class, and the intersection parameters such that for all $d \geq 1$,*

$$\left|\langle f_l, \mathcal{L}^{\text{btw}} f_l\rangle_{L^2(\mathcal{X}_n)}\right| \leq C B_\ell^2 \epsilon_+^4,$$

with probability at least $1 - Cn\exp\left(-cn\epsilon_+^{d+4}\right)$. B_ℓ denotes the regularity constant from (9).

Proof By the definition of the between-manifold Dirichlet energy in (48)

$$\langle f_l, \mathcal{L}^{\text{btw}} f_l\rangle_{L^2(\mathcal{X}_n)} = \frac{\vartheta_\varepsilon}{2n^2} \sum_{j=1}^{N} \sum_{x\in\mathcal{X}_{n,j}} \sum_{k\neq j,\, y\in\mathcal{X}_{n,k}} K_\epsilon^E(x,y)(f_\ell(x) - f_\ell(y))^2.$$

By the $C_\ell B_\ell$-Lipschitz continuity of f_ℓ, $\left(f_\ell(x) - f_\ell(y)\right)^2 \leq C_\ell B_\ell^2 \epsilon_+^2$ whenever $K_\epsilon^E(x,y) \neq 0$. It follows that

$$\left|\langle f_l, \mathcal{L}^{\text{btw}} f_l\rangle_{L^2(\mathcal{X}_n)}\right| \leq C\frac{\vartheta_\varepsilon B_\ell^2 \epsilon_+^2}{n^2} \sum_{j=1}^{N} \sum_{x\in\mathcal{X}_{n,j}} \sum_{k\neq j,\, y\in\mathcal{X}_{n,k}} 1_{\{\|x-y\|\leq\epsilon_+\}}, \tag{105}$$

By Theorem (31), with probability at least $1 - Cn\exp(-cn\epsilon_+^{d+4})$, the number of retained edges between $\mathcal{M}_j$ and $\mathcal{M}_k$ is at most $Cn^2\epsilon_+^{d+4}$. Substituting into (105) yields

$$\left|\langle f_l, \mathcal{L}^{\text{btw}} f_l\rangle_{L^2(\mathcal{X}_n)}\right| \leq C B_\ell^2 \epsilon_+^{-d-2}\epsilon_+^2\epsilon_+^{d+4} = C B_\ell^2 \epsilon_+^4.$$

Finally, the scale condition required by Theorem (31) becomes $r^{2d-1}/\epsilon_+^{d-1} \asymp \epsilon_+^{5d-2}$. Thus, it is enough to require $n\epsilon_+^{5d-2} \to \infty$, completing the proof. ■

C.3 Geodesic Symmetrization for the Density Remainder

Fix $t \in [0,1]$. We define the geodesic-flow symmetrization map $\Phi_t : T\mathcal{M}_j \to T\mathcal{M}_j$ by

$$\Phi_t(x,v) = (\bar{x},\bar{v}) = \left(\exp_x(2t\epsilon_+ v),\ -d(\exp_x)_{2t\epsilon_+ v}(v)\right).$$

Geometrically, Φ_t pairs each tangent vector (x,v) with its time-reversed counterpart along the same geodesic. Since Φ_t is a reparametrization of the geodesic flow, it preserves the canonical volume measure on the tangent bundle. Moreover, $\exp_{\bar{x}}(t\epsilon_+\bar{v}) = \exp_x(t\epsilon_+ v)$ so the Hessian factor is invariant under the pairing $\mathcal{D}_t(\bar{x},\bar{v}) = \mathcal{D}_t(x,v)$:

$$\mathcal{D}_t(x,v) = \nabla_g^2\rho_j(\exp_x(t\epsilon_+ v))\left[d(\exp_x)_{t\epsilon_+ v}(v), d(\exp_x)_{t\epsilon_+ v}(v)\right], \qquad |\mathcal{D}_t(x,v)| \leq C_\rho^{(2)}\|v\|^2.$$

The measure-preserving property of Φ_t symmetrizes the density remainder term as

$$\mathcal{I}_t = \frac{\vartheta_\varepsilon\epsilon_+^{d+3}}{2}\sum_{j=1}^{N} w_j^2 \int_{T\mathcal{M}_j} K_\epsilon^{\mathcal{N}}(v)\mathcal{D}_t(x,v)\Big[f_\ell(x)\rho_j(x)\langle\nabla_g f_\ell(x), v\rangle_g + f_\ell(\bar{x})\rho_j(\bar{x})\langle\nabla_g f_\ell(\bar{x}),\bar{v}\rangle_g\Big]\, dm(x,v),$$

where dm denotes the canonical measure on the tangent bundle, restricted to $\|v\| \le 1$. The advantage of this representation is that the Hessian factor has become common to both terms, allowing the cancellation to occur entirely inside the bracket.

We now estimate the bracket. Adding and subtracting $f_\ell(\bar{x})\rho_j(\bar{x})\langle\nabla_g f_\ell(x), v\rangle_g$ yields

$$\begin{aligned} &|f_\ell(x)\rho_j(x)\langle\nabla_g f_\ell(x), v\rangle_g + f_\ell(\bar{x})\rho_j(\bar{x})\langle\nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g| \\ \le\ &|f_\ell(\bar{x})\rho_j(\bar{x})||\langle\nabla_g f_\ell(x), v\rangle_g + \langle\nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g| + |\langle\nabla_g f_\ell(x), v\rangle_g||f_\ell(x)\rho_j(x) - f_\ell(\bar{x})\rho_j(\bar{x})|. \end{aligned}$$

The first term corresponds to the cancellation of the directional derivatives. Since (x, v) and $(\bar{x}, \bar{v})$ lie on the same geodesic and $\bar{v}$ is the transported negative of v, we invoke the cancellation estimate (to be established below)

$$|\langle\nabla_g f_\ell(x), v\rangle_g + \langle\nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g| \le C\epsilon_+\|v\|^2\|\nabla_g^2 f_\ell\|_{L^\infty(B_g(x,2\epsilon_+))}.$$

The second term measures the variation of the product $f_\ell\rho_j$ between x and $\bar{x}$. Since $d_g(x, \bar{x}) \le C\epsilon_+\|v\|$, the mean value theorem implies

$$|f_\ell(x)\rho_j(x) - f_\ell(\bar{x})\rho_j(\bar{x})| \le C\epsilon_+\|v\|\left(\rho_{\max}\|\nabla_g f_\ell\| + C_\rho^{(1)}\|f_\ell\|\right),$$

where the derivatives are evaluated along the geodesic segment from x to $\bar{x}$.

Combining these estimates and applying Cauchy–Schwarz, we obtain

$$\begin{aligned} |\mathcal{I}_t| &\le C\sum_{j=1}^N w_j^2\vartheta_\varepsilon\epsilon_+^{d+4}\int_{\mathcal{M}_j}\left(|f_\ell(x)|\|\nabla_g^2 f_\ell\|_{L^\infty(B_g(x,2\epsilon_+))} + |\nabla_g f_\ell(x)|^2 + |f_\ell(x)||\nabla_g f_\ell(x)|\right)d\mu_j \\ &\le C\epsilon_+^2\sum_{j=1}^N w_j^2\left(\|f_\ell\|_{L^2(\mathcal{M}_j)}\|\nabla_g^2 f_\ell\|_{L^\infty(\mathcal{M}_j)} + \|\nabla_g f_\ell\|_{L^2(\mathcal{M}_j)}^2 + \|f_\ell\|_{L^2(\mathcal{M}_j)}\|\nabla_g f_\ell\|_{L^2(\mathcal{M}_j)}\right). \end{aligned}$$

The eigenfunction's regularity allows the bracket to be absorbed into the constant. Since the number of manifold components and the mixture weights are fixed, $|\mathcal{I}_t| \le C_\ell B_\ell^2\epsilon_+^2$.

It remains to quantify the cancellation in the symmetrized integrand, we compare the directional derivatives at the paired configuration (x, v) and $(\bar{x}, \bar{v})$, where $\bar{x} := \exp_x(\epsilon_+ v)$ and $\bar{v} := \log_{\bar{x}}(x)/\epsilon_+$. Let $\gamma : [0, 1] \to \mathcal{M}_j$ be the constant-speed minimizing geodesic from x to $\bar{x}$, so that $\dot{\gamma}(0) = \log_x(\bar{x}) = \epsilon_+ v$. Since the tangent vector at its endpoint points opposite to the log map back to its start, $\dot{\gamma}(1) = -\log_{\bar{x}}(x) = -\epsilon_+\bar{v}$. Define $\tilde{f}(s) := f_\ell(\gamma(s))$. Then

$$\tilde{f}'(s) = \langle\nabla_g f_\ell(\gamma(s)), \dot{\gamma}(s)\rangle_g, \qquad \tilde{f}''(s) = \langle\nabla_g^2 f_\ell(\gamma(s))[\dot{\gamma}(s)], \dot{\gamma}(s)\rangle_g.$$

Applying the fundamental theorem of calculus $\tilde{f}'(1) - \tilde{f}'(0) = \int_0^1 \tilde{f}''(s)\mathrm{d}s$ yields

$$-\epsilon_+\left(\langle\nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g + \langle\nabla_g f_\ell(x), v\rangle_g\right) = \int_0^1 \langle\nabla_g^2 f_\ell(\gamma(s))[\dot{\gamma}(s)], \dot{\gamma}(s)\rangle_g \mathrm{d}s.$$

Since $\|\dot{\gamma}(s)\|_g = d_g(x, \bar{x}) \le C\epsilon_+\|v\|$ for all $s \in [0, 1]$, taking the absolute value and bounding the integral gives

$$\epsilon_+\,|\langle\nabla_g f_\ell(\bar{x}), \bar{v}\rangle_g + \langle\nabla_g f_\ell(x), v\rangle_g| \le C\epsilon_+^2\|v\|^2\|\nabla_g^2 f_\ell\|_{L^\infty(B_g(x,\epsilon_+))}.$$

Dividing both sides by ϵ_+ yields the desired estimate.